

A Dual-Layer Framework for Interference and Anomaly Detection in Satellite Systems

Diploma Thesis
Dimitris Charistes

Department of Electrical and Computer Engineering
Faculty of Engineering
ITHACA Laboratory

Advisor: Dr. Panagiotis Sarigiannidis

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Περίληψη

Μια δορυφορική αποστολή μπορεί να αστοχήσει από δύο κατευθύνσεις ταυτόχρονα: από έξω, όταν παρεμβολή αλλοιώνει τη ραδιοζεύξη, και από μέσα, όταν βλάβη υποσυστήματος εμφανίζεται ως ανώμαλη τηλεμετρία λειτουργικής κατάστασης. Η παρούσα διπλωματική εργασία αναπτύσσει ένα πλαίσιο ανίχνευσης δύο επιπέδων που αντιμετωπίζει και τις δύο απειλές ως εκφάνσεις μίας και μόνης αρχής: μοντελοποίηση της κανονικής συμπεριφοράς, ανακατασκευή κάθε παρατήρησης από το μοντέλο αυτό, και σήμανση του υπολοίπου που το μοντέλο δεν μπορεί να εξηγήσει, με χρήση κατωφλίου προσαρμοζόμενου στα δεδομένα. Το πρώτο επίπεδο ανιχνεύει ραδιοσυχνοτικές παρεμβολές, και ειδικότερα την απειλή φέροντος υπό φέρον (Carrier-under-Carrier), σε δορυφορικό αναμεταδότη FDMA, συνδυάζοντας τμηματοποίηση φερόντων με μετασχηματισμό κυματιδίων και ένα χρονικό στάδιο αφαίρεσης καθαρής αναφοράς. Σε αναπαραγώγιμο σύνολο αναφοράς δέκα σεναρίων, με προσομοιωμένο αναμεταδότη 36 MHz σε συνθήκες σχεδόν πλήρους φασματικής κατάληψης, η ανίχνευση αυξάνεται από 7 στα 12 γεγονότα παρεμβολής (ένα μόνο στιγμιότυπο, μηδενικοί ψευδείς συναγερμοί) σε 12 στα 12 (χρονικό στάδιο, μηδενικοί ψευδείς συναγερμοί), συμπεριλαμβανομένων παρεμβολών 22 dB κάτω από το φέρον-ξενιστή τους. Το δεύτερο επίπεδο ανιχνεύει συλλογικές ανωμαλίες στην τηλεμετρία διαστημικού οχήματος μέσω αραιής αναπαράστασης, με λεξικό μαθημένο μέσω K-SVD και από κοινού επίλυση της αραιής κωδικοποίησης και του υπολοίπου με ADMM. Αξιολογείται στο δημόσιο σύνολο αναφοράς SMAP/MSL της NASA έναντι ελεγχόμενης βάσης σύγκρισης ανακατασκευής PCA και δημοσιευμένων αποτελεσμάτων βαθιάς μάθησης: ο αραιός ανιχνευτής επιτυγχάνει το καλύτερο σημειακό F_1 (0,52 έναντι 0,44) και τον χαμηλότερο ρυθμό ψευδών συναγερμών, ενώ η επιλογή της σύμβασης βαθμολόγησης, και όχι η μέθοδος, αποδεικνύεται ότι καθορίζει τον φαινομενικό νικητή. Τα δύο επίπεδα ενοποιούνται τυπικά: ο χρονικός ανιχνευτής ραδιοσυχνοτήτων αποδεικνύεται ότι αποτελεί την εκφυλισμένη περίπτωση ενός ατόμου της ανακατασκευής με αραιό λεξικό, ενώ η αστοχία του κατωφλίου της τηλεμετρίας σε καλά ανακατασκευασμένα δεδομένα επαναπροκύπτει ανεξάρτητα ως το προσθετικό περιθώριο CFAR του ραδιοσυχνοτικού επιπέδου.

Λέξεις-κλειδιά: FDMA, δορυφορικός αναμεταδότης, ραδιοσυχνοτική παρεμβολή, φέρον υπό φέρον, ανάλυση κυματιδίων, τηλεμετρία διαστημικού οχήματος, ανίχνευση ανωμαλιών, αραιή αναπαράσταση, μάθηση λεξικού, K-SVD, ADMM, προσαρμοστική κατωφλίωση, CFAR

Abstract

A satellite mission can fail from two directions at once: from outside, when interference corrupts the radio link, and from inside, when a subsystem fault appears as anomalous housekeeping telemetry. This diploma thesis develops a dual-layer detection framework that addresses both threats as instances of one principle: model normal behaviour, reconstruct each observation from that model, and flag whatever residual the model cannot explain, using a data-adaptive threshold. The first layer detects RF interference — specifically the Carrier-under-Carrier threat — in an FDMA satellite transponder, combining multi-scale wavelet carrier segmentation with a temporal, clean-baseline-differencing stage; on a reproducible ten-scenario benchmark of a simulated 36 MHz transponder operated near full spectral occupancy, detection rises from 7 of 12 interference events (single-snapshot, zero false alarms) to 12 of 12 (temporal stage, zero false alarms), including interferers 22 dB below their host carrier. The second layer detects collective anomalies in spacecraft telemetry by sparse representation — a K-SVD-learned dictionary with an ADMM joint sparse-coding and residual solve — and is evaluated on the public NASA SMAP/MSL benchmark against a controlled PCA reconstruction baseline and published deep-learning results: the sparse detector attains the better point-wise F_1 (0.52 vs. 0.44) and the lower false-alarm rate, while the choice of scoring convention, rather than the method, is shown to determine the apparent winner. The two layers are unified formally: the temporal RF detector is shown to be the degenerate single-atom instance of sparse-dictionary reconstruction, and the telemetry threshold’s failure on well-reconstructed data independently re-derives the RF layer’s additive CFAR margin.

Keywords: FDMA, satellite transponder, RF interference, carrier under carrier, wavelet analysis, spacecraft telemetry, anomaly detection, sparse representation, dictionary learning, K-SVD, ADMM, adaptive thresholding, CFAR

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Dimitrios Charistes
ece02031@uowm.gr

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Chapter 1

Introduction

1.1 Motivation

A satellite mission can be compromised from two directions at once: from outside, when interference corrupts the radio link, and from inside, when a subsystem fault surfaces as anomalous housekeeping telemetry. This thesis builds an automated monitoring system that watches both directions at once; this opening chapter explains why such a system is needed, beginning with the practical experience that prompted it.

As part of the ASPIRE student rocketry project I designed the vehicle's telemetry system end to end, from the physical layer (power amplification, PCB layout, and antenna design in HFSS) through protocol encoding, packetisation and forward error correction. Two problems encountered during the competition shaped the direction of this thesis. First, the heavy use of the 2.4 GHz band by the many participating teams meant that our receiving antenna continually picked up other teams' telemetry as interference, distorting our own link. This was a small-scale but vivid instance of the congestion-driven interference that satellite operators face, and a reminder that a receiver cannot assume the energy in its band belongs to it. Second, and more troubling, nothing checked the telemetry against a model of what normal flight data looks like. Every sensor could pass its power-on checks and report values that were non-zero and individually within range, while the row of readings for a given timestamp still described a flight that was not physically consistent, a velocity that did not follow from the accelerations logged beside it. A per-channel limit check cannot see a fault of this kind, because no single channel leaves its range. What is wrong is a relationship: either between channels at one instant, or

between successive samples of one channel over time. For mission-critical events such as airbrake deployment and initiation of the recovery sequence, we could only confirm *after* the flight whether the data driving those decisions had been sound. In the moment, we could not distinguish a genuine sensor anomaly from corrupted or interfered data. If these failure modes are challenging for a rocket whose flight lasts minutes and whose receiver sits a few kilometres away, they are far more acute for satellite systems, whose signals traverse the full depth of the atmosphere, whose links carry both uplink commands and downlink telemetry, and whose hardware cannot be inspected after the fact. The timescale also changes what detection is worth. The competition allowed no remote intervention and the flight lasted about three minutes, so on that vehicle a check run after recovery would have served much the same purpose as a live one. A spacecraft operates for years, can be commanded from the ground, and develops faults that are often recoverable while they are still forming but not once they are found in post-hoc analysis. Both problems therefore need answering at satellite scale: interference hidden in a crowded band has to be caught, and anomalous telemetry has to be recognised as it arrives rather than in hindsight. A single automated monitoring system can do both.

1.2 Interference and Telemetry Anomalies

Satellites underpin much of modern communication, navigation, Earth observation and defence infrastructure, and the orbital and spectral environment in which they operate has become far more crowded. The number of active satellites passed 10,000 during 2024 and continues to climb, driven largely by large low-Earth-orbit constellations [12], while the usable radio spectrum remains a finite, shared resource under tight regulatory allocation. In Frequency-Division Multiple Access (FDMA) satellite networks many carriers share a single transponder bandwidth, and as occupancy grows the margin that separates legitimate carriers from unwanted emissions shrinks. Radio-Frequency (RF) interference is consequently a persistent and growing operational concern for satellite operators, whether it is accidental (adjacent-carrier spill-over, mispointed terminals, terrestrial sources) or deliberate (jamming and spoofing).

A particularly insidious form of this problem is the *Carrier-under-Carrier* (CUC) situation, in which a weak, often narrower interfering carrier occupies the same centre frequency and bandwidth as a legitimate host carrier. Buried beneath the host, such a signal is easily mistaken for benign channel structure and can slip past conventional energy-threshold detectors. Yet an undetected CUC can quietly degrade a legitimate link, mask unauthorised access, or, in a command-and-telemetry channel, corrupt the very information on which operational decisions depend. Detecting it reliably, on both the uplink and the

downlink, is therefore a prerequisite for trustworthy satellite operation.

Interference on the link, however, is only half of the picture. The information the link carries, the spacecraft's housekeeping telemetry, is itself a safety-critical asset. Temperatures, currents, voltages and sensor states report the vehicle's health, and an anomaly among them may signal an incipient fault whose cost, if missed, ranges from degraded mission return to the loss of the vehicle. Operationally, such anomalies are hard to catch as they happen: telemetry is high-dimensional, strongly correlated and, on orbiting spacecraft, often periodic, so fixed alarm limits either flood operators with false alarms or miss the subtle, *collective* anomalies that unfold gradually across many samples. What the limits miss surfaces only in post-hoc analysis, after the decisive moment has passed: the CNES study behind the NOSTRADAMUS monitoring tool found anomalies that had passed the operational system unnoticed [13].

1.3 Threat Scenarios and Signatures

Figure 1.3.1 places the two threats in the physical scene. An interfering ground terminal, mispointed or unauthorised, transmits into an already-occupied slot of the transponder (step 1); the transponder, a transparent relay, amplifies and re-transmits the composite band (step 2), so by the time it reaches the ground the interference is buried inside a legitimate carrier and appears at the monitoring station's spectrum tap only as a small excess in power density (step 3). Independently of the link, a housekeeping sensor on the spacecraft bus malfunctions (step a); its readings are encoded, frame by frame, into the telemetry stream (step b) and arrive at the station's telemetry tap as values that are individually plausible yet collectively wrong (step c). The monitoring station itself is deliberately passive: it demodulates no customer carrier and commands nothing (Chapter 4). Its two taps correspond one-to-one to the two detection layers this thesis develops.

What the two taps actually record during these events is shown in Figure 1.3.2, built from the same data studied in the remainder of this thesis; each panel contrasts the observation with the naive check that current practice would apply to it. The upper row shows Welch spectra of the simulated 36 MHz FDMA transponder of the RF benchmark (Chapter 4), operated near full occupancy. Panel (a) depicts interference of the familiar kind: a strong emitter switches on inside the band and towers well above the legitimate carriers, so any energy threshold set above the nominal spectral ceiling flags it immediately. Panel (b) shows the same transponder while a carrier-under-carrier event is actually in progress: the reference benchmark scenario E01, in which a 300 ksym/s interferer 14 dB below its host transmits from inside the host's own bandwidth. The composite spectrum

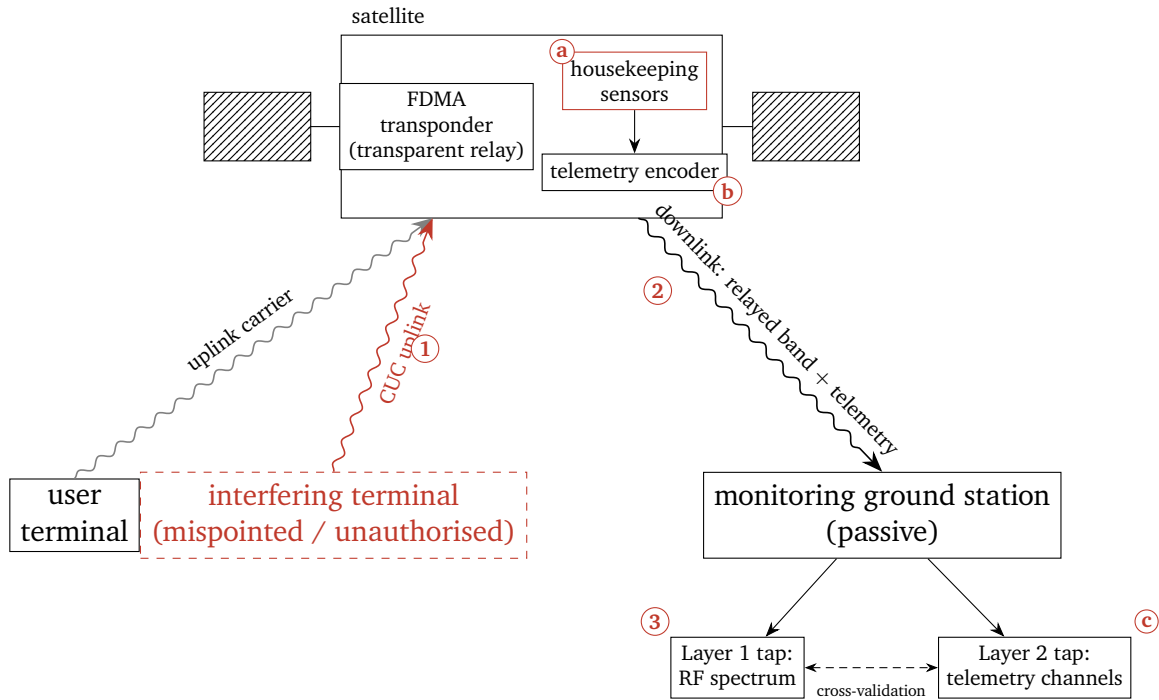


Figure 1.3.1: The physical scene of the two threats. An interfering terminal transmits into an occupied slot of the FDMA transponder (step 1); the transparent transponder relays the composite band (step 2), so the interference reaches the monitoring station's spectrum tap only as a small density excess buried inside a legitimate carrier (step 3). Independently, a housekeeping sensor on the spacecraft bus malfunctions (step a); its readings are encoded into the downlinked telemetry stream (step b) and arrive at the telemetry tap as an in-range, collective anomaly (step c). The station is a passive observer, and its two taps correspond one-to-one to the two detection layers of this thesis; when the telemetry carrier shares the monitored band, an event visible to both taps at once indicates interference corrupting the data itself.

lies on top of its interference-free twin across the entire band; the event surfaces only in the magnified inset, as a power-density excess of ≈ 1.5 dB on the host's flat top. This course of events is what makes the threat dangerous: the interferer switches on, the spectrum keeps its shape, no threshold fires, and the host link degrades while the monitor reports business as usual. Detecting exactly this event class, including bumps three times smaller than E01's, is the task of the detection layer developed in Chapter 5.

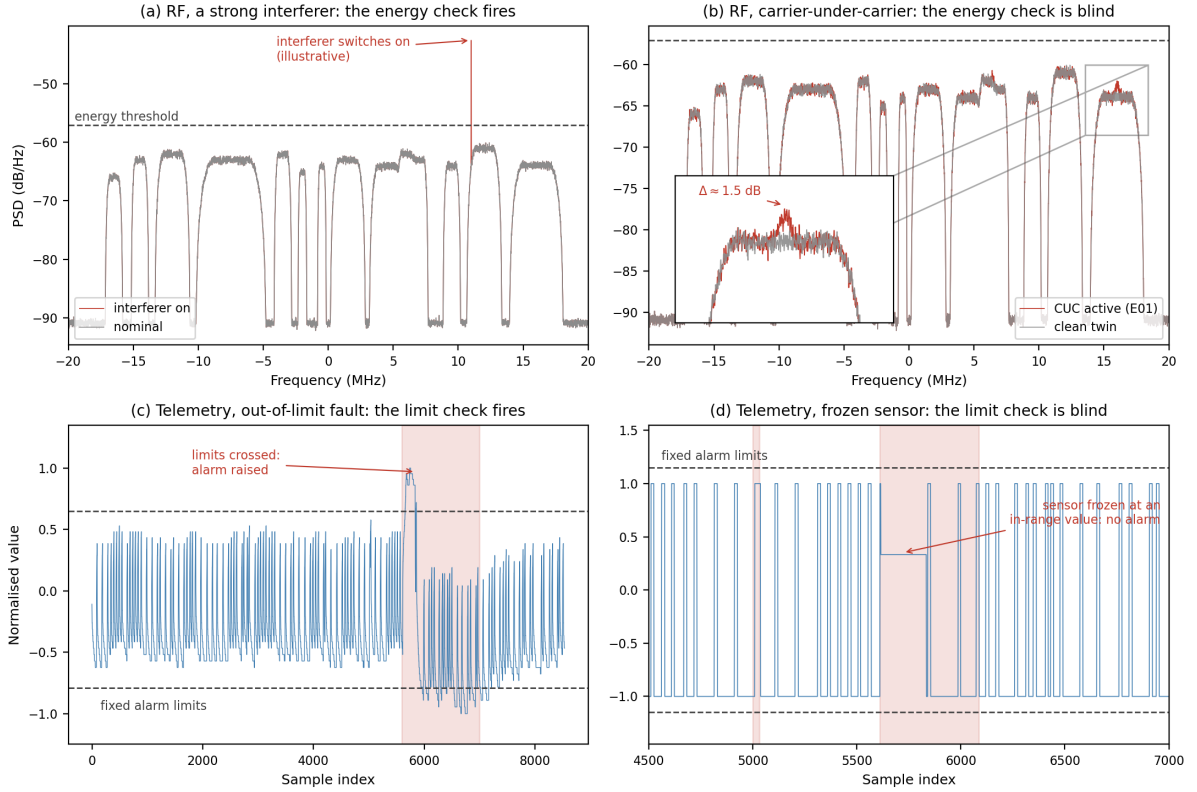


Figure 1.3.2: The two threat classes addressed in this thesis, each contrasted with the naive check of current practice (dashed). *Top*: Welch spectra of the simulated 36 MHz FDMA transponder (Chapter 4). (a) A strong interferer (illustrative tone added to the clean capture) exceeds any energy threshold. (b) A genuine carrier-under-carrier event, benchmark scenario E01: the spectrum with the CUC active (red) is visually identical to its interference-free twin (grey); the inset magnifies the wide host at +16 MHz, where a 300 ksym/s interferer 14 dB below the host raises the flat top by ≈ 1.5 dB. *Bottom*: real SMAP spacecraft telemetry with expert-labelled fault windows shaded. (c) Channel E-2: the fault leaves the nominal envelope and crosses fixed alarm limits. (d) Channel E-1 (magnified view): the sensor freezes at an in-range value; no individual sample violates the limits and the check stays silent.

The lower row shows the second threat on real spacecraft telemetry from the NASA SMAP mission, with the expert-labelled fault windows shaded. In panel (c) the fault announces itself: the signal breaks out of the envelope it has occupied for thousands of samples, and a fixed out-of-limit check (the standard first line of defence in operational telemetry

monitoring) correctly raises an alarm. Panel (d) shows a subtler course of events on a neighbouring channel: a sensor that nominally toggles across its full range freezes at an interior value for over two hundred samples. Every individual reading in the fault window is a perfectly legal value, so the limit check never fires; what is anomalous is not any sample but the collective pattern: the periodic structure that has silently gone missing. Detecting this collective class of anomaly is the task of the layer developed in Chapter 6.

These two concerns, interference on the link and anomalies in the data, are usually handled by separate tools, yet they are two faces of a single question: *is what we are observing normal?* An abnormal observation at the receiver may be a genuine spacecraft fault, corrupted data arising from link errors, or the signature of interference such as a CUC. Resolving which of these it is calls for evidence from both domains. Monitoring the RF spectrum and the telemetry stream in parallel therefore does more than widen anomaly coverage: it enables *cross-validation*. An abnormality that appears in the telemetry but not in the spectrum points to the platform; one that appears in the spectrum points to the link; one that appears in both points to interference corrupting the data. That last reading applies when the telemetry carrier shares the monitored band.

The thesis therefore develops such a system as two complementary detection layers, an RF-interference layer and a telemetry-anomaly layer, unified by a common principle: model what is normal, reconstruct each new observation from that model, and flag the residual that the model cannot explain.

1.4 Problem Statement and Objectives

The problem addressed by this thesis can be stated as a single question posed in two domains: given a satellite communications link, can abnormal conditions be detected automatically and early, whether they arise in the radio-frequency spectrum or in the spacecraft telemetry? The two instances illustrated in Section 1.3 are precisely the hard ones: a weak, narrow CUC raises the host's power density by too little for any single-snapshot threshold to separate from the host's own structure, and a collective telemetry anomaly keeps every individual sample within its nominal range while corrupting the pattern they form together. Treating the two domains in isolation, furthermore, leaves no basis for deciding whether an observed abnormality is interference, corrupted data, or a genuine fault.

Accordingly, the objectives of this thesis are the following.

- O1** Construct a realistic, reproducible FDMA satellite transponder environment with controllable interference (including CUC anomalies), and an accompanying bench-

mark with explicit ground truth, so that detection performance can be measured objectively.

- O2** Detect CUC interference reliably, including the weak and narrow cases that single-snapshot methods miss, with two complementary stages: a multi-scale wavelet carrier segmentation with a flat-top bump test, and a temporal, baseline-referenced integration stage.
- O3** Detect collective anomalies in spacecraft telemetry using sparse representation, specifically K-SVD dictionary learning solved by the Alternating Direction Method of Multipliers (ADMM), evaluated on a public benchmark against established baselines.
- O4** Unify the two detectors under a single reconstruction-residual framework with an adaptive threshold, and evaluate the complete system honestly using metrics appropriate to each domain.

1.5 Contributions

The main contributions of this thesis are the following.

- **A unified reconstruction-residual detection framework.** Both layers are shown to be instances of one principle: model normal behaviour, reconstruct each observation from that model, and threshold the residual it cannot explain. They differ along two axes. The first is the richness of the normality model, from a single clean reference to a learned overcomplete dictionary; the second is the prior built into the threshold, from a data-adaptive Constant-False-Alarm-Rate limit to a periodicity-aware exponentially-weighted band. The temporal RF detector is the single-reference instance; the telemetry detector is the learned-dictionary instance.
- **A wavelet-based RF interference layer with temporal integration.** Building on the multi-scale wavelet technique of [34], a carrier-segmentation and CUC “bump” detector with tunable sensitivity is developed and extended with a multi-frame baseline-differencing stage that recovers weak and narrow CUCs the single-snapshot detector cannot: on the benchmark, detection rises from 7 of 12 CUCs (snapshot, zero false alarms) to 12 of 12 (temporal, zero false alarms).
- **A reproducible FDMA simulation and benchmark.** A scenario-driven GNU Radio transponder model and a ten-scenario benchmark with ground truth support

objective, repeatable evaluation of interference detection, together with drift- and plan-change stress tests of the temporal layer.

- **A sparse-representation telemetry anomaly layer, rigorously evaluated.** The K-SVD-ADMM detector of [18] is implemented from its published equations, two errors in which are identified and corrected here: the data-fidelity term of its on-line objective loses the factor $\frac{1}{2}$ and the square on the Frobenius norm, which breaks the least-squares form of the code update, and its algorithm listing gives the code matrix the wrong dimensions. The detector is then evaluated on the SMAP/MSL benchmark against a controlled PCA-reconstruction baseline and the published LSTM-NDT results, under both point-wise and point-adjusted scoring. The controlled comparison shows that the sparse, overcomplete model buys localisation precision and a lower false-alarm rate, and that the choice of scoring metric, not the method, determines the apparent winner.
- **Honest negative and limiting results.** The thesis documents where intuition fails: data-adaptive entropy weighting proves equivalent to the reference paper's fixed weighting on this band; naive multi-frame averaging cannot separate a weak CUC from deterministic host structure; point-adjusted scoring inflates the apparent performance of broad detectors; and the purely multiplicative telemetry threshold degenerates on well-reconstructed data, which motivates the additive (CFAR) generalisation that unifies it with the RF layer. A controlled ablation of the threshold prior then shows that the periodicity assumption costs performance on the aperiodic majority of the benchmark and only draws level with a structure-free limit on the channels that do carry a period, so the headline configuration sits at the structure-free end.

1.6 Thesis Outline

The remainder of the thesis is organised as follows. Chapter 2 reviews the theoretical background (wavelet analysis, sparse representation and dictionary learning, and the FDMA satellite link and its telemetry) together with related work. Chapter 3 introduces the unified reconstruction-residual framework that underlies both detection layers and positions each within it. Chapter 4 describes the system model and data generation: the FDMA transponder simulation, the signal and power-spectrum model, and the benchmarks used in both domains. Chapter 5 presents Layer 1, the wavelet-based interference detector, from carrier segmentation through the snapshot CUC detector to the temporal baseline-differencing stage. Chapter 6 presents Layer 2, the sparse-representation

telemetry anomaly detector based on K-SVD dictionary learning, ADMM reconstruction and adaptive thresholding. Chapter 7 analyses the performance of both layers with the metrics and comparisons appropriate to each domain. Chapter 8 concludes and outlines directions for future work, including cross-domain fusion of the two detection layers.

Chapter 2

Theoretical Background

2.1 Overview

This chapter develops the theory the two detection layers rest on and reviews the two works from which the thesis takes its methods. It is organised in two parts. Part A (Sections 2.2–2.9) builds the radio-frequency (RF) signal model and the spectral, wavelet and information-theoretic tools used by the interference layer: the FDMA link, the linearly modulated carrier and its pulse-shaping, the Nyquist no-intersymbol-interference condition and the flat-top power spectrum it produces, power-spectral-density estimation, the wavelet transform, and Shannon entropy. Part B develops the foundations of the telemetry layer: the modelling of telemetry anomalies, sparse representation and dictionary learning, and the convex optimisation used to solve the resulting problems. The chapter closes with the related work. The notation introduced here, in particular the symbol set for the carrier model in Sections 2.4–2.5, is used throughout the remainder of the thesis.

2.2 The FDMA Satellite Link

A communications satellite carries a *transponder*: a payload that receives an uplink band from ground terminals, amplifies and frequency-translates it, and retransmits it on a downlink band. In the common *bent-pipe* architecture the transponder performs no demodulation; it relays whatever occupies its passband, so any interfering emission that enters the uplink band is faithfully reproduced on the downlink [25]. This is precisely

why interference must be detected from the spectrum itself rather than assumed away by the payload.

Under *Frequency-Division Multiple Access* (FDMA) a single transponder of bandwidth B_T is shared among N carriers that are separated in frequency. Carrier i is centred at an offset f_i from the band centre and occupies a bandwidth W_i , with the passbands $[f_i - W_i/2, f_i + W_i/2]$ kept disjoint by guard bands. The carriers therefore satisfy

$$\sum_{i=1}^N W_i + \sum (\text{guard bands}) \leq B_T. \quad (2.1)$$

Writing $s_i(t)$ for the complex-baseband waveform of carrier i (defined in Section 2.4), the composite signal presented to the monitoring receiver in complex baseband is

$$r(t) = \sum_{i=1}^N s_i(t) e^{j2\pi f_i t} + w(t), \quad (2.2)$$

where $w(t)$ is additive noise, modelled as a circularly symmetric complex Gaussian process. Equation (2.2) is the signal the detector observes: a sum of frequency-offset carriers on a noise floor.

Two practical features of (2.2) matter later. First, as transponder occupancy grows the guard bands in (2.1) shrink, reducing the margin that separates a legitimate carrier from an unwanted emission and making hidden interference more likely, which is the congestion argument of Chapter 1. Second, a real transponder does not have a perfectly flat frequency response: a gentle *gain tilt* of a few decibels across B_T is typical [25], which appears as a sloped noise floor in the observed spectrum and must be accounted for by the interference detector.

2.3 Noise in the Satellite Link

Every contribution to the composite signal (2.2) other than the carriers is noise, and the term $w(t)$ collects it. Its physical origin is manifold. The dominant part is *thermal* (Johnson–Nyquist) noise, produced by the random motion of charge carriers in the receiver electronics and in every lossy element ahead of them. To it the antenna adds external radiation: the cosmic microwave background at about 2.7 K, diffuse galactic emission, thermal emission from the atmosphere (oxygen and water vapour, rising sharply during rain), and, if they fall in the antenna beam, the Earth and the Sun. These external sources are summarised by an equivalent *antenna noise temperature* T_A and the receiver’s own contribution by a *receiver noise temperature* T_R , which combine into the

system noise temperature

$$T_{\text{sys}} = T_A + T_R. \quad (2.3)$$

By the Johnson–Nyquist relation [30] the available noise power spectral density is flat and equal to

$$N_0 = k_B T_{\text{sys}}, \quad (2.4)$$

with k_B Boltzmann’s constant, so the noise power collected in a bandwidth B is $P_N = N_0 B = k_B T_{\text{sys}} B$.

The statistics of $w(t)$ follow from how this noise is composed. Being the sum of an enormous number of independent microscopic contributions, its amplitude is Gaussian by the central limit theorem; being uncorrelated on the timescales of interest, it is *white*, with the flat spectral density (2.4) across any band far narrower than its correlation bandwidth; and having no preferred phase, its in-phase and quadrature parts are independent with equal power. The channel noise is therefore modelled as additive white Gaussian noise (AWGN): in complex baseband $w(t)$ is a zero-mean, circularly symmetric complex Gaussian process, and its samples satisfy

$$w[n] = w_I[n] + j w_Q[n], \quad w_I, w_Q \sim \mathcal{N}(0, \frac{\sigma^2}{2}) \text{ i.i.d.}, \quad \mathbb{E}[w[n] w^*[m]] = \sigma^2 \delta[n - m], \quad (2.5)$$

with $\mathbb{E}[w[n]] = 0$, $\mathbb{E}[w[n]w[m]] = 0$, and $\sigma^2 = N_0 B$ the noise power in the observed band. Circular symmetry (invariance of the distribution to phase rotation, equivalently $\mathbb{E}[w^2] = 0$) is the appropriate model for passband thermal noise translated to baseband. In the frequency domain white noise has a flat *expected* power spectral density equal to N_0 : this is the *noise floor* on which the carriers of (2.2) sit. A single periodogram estimate of that floor is, however, highly variable: each bin is an exponential (chi-square with two degrees of freedom) random variable whose standard deviation equals its mean. This is the concrete reason the Welch averaging of Section 2.6 is indispensable. Averaging K frames reduces the floor’s variance by a factor K and turns a noisy estimate into a smooth line against which carrier structure can be read. In a real transponder the flat N_0 is further shaped by the frequency-dependent gain of Section 2.2, so the observed floor is a gently tilted line rather than a constant; estimating that tilted floor robustly, in a band crowded with carriers, is a prerequisite for the interference layer.

The strength of a carrier relative to this floor is quantified by the carrier-to-noise ratio $C/N = P_c/(N_0 B)$, or per information bit by E_b/N_0 , and the receiving station’s sensitivity by the figure of merit G/T (antenna gain over system noise temperature). Beyond setting the link’s operating point, the noise floor bounds detection: the Carrier-under-Carrier excess of Section 2.7 must be separated not only from the host’s deterministic structure but from this random floor. The two behave oppositely under integration: the deterministic

host and CUC persist while the Gaussian noise averages down as $1/\sqrt{K}$ (Section 2.6). It is exactly this asymmetry that the temporal layer of the framework (Chapter 3) exploits to lift a weak CUC out of the noise once the host has been cancelled.

2.4 Linearly Modulated Carriers: I/Q Representation and Pulse Shaping

2.4.1 Complex-baseband (I/Q) representation

A real bandpass signal $x(t)$ centred at a carrier frequency f_c can be written in terms of a complex *baseband* (equivalent-lowpass) signal $x_{\text{bb}}(t)$ as

$$x(t) = \Re\{x_{\text{bb}}(t) e^{j2\pi f_c t}\}, \quad x_{\text{bb}}(t) = x_I(t) + j x_Q(t), \quad (2.6)$$

where $x_I(t)$ and $x_Q(t)$ are the in-phase (I) and quadrature (Q) components. All processing in this thesis operates on the sampled complex baseband

$$x_{\text{bb}}[n] = x_{\text{bb}}(nT_s), \quad T_s = 1/f_s, \quad (2.7)$$

with f_s the sampling rate; each sample is one complex number $x_I[n] + j x_Q[n]$, stored as a pair of `float32` values. This is the format of the recordings analysed in later chapters.

2.4.2 Linear modulation

A digitally modulated carrier is formed by linearly modulating a stream of data symbols onto a shaping pulse. Let $\{a_k\}$ be the symbol sequence, drawn from a finite constellation $\mathcal{A}$, transmitted every T seconds; T is the symbol period and $R_s = 1/T$ the symbol rate. With $g(t)$ the transmit pulse, the complex-baseband carrier is

$$s(t) = \sum_k a_k g(t - kT). \quad (2.8)$$

For the quaternary phase-shift keying (QPSK) used throughout, $\mathcal{A} = \{(\pm 1 \pm j)/\sqrt{2}\}$, so $|a_k| = 1$ and each symbol carries two bits. The symbols are modelled as zero-mean and uncorrelated,

$$\mathbb{E}[a_k] = 0, \quad \mathbb{E}[a_k a_\ell^*] = \sigma_a^2 \delta[k - \ell], \quad (2.9)$$

with σ_a^2 the symbol variance and $\delta[\cdot]$ the Kronecker delta.

2.4.3 Pulse shaping: the raised-cosine family

The pulse $g(t)$ is chosen to confine the carrier to its allotted bandwidth while avoiding interference between successive symbols (Section 2.5). The standard choice is the *raised-cosine* (RC) family, defined most simply in the frequency domain. With roll-off factor $\beta \in [0, 1]$, the RC spectrum is

$$G_{\text{RC}}(f) = \begin{cases} T, & |f| \leq \frac{1-\beta}{2T}, \\ \frac{T}{2} \left[1 + \cos \left(\frac{\pi T}{\beta} \left(|f| - \frac{1-\beta}{2T} \right) \right) \right], & \frac{1-\beta}{2T} < |f| \leq \frac{1+\beta}{2T}, \\ 0, & |f| > \frac{1+\beta}{2T}. \end{cases} \quad (2.10)$$

The pulse thus occupies a two-sided bandwidth

$$W = \frac{1+\beta}{T} = (1+\beta) R_s, \quad (2.11)$$

so the roll-off β is the fractional excess bandwidth over the minimum R_s ; the simulations use $\beta = 0.35$.

In practice the shaping is split between transmitter and receiver by a *matched filter*: the transmit and receive filters are each the *root-raised-cosine* (RRC) pulse $g_{\text{RRC}}(t)$, whose spectrum satisfies

$$G_{\text{RRC}}(f) = \sqrt{G_{\text{RC}}(f)}. \quad (2.12)$$

The cascade of the two, which is the pulse that determines intersymbol interference, has spectrum

$$P(f) = G_{\text{RRC}}(f) G_{\text{RRC}}^*(f) = |G_{\text{RRC}}(f)|^2 = G_{\text{RC}}(f), \quad (2.13)$$

that is, the overall pulse $p(t)$ is a raised-cosine pulse. Two facts follow from (2.13) that are central to the thesis: the *end-to-end* pulse $p(t)$ is RC, so it is free of intersymbol interference (Section 2.5); and the *transmitted* carrier is shaped by g_{RRC} , so its power spectrum has the magnitude $|G_{\text{RRC}}(f)|^2 = G_{\text{RC}}(f)$, a flat-topped shape whose consequences we derive next.

2.4.4 Time–frequency localisation

The occupied bandwidth (2.11) is not an accident of the raised-cosine choice but an instance of a fundamental limit: a signal cannot be made arbitrarily narrow in time and in frequency at the same time. This reciprocity fixes the bandwidth of every carrier in this thesis and recurs in the spectral estimator and the wavelet analysis that follow. Let

$g(t)$ have unit energy, $\int |g(t)|^2 dt = 1$, and be centred at the origin in both domains, $\int t |g(t)|^2 dt = 0$ and $\int f |G(f)|^2 df = 0$. Define its time and frequency spreads as the root-mean-square widths

$$\Delta t^2 = \int t^2 |g(t)|^2 dt, \quad \Delta f^2 = \int f^2 |G(f)|^2 df. \quad (2.14)$$

The time–bandwidth (Gabor–Heisenberg) inequality bounds the product of these spreads [24]: with the definitions (2.14),

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}, \quad (2.15)$$

with equality if and only if $g(t)$ is a Gaussian pulse.

The consequence for modulation is immediate. To carry symbols at rate $R_s = 1/T$ the shaping pulse must be localised in time to a duration on the order of T ; by (2.15) its spectrum must then spread over a bandwidth on the order of $1/T = R_s$. A carrier signalling at R_s therefore cannot occupy less than about R_s of bandwidth, the Nyquist minimum recovered as $\beta \rightarrow 0$ in (2.11). The root-raised-cosine excess βR_s is the price paid for pulses with realisable, fast-decaying tails. The same reciprocity governs two constructions used later. In spectral estimation a length- N block localises the observation to a duration NT_s and so cannot resolve frequency more finely than $\Delta f = f_s/N$ (Eq. (2.24)): the resolution bandwidth is this inequality applied to the analysis window, and windowing (Section 2.6) chooses where to sit within the bound, trading a wider main lobe for lower sidelobes. In interference, the same reciprocity sets the width of a CUC’s spectral footprint: a low symbol rate means long pulses and a narrow band. The height of that footprint then follows from power conservation, since a fixed power spread over a narrower band gives a higher density. This is why the *bandwidth* ratio, and not merely the power ratio, drives the bump height in (2.31).

2.5 Intersymbol Interference and the Nyquist Criterion

2.5.1 Intersymbol interference

Consider the matched-filtered signal sampled at the symbol instants $t = mT$. From (2.8) with end-to-end pulse $p(t)$,

$$y[m] = \sum_k a_k p((m-k)T) + n[m] = \underbrace{a_m p(0)}_{\text{desired symbol}} + \underbrace{\sum_{k \neq m} a_k p((m-k)T)}_{\text{intersymbol interference (ISI)}} + n[m], \quad (2.16)$$

where $n[m]$ is filtered noise. The middle term is the *intersymbol interference* (ISI): the leakage of neighbouring symbols into the decision for a_m . It vanishes for every symbol sequence if and only if the pulse has equally spaced zero crossings at the non-zero symbol instants,

$$p(kT) = p(0) \delta[k], \quad k \in \mathbb{Z}. \quad (2.17)$$

Normalising $p(0) = 1$, condition (2.17) states that the sampled pulse is a unit impulse. The following criterion characterises it in the frequency domain.

2.5.2 The Nyquist no-ISI criterion

The zero-ISI condition (2.17) has a classical frequency-domain characterisation, the *Nyquist criterion* [30]: a pulse $p(t)$ normalised to $p(0) = 1$ satisfies (2.17) if and only if its *folded spectrum* is constant,

$$\sum_n P\left(f - \frac{n}{T}\right) = T \quad \text{for all } f, \quad (2.18)$$

a result of applying the Poisson summation formula to the samples $\{p(kT)\}$.

Equation (2.18) is the precise statement of the “flat spectrum” condition: it is the *folded* (aliased) spectrum, not the raw passband, that must be flat.

2.5.3 The raised-cosine pulse satisfies the criterion

The raised-cosine spectrum (2.10) satisfies the Nyquist criterion (2.18): it is band-limited to $|f| \leq (1 + \beta)/2T < 1/T$, so at any frequency at most two aliases overlap, and its band-edge (vestigial) symmetry about $f = 1/2T$, namely $G_{\text{RC}}(\frac{1}{2T} + x) + G_{\text{RC}}(\frac{1}{2T} - x) = T$, makes them sum to T [30]. The RC pulse is therefore free of ISI, and by (2.13) the matched RRC transmit/receive pair produces exactly this RC end-to-end pulse, so an RRC link is ISI-free.

2.5.4 The flat-top power spectrum

The property that the interference detector actually observes is not the folded spectrum but the *power spectral density* (PSD) of a single transmitted carrier. For the linearly modulated signal (2.8) with symbol statistics (2.9), the PSD is

$$S_s(f) = \frac{\sigma_a^2}{T} |G(f)|^2. \quad (2.19)$$

Derivation of (2.19). The signal (2.8) is cyclostationary with period T . Its statistical autocorrelation is

$$R_s(t + \tau, t) = \mathbb{E}[s(t + \tau)s^*(t)] = \sum_k \sum_\ell \mathbb{E}[a_k a_\ell^*] g(t + \tau - kT) g^*(t - \ell T).$$

Using (2.9), $\mathbb{E}[a_k a_\ell^*] = \sigma_a^2 \delta[k - \ell]$, so only $k = \ell$ survives. Averaging over one symbol period to obtain the autocorrelation of the equivalent stationary process,

$$\begin{aligned} \bar{R}_s(\tau) &= \frac{1}{T} \int_0^T R_s(t + \tau, t) dt = \frac{\sigma_a^2}{T} \sum_k \int_0^T g(t + \tau - kT) g^*(t - kT) dt \\ &= \frac{\sigma_a^2}{T} \int_{-\infty}^{\infty} g(u + \tau) g^*(u) du = \frac{\sigma_a^2}{T} \phi_g(\tau), \end{aligned}$$

where the sum over k stitches the per-period integrals into a single integral over all time, and $\phi_g(\tau) = \int g(u + \tau)g^*(u) du$ is the pulse autocorrelation. By the Wiener–Khinchin theorem the PSD is the Fourier transform of $\bar{R}_s(\tau)$, and since $\phi_g(\tau) \leftrightarrow |G(f)|^2$, we obtain (2.19). (A non-zero symbol mean would add discrete spectral lines at multiples of $1/T$; for the zero-mean constellation (2.9) these vanish.) $\square$

For a carrier transmitted through the RRC filter, $G(f) = G_{\text{RRC}}(f)$ and $|G_{\text{RRC}}(f)|^2 = G_{\text{RC}}(f)$ by (2.13), so its PSD is

$$S_s(f) = \frac{\sigma_a^2}{T} G_{\text{RC}}(f) = \begin{cases} \sigma_a^2, & |f| \leq \frac{1 - \beta}{2T}, \\ \text{cosine roll-off}, & \frac{1 - \beta}{2T} < |f| \leq \frac{1 + \beta}{2T}, \\ 0, & \text{otherwise.} \end{cases} \quad (2.20)$$

The carrier’s power spectrum is therefore *flat* at level σ_a^2 across its central passband $|f| \leq (1 - \beta)/2T$, falling off only over the transition band. This *flat top* is the spectral signature of a correctly pulse-shaped, ISI-free carrier. In the FDMA composite (2.2) each carrier contributes a flat-topped $S_{s_i}(f - f_i)$ on the noise floor, and, as developed in Section 2.7, a Carrier-under-Carrier interferer betrays itself as a small, localised excess riding on top of a host carrier’s otherwise flat top. Detecting that excess against the host’s own structure is the task of the interference layer.

2.6 Power Spectral Density Estimation

The interference layer never observes the carrier PSD (2.20) directly; it must *estimate* the power spectrum from a finite record of I/Q samples (2.7). This section sets out the

estimator used throughout, Welch’s averaged periodogram with a Blackman window, together with the resolution, variance and leakage considerations that dictate its parameters.

2.6.1 The DFT and the periodogram

Given a length- N block of complex-baseband samples $x[0], \dots, x[N-1]$, the discrete Fourier transform (DFT) is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, \dots, N-1, \quad (2.21)$$

computed in $\mathcal{O}(N \log N)$ operations by the fast Fourier transform (FFT). Bin k corresponds to the frequency

$$f_k = k \frac{f_s}{N}, \quad k = 0, \dots, N-1, \quad (2.22)$$

which the `fftshift` operation re-indexes to the symmetric range $[-f_s/2, f_s/2)$ so that direct current ($f = 0$) lies at the centre of the axis. The *periodogram* estimate of the PSD is the normalised squared magnitude

$$\hat{S}(f_k) = \frac{1}{N} |X[k]|^2, \quad (2.23)$$

which the thesis reports in decibels as $P(f_k) = 10 \log_{10} \hat{S}(f_k)$.

The bin spacing in (2.22) fixes the *resolution bandwidth*

$$\Delta f = \frac{f_s}{N}; \quad (2.24)$$

two features closer in frequency than Δf fall in the same bin and cannot be separated. The specific sampling rate and block length adopted in this thesis, and the resulting resolution bandwidth and frequency span, are stated with the system model in Chapter 4.

2.6.2 Bias, variance and Welch’s method

The periodogram (2.23) is asymptotically unbiased but *inconsistent*: its variance tends to $S(f)^2$ as $N \rightarrow \infty$ and does not diminish with record length [36], so a single periodogram is far too noisy to read a ~ 0.5 dB feature from. The remedy is averaging. Welch’s method [39] partitions the record into K blocks $x_i[n]$, $i = 0, \dots, K-1$, applies a window

$w[n]$ to each, and averages the resulting modified periodograms,

$$\hat{S}_W(f_k) = \frac{1}{K} \sum_{i=0}^{K-1} \frac{1}{NU} \left| \sum_{n=0}^{N-1} w[n] x_i[n] e^{-j2\pi kn/N} \right|^2, \quad U = \frac{1}{N} \sum_{n=0}^{N-1} w[n]^2, \quad (2.25)$$

where the window power U normalises the estimate so that it remains unbiased in total power. For approximately independent blocks the variance of $\hat{S}_W$ falls as $1/K$, trading a factor of K in variance against the data consumed. The implementation uses non-overlapping blocks, and the block count K is the single knob that sets the smoothness of the estimate.

This averaging is also the mechanism the temporal layer of Chapter 5 exploits, for the reason given in Section 2.3: the random component falls as $1/\sqrt{K}$ while a deterministic interference excess does not.

2.6.3 Spectral leakage and the Blackman window

Observing only N samples is equivalent to multiplying the infinite signal by a length- N window. Multiplication in time is convolution in frequency,

$$x_w[n] = w[n] x[n] \iff X_w(f) = (X * W)(f), \quad (2.26)$$

so every true spectral component is smeared by the window's transform $W(f)$. For the implicit *rectangular* window this transform is a Dirichlet (periodic sinc) kernel whose first sidelobe lies only about 13 dB below the main lobe; energy from a strong component therefore *leaks* through these sidelobes into distant bins. In a crowded transponder this is fatal: the sidelobes of a strong host carrier, or of its strong neighbours, can spill onto a flat top and either mask a weak CUC bump or manufacture a spurious one.

Tapering the block with a smooth window suppresses the sidelobes at the cost of a wider main lobe, i.e. of resolution. The thesis uses the *Blackman* window,

$$w[n] = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right), \quad n = 0, \dots, N-1, \quad (2.27)$$

whose peak sidelobe lies about 58 dB below the main lobe, roughly 45 dB lower than the rectangular window, while widening the main lobe by about a factor of three [17]. This is precisely the trade the CUC problem calls for: the detector must read a ~ 0.5 –1 dB power-density bump sitting on a flat top, so it matters far more that a strong carrier's leakage lie tens of decibels beneath that bump than that two close tones be resolved. Under Blackman weighting a strong carrier's leakage onto a neighbouring flat top lies

tens of decibels beneath the bumps being read, so a bump can be attributed to a buried carrier rather than to leakage. The attendant loss of resolution is immaterial here, since $\Delta f \approx 4.88$ kHz is already far finer than the features of interest.

2.7 The Carrier-under-Carrier Interference Model

Among the interference types an FDMA monitor must contend with, the *Carrier-under-Carrier* (CUC) is the most difficult [34]. A CUC is an unauthorised carrier that shares the centre frequency of, and lies within the bandwidth of, a legitimate *host* carrier. It may be deliberate (an attempt to hide a pirate link, or to spoof under the cover of a legitimate signal) or accidental, as when a misconfigured terminal transmits on an already-occupied slot. Because it sits at the host's frequency rather than in a guard band, a CUC creates no new spectral edges and adds no energy where the noise floor is expected: a detector that segments the band into carriers and flags unexpected occupancy will not see it. Its only signature is a change in the *shape* of the host carrier's own spectrum.

Model the occupied region as the superposition of the host and the CUC at a common centre frequency f_c ,

$$s(t) = s_h(t) + s_c(t), \quad (2.28)$$

where the host has total power P_h and symbol rate $R_{s,h}$, and the CUC has power $P_c \ll P_h$ and symbol rate $R_{s,c} < R_{s,h}$, both linearly modulated as in Section 2.4. Being statistically independent, their power spectra add,

$$S(f) = S_h(f) + S_c(f), \quad (2.29)$$

so within the CUC's narrower band the host's flat top is raised by the CUC's own flat-top density, while outside that band the host is unchanged. The CUC therefore appears as a *localised bump* on the host's flat top, of width equal to the CUC bandwidth.

The height of that bump is governed by two ratios. From the flat-top result (2.20), a carrier of total power P has a flat-top density

$$S_{\text{flat}} \approx \frac{P}{B_{\text{eff}}}, \quad B_{\text{eff}} \approx R_s, \quad (2.30)$$

where the effective bandwidth $B_{\text{eff}} = P/S_{\text{flat}}$ is, for a root-raised-cosine carrier, of the order of the symbol rate. Substituting the host and CUC densities into (2.29), the excess of the composite flat top over the host alone, expressed in decibels, is

$$\Delta = 10 \log_{10} \left(\frac{S_{h,\text{flat}} + S_{c,\text{flat}}}{S_{h,\text{flat}}} \right) = 10 \log_{10} \left(1 + \frac{P_c}{P_h} \cdot \frac{R_{s,h}}{R_{s,c}} \right). \quad (2.31)$$

The bump grows with the *power* ratio P_c/P_h and, just as strongly, with the *bandwidth* ratio $R_{s,h}/R_{s,c}$: a narrow CUC concentrates its power into a higher density and so lifts the host flat top more, even when its total power is small. As an illustration, a CUC 13 dB below its host ($P_c/P_h \approx 0.05$) occupying about 30% of the host bandwidth ($R_{s,h}/R_{s,c} \approx 3.3$) produces $\Delta = 10 \log_{10}(1 + 0.05 \cdot 3.3) \approx 0.7$ dB.

Equation (2.31) also exposes the difficulty. A weak but wide CUC ($R_{s,c} \rightarrow R_{s,h}$) lifts the entire host top by a fraction of a decibel with no localised feature, and is essentially invisible. A weak but narrow CUC does produce a localised bump, but one whose height, often only 0.5 to 1 dB, is comparable to the host's own spectral structure: the curvature of the host's raised-cosine shoulders and the estimation variance of the Welch spectrum (Section 2.6). Separating such a bump from the host's intrinsic shape is beyond a single power-spectrum snapshot, and is the central challenge taken up by the interference layer (Chapters 3 and 5). A standalone unauthorised carrier falling in a guard band, by contrast, deposits energy where only noise is expected and is comparatively easy to flag; the CUC is hard precisely because it hides inside a legitimate signal.

2.8 Wavelet Analysis

2.8.1 From Fourier to wavelets

The power spectrum of Section 2.6 reveals which frequencies are present, but locating the carrier structure within it (edges, flat tops, bumps) calls for analysis that is itself localised in frequency, and at a resolution appropriate to each feature's width. The Fourier transform offers no localisation, and the short-time Fourier transform imposes a single fixed window and hence, by the time–frequency bound (2.15), one fixed resolution at all frequencies. The *wavelet transform* instead analyses a signal with dilated copies of a single prototype, giving fine localisation for narrow features and coarse, wide-support analysis for broad ones, at a resolution that adapts with scale. In this thesis the wavelet transform is applied to the spectrum $P(f)$ itself: the “signal” is the power-versus-frequency curve, so translations index frequency position and scales index the width of spectral features. It is the tool that locates carrier band edges and characterises the flat-top structure on which the interference detector operates.

2.8.2 The continuous wavelet transform

A mother wavelet $\psi(t)$ is a zero-mean, finite-energy function, $\int \psi(t) dt = 0$; its dilations and translations

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right), \quad a > 0, b \in \mathbb{R}, \quad (2.32)$$

form the analysing family, with a a *scale* (large a meaning wide support and low frequency) and b a translation. The continuous wavelet transform (CWT) of a signal x is the family of inner products

$$W_x(a, b) = \int x(t) \psi_{a,b}^*(t) dt = \frac{1}{\sqrt{a}} \int x(t) \psi^*\left(\frac{t-b}{a}\right) dt. \quad (2.33)$$

Provided ψ satisfies the admissibility condition $C_\psi = \int_0^\infty |\Psi(f)|^2 / f df < \infty$, which forces the zero-mean property, the signal can be recovered from W_x , so (2.33) discards no information [24]. Each coefficient measures the similarity of x , near position b , to the wavelet at scale a ; choosing the scale therefore selects, in the sense of the bound (2.15), a particular operating point on the time–frequency trade-off.

2.8.3 Derivative-of-Gaussian wavelets and edge detection

The interference layer must locate the *edges* of carriers: the transitions between the noise floor and a flat top. A wavelet matched to this task is the first derivative of a Gaussian (the *gaus1* wavelet),

$$\psi(t) = -t e^{-t^2/2} \propto \frac{d}{dt} e^{-t^2/2}, \quad (2.34)$$

for which the CWT (2.33) is, up to scaling, a *smoothed derivative* of the signal at scale a : $W_x(a, b) \propto a \frac{d}{db} (x * \theta_a)(b)$, where θ_a is a Gaussian of width a [24]. A coefficient is therefore large in magnitude exactly where the smoothed signal has a steep slope, that is at an edge, and its *sign* encodes the *direction* of the transition: a rising edge (the floor stepping up to a carrier top) and a falling edge (a carrier top stepping down to the floor) produce coefficients of opposite sign. The specific sign convention, and its mapping to carrier starts and ends, is fixed with the segmentation algorithm in Chapter 5. This signed, multi-scale edge response is sharp at small scales for narrow carriers and well-defined only at large scales for wide ones. It is the raw material the segmentation stage of the interference layer builds on. Because a true edge produces a coefficient of consistent sign across a range of scales whereas noise does not, combining scales improves reliability; how the scales are weighted is taken up, using Shannon entropy, in Section 2.9.

2.8.4 The discrete wavelet transform and multiresolution

For denoising it is convenient to sample the CWT on a dyadic grid, $a = 2^j$ and $b = k 2^j$, giving the discrete wavelet transform (DWT). By Mallat's multiresolution analysis [24] the DWT is computed with a filter bank: a scaling function φ with its low-pass filter h produces a coarse *approximation*, while the wavelet ψ with its high-pass filter g produces the *detail*, each followed by downsampling by two; iterating on the approximation yields a level- J decomposition

$$x \longrightarrow \{c_{A,J}, c_{D,J}, c_{D,J-1}, \dots, c_{D,1}\}, \quad (2.35)$$

with $c_{A,J}$ the approximation coefficients and $c_{D,j}$ the detail coefficients at level j . The thesis uses the *symlet* `sym4`, an orthogonal, near-symmetric wavelet with four vanishing moments [8]. Its near-symmetry gives an almost linear phase, so that denoising does not displace the very carrier edges the next stage must locate.

2.8.5 Wavelet denoising

Denoising exploits the fact that additive noise spreads its energy thinly across all detail coefficients, whereas a piecewise-smooth signal concentrates into a few large ones. One takes the DWT, shrinks the detail coefficients toward zero, and inverts. The thesis uses *soft* thresholding [9],

$$\eta_\lambda(c) = \text{sign}(c) \max(|c| - \lambda, 0), \quad (2.36)$$

with the *universal* (VisuShrink) threshold of Donoho and Johnstone [10],

$$\lambda = \hat{\sigma} \sqrt{2 \ln N}, \quad (2.37)$$

where N is the signal length and $\hat{\sigma}$ is a robust estimate of the noise standard deviation formed from the finest-scale detail coefficients via the median absolute deviation (MAD),

$$\hat{\sigma} = \frac{\text{median}(|c_{D,1} - \text{median}(c_{D,1})|)}{0.6745}. \quad (2.38)$$

The constant $0.6745 = \Phi^{-1}(3/4)$ converts the MAD to a Gaussian standard deviation, and the MAD is preferred over the sample standard deviation because it is insensitive to the few large coefficients that carry genuine edges. Applying (2.36)–(2.38) to the detail levels while leaving the approximation intact flattens the noise floor of the spectrum without blurring the carrier edges, yielding the denoised spectrum on which segmentation and the flat-top bump test operate.

2.9 Shannon Entropy as a Concentration Measure

The final tool the interference layer needs is a way to judge, from the data alone, which of the wavelet scales of Section 2.8 to trust. Shannon entropy [35] provides it. For a discrete probability distribution $p = (p_1, \dots, p_M)$ with $p_i \geq 0$ and $\sum_i p_i = 1$, the Shannon entropy is

$$H(p) = - \sum_{i=1}^M p_i \ln p_i, \quad (2.39)$$

with the convention $0 \ln 0 = 0$. Entropy measures how *dispersed* the distribution is: it is maximised, $H = \ln M$, by the uniform distribution $p_i = 1/M$, and minimised, $H = 0$, when all the mass falls on a single outcome. Low entropy therefore signals *concentration*; high entropy signals diffuseness.

Applied to the wavelet coefficients this becomes a per-scale reliability measure. At scale j the coefficient energy across the frequency bins defines a distribution

$$p_j(i) = \frac{|c_j(i)|^2}{\sum_{i'} |c_j(i')|^2}, \quad (2.40)$$

whose entropy $H_j = H(p_j)$ is small when that scale's response is focused at a few bins, the signature of a genuine, localised edge, and large when the response is smeared across the band, as for a scale reacting mostly to noise or poorly matched to the feature. The multi-scale edge votes of Section 2.8 can then be combined with weights that favour the low-entropy (focused) scales; the thesis uses a soft, data-adaptive weight $w_j \propto 1/\sqrt{H_j}$, in contrast to a fixed scale weighting, a choice examined in the interference layer (Chapter 5).

2.10 Satellite Telemetry and Anomaly Types

The second layer operates not on the RF spectrum but on the spacecraft's *housekeeping telemetry*: the stream of temperatures, currents, voltages, and sensor or actuator states that report the vehicle's health, relayed to the ground over the telemetry, tracking and command (TT&C) link. Such telemetry is a multivariate time series with three properties that shape any detector: it is high-volume and high-dimensional; its channels are strongly correlated; and many of them are *periodic*, driven by orbital and thermal cycles, while others follow commands and change only when the spacecraft is told to. A fixed alarm limit on each channel, the classical approach in flight operations [13], copes badly with all three: it ignores correlations, and on periodic data it must be set high enough to clear the normal cyclic excursions, at which point it misses the subtle deviations that

matter.

It is useful to distinguish three kinds of anomaly [4]. A *point* anomaly is a single sample far from the normal range. A *contextual* anomaly is a sample whose value is normal in general but abnormal in its context (for instance, at that phase of the orbit). A *collective* anomaly is a *subsequence* that is anomalous as a whole even though its individual samples may each look ordinary: a stretch of telemetry with the wrong shape, rate, or correlation. Many spacecraft faults present this way (a slow drift, a frozen sensor, a lost oscillation), and collective anomalies are the target of the telemetry layer. One caution about terminology follows from this. Public telemetry benchmarks label an anomaly's *extent* as an index range, but classify it by the *nature* of the deviation, typically as point or contextual in the sense above. A labelled range may therefore be collective in extent while carrying a point or contextual class, and Section 4.5.1 states how the benchmark used here is annotated.

Because a collective anomaly lives in a subsequence rather than a sample, the series must be analysed in windows. Given a scalar telemetry channel $\{t_1, t_2, \dots, t_N\}$, a sliding window of length w and unit shift produces the overlapping segments

$$s_i = [t_i, t_{i+1}, \dots, t_{i+w-1}]^\top \in \mathbb{R}^w, \quad i = 1, \dots, L, \quad L = N - w + 1, \quad (2.41)$$

which are stacked as the columns of a *sample matrix* $S = [s_1 | s_2 | \dots | s_L] \in \mathbb{R}^{w \times L}$. This embedding turns the question “is this subsequence abnormal?” into “is this vector s_i well explained by a model of normal windows?”, which is precisely the question sparse representation answers. The columns of the *anomaly-free* training portion of S serve as the training signals for the dictionary learning of the next section.

2.11 Sparse Representation and Dictionary Learning

2.11.1 Sparse representation of a signal

Let $D = [d_1 | \dots | d_m] \in \mathbb{R}^{n \times m}$ be a *dictionary* whose columns d_j are unit-norm *atoms*. The dictionary is *overcomplete* when $m > n$, so it spans $\mathbb{R}^n$ in infinitely many ways; sparse representation selects, among these, a representation of a signal $y \in \mathbb{R}^n$ that uses as few atoms as possible,

$$y \approx Dx, \quad \text{with } x \in \mathbb{R}^m \text{ sparse.} \quad (2.42)$$

Sparsity is measured by the ℓ_0 pseudo-norm $\|x\|_0$, the number of non-zero entries of x . Two equivalent problem statements are used: the error-constrained form and the

sparsity-constrained form,

$$\min_x \|x\|_0 \text{ s.t. } \|y - Dx\|_2 \leq \varepsilon \quad \text{or} \quad \min_x \|y - Dx\|_2^2 \text{ s.t. } \|x\|_0 \leq T_0, \quad (2.43)$$

where ε bounds the reconstruction error and T_0 the number of active atoms. Both are combinatorial and NP-hard in general [26]: solving them exactly requires searching over subsets of atoms. Two practical routes avoid the search, a greedy pursuit and a convex relaxation.

2.11.2 Greedy pursuit: orthogonal matching pursuit

Orthogonal matching pursuit (OMP) [28] builds the support one atom at a time, at each step choosing the atom most correlated with the current residual and then re-solving a least squares problem on the chosen atoms. With residual r , support set Λ , and D_Λ the sub-dictionary of selected columns:

1. Initialise $r_0 = y$, $\Lambda_0 = \emptyset$, $t = 0$.

2. **Atom selection.** Choose the atom best aligned with the residual,

$$j_t = \arg \max_j |d_j^\top r_{t-1}|, \quad \Lambda_t = \Lambda_{t-1} \cup \{j_t\}. \quad (2.44)$$

3. **Coefficient update.** Re-estimate all active coefficients by least squares,

$$x_{\Lambda_t} = \arg \min_z \|y - D_{\Lambda_t} z\|_2 = (D_{\Lambda_t}^\top D_{\Lambda_t})^{-1} D_{\Lambda_t}^\top y = D_{\Lambda_t}^+ y. \quad (2.45)$$

4. **Residual update.** $r_t = y - D_{\Lambda_t} x_{\Lambda_t}$; increment t and repeat until $\|x\|_0 = T_0$ (or $\|r_t\|_2 \leq \varepsilon$).

The name is “orthogonal” because the least squares step (2.45) makes the new residual orthogonal to the span of all selected atoms, $r_t \perp \text{span}(D_{\Lambda_t})$, so no atom is ever chosen twice and the residual energy decreases monotonically. OMP is the sparse-coding engine used inside the dictionary learning below.

2.11.3 Convex relaxation

The alternative replaces the non-convex ℓ_0 count by the convex ℓ_1 norm $\|x\|_1 = \sum_j |x_j|$ which is the tightest convex surrogate of ℓ_0 on the unit cube. This yields either basis pursuit or, in its error-penalised form, the LASSO / basis-pursuit denoising problem [6,

38]

$$\min_x \frac{1}{2} \|y - Dx\|_2^2 + \lambda \|x\|_1, \quad (2.46)$$

which is convex and therefore has no spurious local minima. Under incoherence or restricted-isometry conditions on D the ℓ_1 solution coincides with the ℓ_0 solution [3], so (2.46) recovers the sparse representation while remaining tractable. Its non-smooth ℓ_1 term is what the optimisation of Section 2.12 handles; the same relaxation underlies the online telemetry problem of Chapter 6.

2.11.4 Fixed and learned dictionaries

The power of (2.42) rests on the dictionary. A *fixed* dictionary built from analytic atoms, a discrete cosine or wavelet basis, is universal but not tailored to the data; a *learned* dictionary, adapted to a corpus of representative signals, typically represents them far more sparsely [33] and is the choice made here. Learning the dictionary from *normal* telemetry is what makes the residual of Section 2.11, and hence the anomaly score, meaningful: a normal window is sparsely representable, an anomalous one is not.

2.11.5 The dictionary-learning problem

Given a set of training signals collected as the columns of $Y = [y_1 | \cdots | y_L] \in \mathbb{R}^{n \times L}$ (for the telemetry layer, the anomaly-free windows of Section 2.10), dictionary learning seeks a dictionary D and sparse codes $X = [x_1 | \cdots | x_L]$ that jointly minimise the reconstruction error,

$$\min_{D, X} \|Y - DX\|_F^2 \quad \text{s.t.} \quad \|x_i\|_0 \leq T_0 \quad \forall i, \quad \|d_j\|_2 = 1 \quad \forall j, \quad (2.47)$$

where $\|\cdot\|_F$ is the Frobenius norm and the unit-norm constraint removes the scale ambiguity between an atom and its coefficients. Problem (2.47) is non-convex in (D, X) jointly but becomes tractable in each variable separately, which suggests alternating minimisation: with D fixed, (2.47) decouples over the columns into L sparse-coding problems (2.43) solved by OMP; with X fixed, the dictionary is updated. The K-SVD algorithm of Aharon et al. [1] is a particularly effective way to perform the second step.

2.11.6 K-SVD

K-SVD updates the dictionary *one atom at a time*, and it updates each atom's coefficients along with it. Let x_T^k denote the k -th row of X , i.e. the coefficients that multiply atom d_k

across all signals. Isolating atom k in the error,

$$\|Y - DX\|_F^2 = \left\| Y - \sum_j d_j x_T^j \right\|_F^2 = \left\| \underbrace{\left(Y - \sum_{j \neq k} d_j x_T^j \right)}_{E_k} - d_k x_T^k \right\|_F^2, \quad (2.48)$$

where E_k is the representation error of all signals *with atom k removed*. The update of (d_k, x_T^k) is thus the best rank-one approximation of E_k . Approximating E_k directly, however, would fill in the currently-zero entries of x_T^k and destroy the sparsity established in the coding step. K-SVD avoids this by restricting attention to the signals that actually use atom k ,

$$\omega_k = \{i : x_T^k(i) \neq 0\}, \quad E_k^R = E_k(:, \omega_k), \quad (2.49)$$

so that only the already-active coefficients are modified. The restricted update is the rank-one problem

$$\min_{d_k, x_R^k} \|E_k^R - d_k (x_R^k)^\top\|_F^2, \quad (2.50)$$

whose solution is characterised by the Eckart–Young theorem [16].

Proposition 2.1 (Eckart–Young). *Let M have the singular value decomposition $M = U\Sigma V^\top$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ and left/right singular vectors u_i, v_i . Then the best rank- r approximation of M in Frobenius norm, $\min_{\text{rank}(B) \leq r} \|M - B\|_F$, is the truncated SVD $B^* = \sum_{i=1}^r \sigma_i u_i v_i^\top$.*

Applying the proposition to (2.50) with rank $r = 1$ and $M = E_k^R = U\Sigma V^\top$, the optimal atom and coefficients are

$$d_k \leftarrow u_1, \quad x_R^k \leftarrow \sigma_1 v_1, \quad (2.51)$$

with u_1, v_1 the first left and right singular vectors and σ_1 the largest singular value; $\|d_k\|_2 = 1$ holds automatically since u_1 is a unit vector. The full algorithm alternates the two steps to convergence:

1. **Sparse coding.** With D fixed, solve $x_i = \arg \min \|y_i - Dx_i\|_2^2$ s.t. $\|x_i\|_0 \leq T_0$ for each column, by OMP.
2. **Dictionary update.** For each atom $k = 1, \dots, m$: form E_k^R via (2.49) and update (d_k, x_R^k) by the rank-one SVD (2.51).

Because (2.51) refits the coefficients while updating the atom, K-SVD converges faster than the *method of optimal directions* [11], which re-solves for the whole dictionary by least squares while holding the codes fixed. The learned D and the sparse-coding machinery are exactly what the telemetry layer applies to the windows of (2.41). Its online,

robust variant, which adds an explicit residual term to absorb anomalies and is solved by the method of the next section, is developed in Chapter 6.

2.12 Convex Optimisation for Sparse Recovery

The relaxed problem (2.46), and the online telemetry problem it leads to, are convex but non-smooth because of the ℓ_1 (or block- ℓ_1) penalty. Two ingredients solve them efficiently: the proximal operator, and the alternating direction method of multipliers.

2.12.1 Proximal operator and soft-thresholding

The proximal operator [27] of a function g with parameter λ is

$$\text{prox}_{\lambda g}(v) = \arg \min_x g(x) + \frac{1}{2\lambda} \|x - v\|_2^2. \quad (2.52)$$

For $g(x) = \|x\|_1$ the minimisation separates coordinate-wise, and each coordinate solves $\min_x |x| + \frac{1}{2\lambda}(x - v)^2$, whose solution is the *soft-thresholding* operator

$$\text{prox}_{\lambda \|\cdot\|_1}(v) = \eta_\lambda(v) = \text{sign}(v) \max(|v| - \lambda, 0), \quad (2.53)$$

applied element-wise. This is the same operator (2.36) met in wavelet denoising. That the ℓ_1 proximal map is soft-thresholding is the reason ℓ_1 -penalised problems reduce to simple shrinkage steps.

2.12.2 Group sparsity and block soft-thresholding

Some problems require sparsity not over individual entries but over *groups* of them. This is central to the telemetry layer, whose robust formulation (Chapter 6) represents the data as a sparse reconstruction plus a residual matrix E whose *columns* correspond to windows: an anomalous window is a non-zero column of E , and one expects only a few of them. Column sparsity of a matrix E with columns $E(:, i)$ is measured by the mixed $\ell_{2,0}$ pseudo-norm, the number of non-zero columns, and by its convex surrogate, the $\ell_{2,1}$ norm [42],

$$\|E\|_{2,1} = \sum_i \|E(:, i)\|_2, \quad (2.54)$$

which sums the Euclidean lengths of the columns and therefore drives *whole* columns to zero while leaving the entries within a surviving column untouched. (By contrast

$\|X\|_{1,1} = \sum_{i,j} |X_{ij}|$ is the ordinary element-wise ℓ_1 norm, whose proximal map is the element-wise soft-thresholding above.)

The proximal operator of the column- ℓ_2 penalty acts column-wise as *block* (or vector) soft-thresholding: for a single column v ,

$$\text{prox}_{\beta\|\cdot\|_2}(v) = \max\left(0, 1 - \frac{\beta}{\|v\|_2}\right) v, \quad (2.55)$$

which shrinks the column toward the origin along its own direction and sets it exactly to zero when $\|v\|_2 \leq \beta$. Block soft-thresholding is to the $\ell_{2,1}$ norm what element-wise soft-thresholding (2.53) is to the ℓ_1 norm; it is the operation the telemetry solver applies to the residual E to decide, one column at a time, which windows are anomalous.

2.12.3 The alternating direction method of multipliers

ADMM [2] solves problems that split into two terms coupled by a linear constraint,

$$\min_{x,z} f(x) + g(z) \quad \text{s.t.} \quad Ax + Bz = c, \quad (2.56)$$

by working on the (scaled) augmented Lagrangian

$$L_\rho(x, z, u) = f(x) + g(z) + \frac{\rho}{2} \|Ax + Bz - c + u\|_2^2, \quad (2.57)$$

where u is the scaled dual variable and $\rho > 0$ the penalty parameter. It then alternates minimisation over x and z with a dual ascent step,

$$x^{k+1} = \arg \min_x f(x) + \frac{\rho}{2} \|Ax + Bz^k - c + u^k\|_2^2, \quad (2.58)$$

$$z^{k+1} = \arg \min_z g(z) + \frac{\rho}{2} \|Ax^{k+1} + Bz - c + u^k\|_2^2, \quad (2.59)$$

$$u^{k+1} = u^k + Ax^{k+1} + Bz^{k+1} - c. \quad (2.60)$$

For convex f and g the iterates converge to a solution of (2.56). The method is attractive here because it *decouples* a hard composite objective into subproblems that are individually easy: the x -update is typically a linear least-squares solve, while the z -update, when g is an ℓ_1 or $\ell_{2,1}$ penalty, is a closed-form proximal step, element-wise (2.53) or block (2.55) soft-thresholding, respectively. This structure, and its ability to scale to the large sample matrices of Section 2.10, are why the telemetry layer solves its sparse reconstruction by ADMM. The specific splitting, which introduces a column-sparse residual matrix alongside the sparse code, is derived in Chapter 6.

2.13 Adaptive Thresholding

Both layers reduce a signal to a scalar *anomaly score* per position and must then decide, by a threshold, which scores are anomalous. A single global threshold is inadequate when the score's baseline varies, most acutely for periodic telemetry, where the recurring cyclic structure produces recurring peaks in the score, so a fixed limit set above them misses genuine anomalies while one set below them raises false alarms at every cycle. *Adaptive* thresholds address this by tracking the local statistics of the score. Two mechanisms recur in this thesis. The exponentially weighted moving average (EWMA) [32] maintains a running estimate of the score's trend,

$$U_t = \lambda U_{t-1} + (1 - \lambda) a_t, \quad \lambda \in (0, 1), \quad (2.61)$$

from which a band is placed around U_t to flag departures from the recent trend. The constant-false-alarm-rate (CFAR) principle, borrowed from radar [31], sets the threshold from a robust local estimate of the background level, for example $\text{median} + k \cdot \text{MAD}$, so that the false-alarm probability stays constant as the background changes. Both are data-adaptive; they differ in the prior they assume about the background, and the framework of Chapter 3 shows that the CFAR limit is the non-periodic generalisation of the period-referenced EWMA band.

2.14 Related Work

This thesis takes its two detection layers directly from two prior works: a multi-scale wavelet method for interference detection in FDMA satellite networks, and a sparse-representation method for anomaly detection in satellite telemetry. Each is reviewed below, together with the ways this thesis extends or departs from it, and with the neighbouring lines of work each sits in.

2.14.1 Multi-scale wavelet interference detection

The interference layer builds on the automated interference-detection system of Sai et al. [34]. That work targets the same setting as Section 2.2, an FDMA transponder carrying heterogeneous carriers of differing bandwidths and power levels, where a single energy threshold is unreliable, flagging weak legitimate carriers while missing anomalies near strong ones. Its response is to analyse the power spectrum with a multi-scale wavelet transform (Section 2.8): carrier boundaries are located from the wavelet edge response

across several scales, and the per-scale edge evidence is combined by a fixed cross-scale weighting that emphasises the mid-scales. Segmenting the band into carriers in this way, rather than thresholding raw energy, lets the method accommodate the heterogeneity of the carriers and reduces false alarms relative to conventional energy detection.

This thesis adopts that multi-scale wavelet segmentation as the front end of its snapshot interference detector, but differs in four respects. First, it targets specifically the Carrier-under-Carrier threat of Section 2.7: an interferer that produces no new edge and no guard-band energy, and so is invisible to segmentation alone, revealing itself only as a localised power-density bump on a host carrier’s flat top. Detecting that bump, rather than segmenting carriers, is the snapshot layer’s objective. Second, because a weak or narrow CUC produces a bump comparable to the host’s own structure and beyond the reach of any single spectrum, the thesis adds a temporal, clean-baseline-referenced stage (Chapters 3 and 5) that recovers the cases the snapshot detector cannot. Third, it replaces the fixed cross-scale weighting with the data-adaptive, entropy-based weighting of Section 2.9; a controlled comparison (Chapter 5) finds the two equivalent on this band, a negative result reported rather than overstated. Finally, the thesis contributes a reproducible FDMA transponder simulation and a labelled benchmark, absent from the reference work, that make the detection performance measurable.

A parallel line of work approaches spectrum monitoring with data-driven deep learning rather than model-based signal processing. Kim et al. [22], for instance, cast wideband spectrum monitoring as an object-detection problem on time–frequency spectrograms, training deep neural networks to localise abnormal or interfering emissions and reporting strong detection even for low-power signals. Such methods can be highly effective, but they require labelled training data and yield decisions that are hard to interpret. The interference layer of this thesis deliberately remains model-based: its wavelet segmentation and temporal baseline-differencing need no training, rest on the explicit signal model developed in this chapter, and make transparent exactly why a detection is raised. A third line reads what the power spectrum discards. A modulated carrier is *cyclostationary*, with statistics that repeat at multiples of its symbol rate, and the spectral-correlation methods of Gardner [15] read those cycle frequencies directly. A CUC signals at its own rate, so it occupies cycle frequencies the host does not. What rules the route out here is its cost: the spectral correlation function is estimated over a two-dimensional frequency/cycle-frequency plane, where this interference layer is one FFT pipeline and a linear pass over 8192 bins.

2.14.2 Sparse-representation telemetry anomaly detection

Dimensionality reduction is the older data-driven route here. Yairi et al. [41] model normal housekeeping behaviour by probabilistic clustering over a reduced set of directions, scoring each observation by how badly that model reconstructs it, which is the same shape as the PCA baseline of Chapter 7 and what makes that baseline a method rather than a straw man. Sparse representation is by now an equally established approach. Most directly related to this thesis, Pilastre et al. [29] address anomaly detection in *mixed* satellite telemetry (continuous measurements together with discrete status words) by learning a dictionary of normal patterns and sparsely reconstructing each windowed observation, flagging as anomalous those windows the dictionary cannot represent, that is, those that leave a large reconstruction residual. Their formulation explicitly captures the correlations among heterogeneous parameters and detects both point and collective anomalies, and it shares the reconstruction-residual principle that this thesis places at the centre of its framework (Chapter 3). The telemetry layer developed here builds specifically on the closely related K-SVD–ADMM formulation of He et al. [18]. That work addresses collective anomalies in satellite telemetry (Section 2.10), where periodicity and strong inter-channel correlation defeat fixed alarm limits. It proceeds in two stages. Offline, a dictionary of normal local patterns is learned from anomaly-free telemetry by K-SVD (Section 2.11.6) over sliding windows. Online, each new window is sparsely reconstructed from that dictionary by an ADMM solver (Section 2.12) in which a column-sparse residual matrix absorbs the part of the signal the normal dictionary cannot explain; the norm of that residual is the anomaly score. A period-referenced exponentially-weighted threshold (Section 2.13) then flags anomalies while tolerating the recurring peaks that periodic telemetry induces. On a satellite antenna-component telemetry case study the method is reported to outperform one-class SVM, a meta-feature approach, and an LSTM-based detector, particularly on collective anomalies.

This thesis implements that method and extends its evaluation and framing. Where the original work validates on a single private dataset, the thesis evaluates the detector on the public SMAP/MSL benchmark against a controlled Principal Component Analysis reconstruction baseline, identical in pipeline but dense and linear rather than sparse and overcomplete, and against the published LSTM-based results, under both point-wise and point-adjusted scoring (Chapter 7). That comparison isolates what the sparse, overcomplete dictionary contributes, and what the scoring convention contributes on its own. Herrmann et al. [19] re-evaluate several deep anomaly detectors on these benchmarks and show that point adjustment overestimates performance enough to make trivial baselines look competitive, which is why both conventions are reported here.

Chapter 3

A Unified Reconstruction-Residual Detection Framework

3.1 Motivation: Two Problems, One Principle

The two surveillance problems addressed by this thesis, interference in the radio-frequency spectrum and anomalies in spacecraft telemetry, are conventionally treated by two different toolboxes. Yet they share the same obstacle: in both domains, *a fixed threshold applied to the raw signal cannot work*, because the normal signal is itself large, structured, and inhomogeneous. In the RF domain, the transponder carries legitimate carriers of widely differing bandwidths and powers. The Carrier-under-Carrier threat of Section 2.7 hides beneath an authorised carrier, so that the energy at the interfered frequencies is dominated by energy that is *supposed* to be there. In the telemetry domain (Section 2.10), nominal channels swing periodically through large excursions, while the most dangerous faults are collective anomalies whose individual samples remain within nominal range; fixed alarm limits are blind to the fault and triggered by the health. Both situations were shown on real data in Figure 1.3.2.

In both domains, then, the *signal* carries too much legitimate structure to be thresholded. The resolution adopted throughout this thesis is to stop thresholding the signal and start thresholding what a model of normal behaviour *cannot actually explain* about it. If normality can be modelled, by a reference spectrum or by a learned library of normal patterns, then every observation can be reconstructed from that model, and whatever part of the observation resists reconstruction is the abnormal part. The legitimate structure, however large, is absorbed by the model; the detection evidence is the residual.

This chapter develops that principle once, formally, and then shows that the two detection layers of the thesis are its two instantiations. They are not two methods that happen to coexist, but two points on a single, explicit design axis. The chapter defines the framework (Section 3.2) and its design axes (Section 3.3), maps each layer onto it (Sections 3.4 and 3.5), presents the unifying argument (Section 3.6), and states plainly what the framework does *not* cover (Section 3.7).

3.2 The General Framework

The principle itself is not new. Reconstructing an observation from a model of normality and thresholding what is left over underlies the sparse telemetry detectors of Section 2.14.2, and in a plainer form it underlies every baseline-subtraction method in spectrum monitoring. What this chapter adds is a way of organising it. The statement below is general enough that the two layers of this thesis are *instances* of it rather than analogues; it spans a two-dimensional space in which each detector sits at an identifiable point; and Section 3.6 gives the reductions that connect those points.

Let $\mathbf{y} \in \mathbb{R}^N$ denote an *observation*: one slice of the monitored signal, presented in whatever domain the layer operates in. The framework consists of five stages, illustrated in Figure 3.2.1.

1. **Reference data.** A body of data $\mathcal{Y}_{\text{ref}}$ known (or trusted) to be free of the condition to be detected as an abnormality. In the RF layer this is a clean capture of the authorised carrier plan; in the telemetry layer it is the anomaly-free training telemetry.
2. **Normality model.** From the reference data, a model $\mathcal{M}$ of normal behaviour is constructed. The model may be as simple as a single stored template or as rich as a learned overcomplete dictionary; this choice is the first design axis of Section 3.3.
3. **Reconstruction.** Each observation is reconstructed from the model,

$$\hat{\mathbf{y}} = \mathcal{R}(\mathbf{y}; \mathcal{M}), \quad (3.1)$$

where $\mathcal{R}$ returns the model's best explanation of $\mathbf{y}$ using only normal structure. For a fixed template the reconstruction is the template itself; for a dictionary it is the solution of a sparse-coding problem.

4. **Residual score.** The detection evidence is the part of the observation the model could not explain,

$$r = d(\mathbf{y}, \hat{\mathbf{y}}), \quad (3.2)$$

where $d(\cdot, \cdot)$ is a domain-appropriate discrepancy: a per-bin difference in decibels for spectra, a residual-column norm for telemetry windows. Note that r is small wherever the observation is normal *regardless of how large or structured the normal signal is*, because that structure has been absorbed by $\hat{y}$.

5. **Adaptive threshold and decision.** The residual is compared against a threshold τ derived from the residual's *own* statistics rather than from fixed engineering limits, and the detector flags, and localises, wherever $r > \tau$. The prior knowledge built into τ , none or an assumed periodicity, is the second design axis.

Nothing in the five stages refers to radio spectra or to telemetry: the framework is domain-agnostic. All domain knowledge enters through three choices: what an observation y is, how rich the model $\mathcal{M}$ is, and what prior the threshold τ assumes. The first choice is dictated by the problem; the remaining two form a two-dimensional design space in which every detector of this thesis is a point.

3.3 The Two Design Axes

Axis 1: richness of the normality model. At one extreme the model is a *single fixed reference*: one stored template of normal behaviour, against which every observation is differenced. This is cheap, transparent, and requires only that one clean reference exist; but it can only describe a normality that is essentially *one* pattern. At the other extreme the model is a *learned overcomplete dictionary*: many prototype patterns extracted from training data, of which each observation may combine a few (Section 2.11). This describes a normality with many faces, meaning normal behaviour that legitimately takes many different shapes, as telemetry does across the phases of an orbit or the operating modes of an instrument, each shape assembled from a few of the stored prototypes. The price is anomaly-free training data and an optimisation problem per observation. Between the extremes lies the *linear subspace*: a principal-component model that reconstructs every observation by orthogonal projection onto a few dense directions. The thesis instruments this midpoint: a working detector sits at exactly this point of the axis, a PCA-reconstruction baseline identical to the telemetry detector in every pipeline stage except the model, and it serves as the controlled baseline of Chapter 7. The continuum of model richness is therefore realised at three concrete points: fixed reference, linear subspace, learned dictionary. Only the last two form a controlled comparison. The PCA baseline and the K-SVD detector share every pipeline stage and run on the same telemetry, so any difference between them is attributable to the model alone (Chapter 7). The

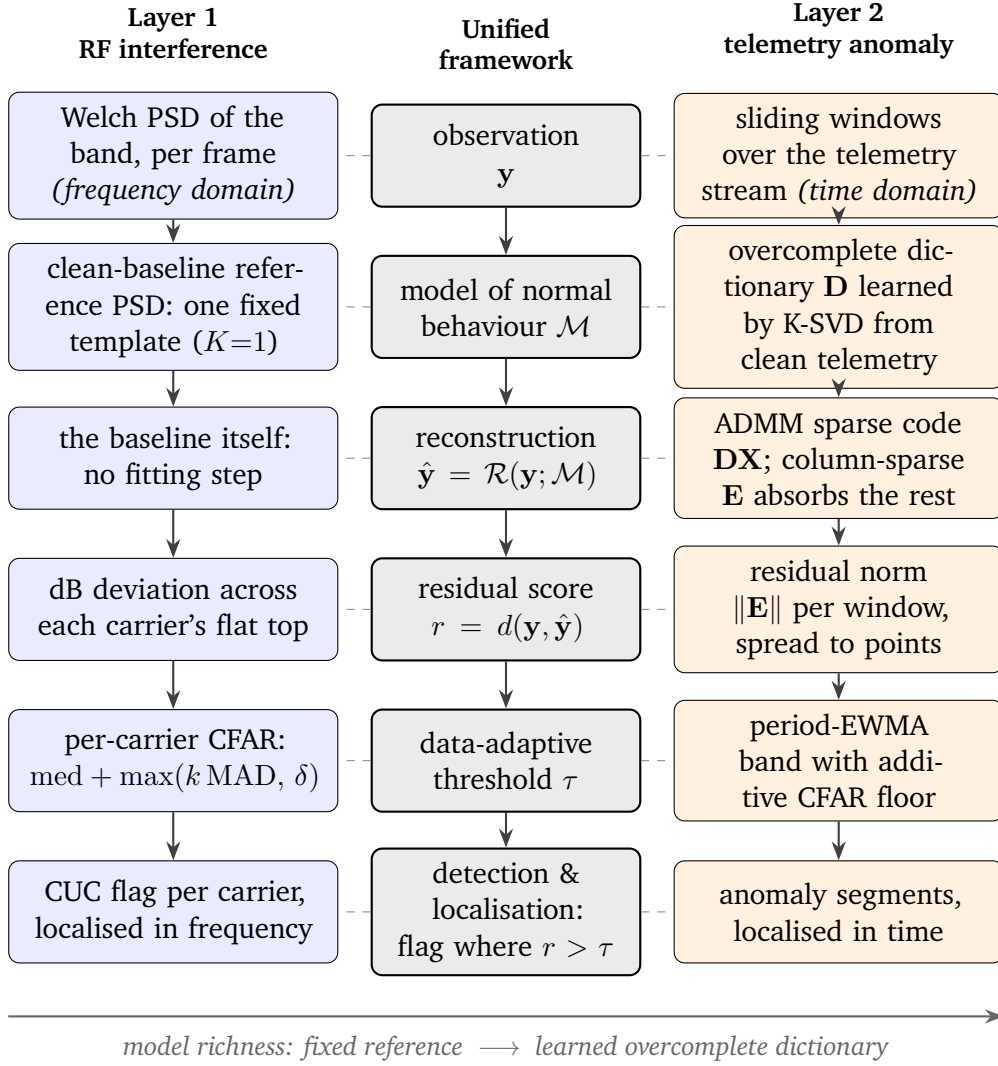


Figure 3.2.1: The unified reconstruction-residual detection framework (center) and its two instantiations. Both layers score an observation by how poorly a model of normal behaviour reconstructs it, and compare that residual against a data-adaptive threshold. The layers differ in the domain of the observation (frequency versus time) and, chiefly, in the richness of the normality model: the interference layer uses a single fixed clean-baseline reference (the degenerate, size-one dictionary), so its reconstruction step is trivial, whereas the telemetry layer learns an overcomplete dictionary by K-SVD and must solve a sparse-coding problem (ADMM) to reconstruct. The residual-scoring and additive adaptive-thresholding stages are common machinery, differing only in parameters.

fixed-reference point sits on the same axis in the other domain, which locates the RF detector on the continuum without entering that comparison.

Axis 2: prior knowledge in the threshold. The threshold τ is always derived from the residual's own statistics, but detectors differ in what they assume about the monitored process *before* any residual is observed, their *prior*, and hence in the structure their rule can exploit. The structure-free choice is a *Constant False Alarm Rate* (CFAR) rule: an estimate of the residual's location plus a robust scale multiple and an additive floor, $\tau = \text{med}(r) + \max(k \text{MAD}(r), \delta)$, which adapts to whatever floor the residual presents without assuming anything about its evolution (Section 2.13). When the monitored process is known in advance to be periodic, as orbital and thermal cycles make much telemetry, the threshold can embed that prior, building the known period into the rule itself: the period-referenced exponentially-weighted moving average (EWMA) tracks what the residual normally scores *at this phase of the cycle*, one period earlier, and sets a band around it. A nominal *residual* peak that recurs every cycle therefore lies inside the band, expected at that phase, and raises no alarm, while the same excursion at an unusual phase falls outside it. This tolerance of recurring nominal peaks is what the periodic prior buys; the CFAR rule, assuming nothing, buys generality instead. Section 3.6 shows that the two are not merely analogous but related by generalisation: the additive CFAR margin is precisely what the periodic band needs when its own assumptions fail.

Figure 3.3.1 places the detectors of this thesis on the plane spanned by the two axes.

3.4 Instance 1: The Temporal RF Detector as the Degenerate Case

The temporal stage of the interference layer (developed in full in Chapter 5) maps onto the framework as follows. The observation y is the N -frame Welch-averaged power spectral density of the transponder band (Section 2.6), expressed in decibels. The reference data is a single clean capture of the same authorised carrier plan, a *baseline twin*, taken at a known interference-free instance of the band; averaged over the same number of frames, it yields the baseline spectrum b . The normality model is that baseline and nothing more: $\mathcal{M} = \{b\}$. Reconstruction is correspondingly trivial,

$$\hat{y} = b, \tag{3.3}$$

up to a registration step. The baseline is first frequency-aligned to the current capture. The *difference spectrum*, the bin-by-bin difference $y - b$ between the current spectrum

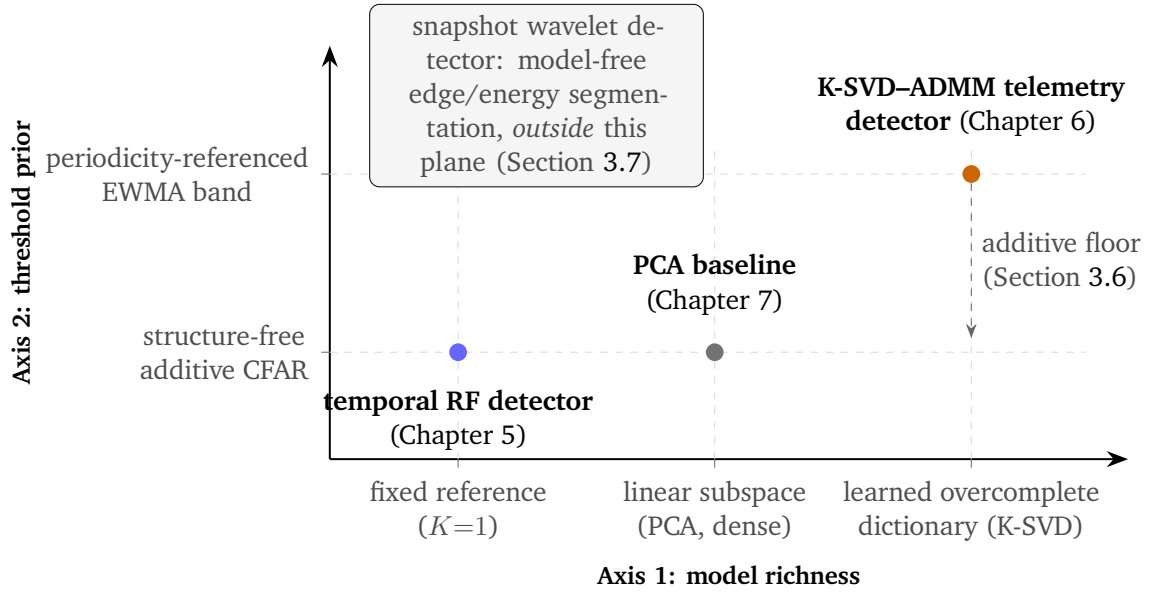


Figure 3.3.1: The detectors of this thesis as points in the two-dimensional design space of the framework. Axis 1 is the richness of the normality model, experimentally sampled at three points: the temporal RF detector (single fixed reference), the controlled PCA baseline of Chapter 7 (dense linear subspace), and the K-SVD-ADMM telemetry detector (learned overcomplete dictionary). Axis 2 is the prior built into the adaptive threshold: the structure-free additive CFAR rule versus the periodicity-referenced EWMA band, sampled at three points by the ablation of Section 7.2.4. The K-SVD-ADMM detector is drawn at its designed position, the periodicity end. Two effects pull it downward. The additive floor of Section 3.6 moves it structurally, and the configuration actually evaluated in Chapter 7 uses a structure-free threshold, which places the *measured* detector near the same height as the other two. The snapshot wavelet detector of Layer 1 is a model-free segmentation method and deliberately sits outside this plane.

and the aligned baseline in decibels, is then detrended by subtracting a straight line fitted to it across frequency. This step does fit a few parameters: the frequency shift, and the offset and slope of that line, that is a residual gain and tilt. They describe the *measurement*, not the normality: they compensate the instrumentation drift that accumulates between the two captures, so that like is compared with like. In statistical terms they are *nuisance* parameters; the model itself still contains exactly one normal pattern, and nothing is fitted that selects among alternative patterns. That aligned and detrended difference spectrum is the residual, integrated over a band comparable to the expected CUC bandwidth (a rectangular matched filter that trades frequency resolution for per-bin noise), and the threshold is the CFAR rule

$$\tau = \text{med}(r) + \max(k \text{MAD}(r), \delta), \quad (3.4)$$

whose adaptive term tracks the integrated residual floor (which falls as integration deepens) while the additive floor δ guards against the vanishing-residual regime at very large N . Every excursion of r above τ is reported with its frequency, excess, and margin.

The claim that this detector is an instance of sparse-representation reconstruction, and indeed *the degenerate instance*, can be made exact. Sparse coding (Section 2.11) reconstructs an observation as $\hat{y} = \mathbf{D}\mathbf{x}$ with $\mathbf{D}$ a dictionary of K atoms and $\mathbf{x}$ a sparse coefficient vector. Set $K = 1$ with the single atom $\mathbf{d}_1 = \mathbf{b}$, and fix the coefficient $x_1 = 1$: then $\hat{y} = \mathbf{D}\mathbf{x} = \mathbf{b}$, which is exactly Equation (3.3). The sparse-coding *optimisation* disappears because a one-atom dictionary with a pinned coefficient leaves nothing to optimise, which is precisely why the RF reconstruction step is computationally free. Pinning the coefficient is the step that makes the reduction exact. Left free, x_1 would act as a gain on the baseline, which in decibels is an offset, and that offset is one of the nuisance parameters the detrending step already removes. The correspondence also explains the method’s data requirement from the framework side: the clean baseline twin is nothing other than the *training set* of a size-one dictionary. Where the telemetry layer needs a corpus of anomaly-free windows to learn many atoms, the RF layer needs exactly one clean capture to “learn” its single atom.

3.5 Instance 2: The Telemetry Detector as the Rich Case

The telemetry layer (analysed in full in Chapter 6) occupies the other end of the model-richness axis. The observation is a matrix $\mathbf{S} \in \mathbb{R}^{w \times L}$ of L sliding windows of length w over a telemetry channel (Section 2.10). The normality model is an overcomplete dictionary $\mathbf{D} \in \mathbb{R}^{w \times K}$, $K > w$, learned by K-SVD (Section 2.11.6) from anomaly-free

training windows. Reconstruction is no longer trivial: it is the solution of the convex program

$$\min_{\mathbf{X}, \mathbf{E}} \frac{1}{2} \|\mathbf{S} - \mathbf{D}\mathbf{X} - \mathbf{E}\|_F^2 + \alpha \|\mathbf{X}\|_{1,1} + \beta \|\mathbf{E}\|_{2,1}, \quad (3.5)$$

solved by ADMM (Section 2.12), in which the column-sparse matrix $\mathbf{E}$ absorbs, *inside the optimisation itself*, whatever the normal dictionary cannot explain. The residual score of window j is $r_j = \|\mathbf{E}(:, j)\|_2$, spread back to time points, and the threshold is the period-referenced EWMA band of Section 2.13, augmented, for reasons developed in Section 3.6, with an additive floor.

Model richness is bought, not given, and the framework makes its price explicit. Where the RF instance needs one clean capture, this instance needs a *corpus* of anomaly-free telemetry to learn from; where the RF reconstruction is a stored template, this reconstruction is an iterative optimisation per batch of windows; and where the RF detector has two threshold parameters, this detector adds the dictionary size, the sparsity weights α and β , and the EWMA parameters. What the price purchases is the ability to model a normality with many faces (periodic cycles, command responses, mode changes) that no single template could describe, and Chapter 7 quantifies the return on that price against the deliberately cheaper PCA midpoint.

Figure 3.5.1 traces both instances end-to-end at the level of concrete data: what enters each processing stage, in what form, and what leaves it.

3.6 The Unifying Argument

That two detectors can be drawn in one diagram does not by itself make them one method. This section presents three results that justify the claim of unification on technical grounds: one formal, and two found empirically during this work.

3.6.1 Endpoints of one continuum

Section 3.4 showed the exact reduction: baseline differencing is sparse-dictionary reconstruction with $K = 1$ and a pinned coefficient. The two layers therefore do not merely resemble each other. They are the two endpoints of the model-richness axis, connected through the instrumented midpoint of the PCA baseline. Reading the axis left to right: a single template ($K=1$, no fitting), a dense linear subspace (a few principal directions, fitting by orthogonal projection), a learned overcomplete dictionary ($K > w$ atoms, fitting by sparse coding). Each step to the right buys expressiveness and pays in reference data, computation, and parameters. The continuum view also explains *when* each endpoint is appropriate: RF normality under a fixed carrier plan genuinely *is* one pattern, so the

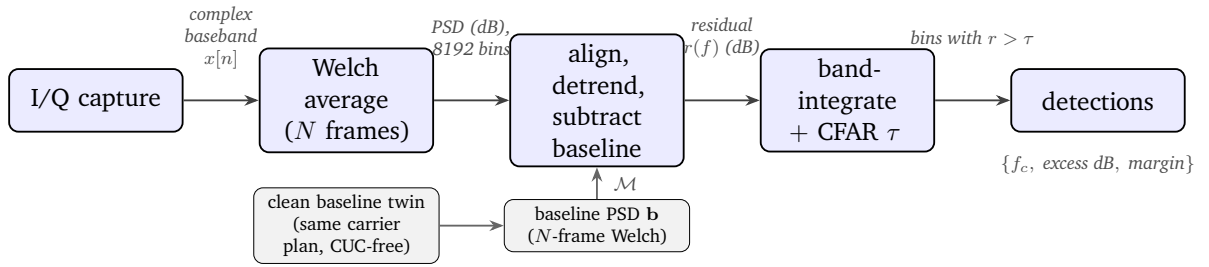
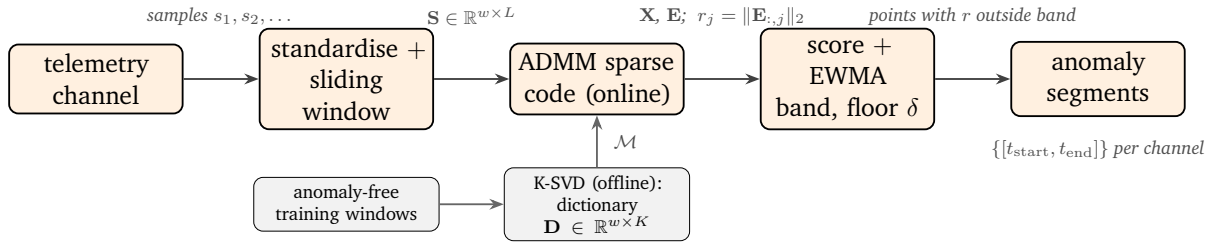
Layer 1: RF interference (Chapter 5)**Layer 2: telemetry anomaly (Chapter 6)**

Figure 3.5.1: Processing sequence and concrete data flow of the two instantiations. Each lane runs left to right from raw input to localised detections; the labels on the arrows give the form of the data passed between stages. The lower branch of each lane supplies the normality model $\mathcal{M}$: a single clean baseline spectrum in Layer 1, a K-SVD dictionary learned offline from anomaly-free windows in Layer 2. The two lanes are stage-for-stage the same pipeline; they differ in the domain of the data and in how much work the model-dependent stages (shaded branches and third box) must do.

degenerate model suffices and anything richer would add cost without expressiveness; telemetry normality is many-patterned, so the rich model earns its price.

3.6.2 The threshold generalisation discovery

The two threshold rules were adopted independently: CFAR from the radar tradition for the RF layer, the period-referenced EWMA band (Section 2.13) from the reference telemetry method [18] for the telemetry layer. Their unification was not designed; it was forced by a failure. The reference band is purely *multiplicative*: its upper and lower limits are $U_t(1 + \eta)$ and $U_t(1 - \gamma)$, proportional to the tracked statistic U_t . That statistic is the exponentially weighted moving average of the residual score, Equation (2.61), which the telemetry layer references one period back so that it tracks what the score normally is at this phase of the cycle (the band as implemented is Equation (6.9)). When the learned dictionary reconstructs normal telemetry almost perfectly, which the K-SVD dictionary of Chapter 6 does, the residual baseline, and with it U_t , approaches zero; every multiplicative band then collapses to zero width, and the detector over-flags on noise. The repair implemented in this thesis is an *additive floor* on the band width, so that the threshold never tightens beyond a fixed margin. But an additive robust margin is exactly the CFAR rule's $\max(k \text{ MAD}, \delta)$ term. The telemetry layer's own failure regime pushed it to the threshold the RF layer already used. The relationship is therefore one of generalisation, not analogy: the additive CFAR rule is what the periodic EWMA band reduces to when its structural assumptions (a periodic process, a non-vanishing residual baseline) are withdrawn, and any deployment of the telemetry detector on well-reconstructed or non-periodic channels *requires* the CFAR component to remain sound. That last clause is a prediction, and Section 7.2.4 tests it: on the benchmark's aperiodic majority the structure-free limit outscores the periodic band, and on the channels that do carry a period the two draw level. On this benchmark the reduction is therefore as good as or better than the band it replaces, wherever it is applied.

3.6.3 The shared failure mode and its common remedy

Both layers, in the course of this work, ran into the same problem: *deterministic structure does not average down*. On the RF side, a weak CUC's sub-decibel density bump is of the same order as the host carrier's own root-raised-cosine spectral shape; because both are deterministic, no amount of Welch averaging improves their ratio, and the snapshot detector's miss list is irreducible by longer integration alone. On the telemetry side, periodic telemetry drives the reconstruction residual through recurring nominal peaks that a threshold referenced to the immediate past mistakes for anomalies. The two

layers escape through different doors, but the doors share a frame: each references the residual to a structure that *shares* the deterministic component, so that the subtraction cancels it. The RF layer differences against a clean baseline twin (same carrier plan, same deterministic shape); the telemetry threshold references the statistic one *period* back (same phase of the cycle, same nominal peak). In framework terms, both are the same manoeuvre, aligning the reference with the deterministic structure of the observation, applied once to the reconstruction model and once to the threshold.

Table 3.6.1 collects the complete mapping.

Table 3.6.1: Complete mapping of the two detection layers onto the unified framework. The rows above the rule are the framework stages of Section 3.2; the rows below it are the coordinates of each detector on the two design axes of Section 3.3.

Framework element	Layer 1: temporal RF	Layer 2: telemetry
domain	PSD over frequency	sensor value over time
observation $\mathbf{y}$	N -frame Welch PSD (dB)	window matrix $\mathbf{S} \in \mathbb{R}^{w \times L}$
reference data	one clean baseline capture	anomaly-free training windows
model $\mathcal{M}$	baseline spectrum $\mathbf{b}$	K-SVD dictionary $\mathbf{D}$, $K > w$
reconstruction	$\hat{\mathbf{y}} = \mathbf{b}$ (aligned, detrended)	$\mathbf{DX}$ by ADMM, Eq. (3.5)
residual r	band-integrated $\mathbf{y} - \mathbf{b}$	$r_j = \ \mathbf{E}(:, j)\ _2$
threshold τ	med + $\max(k \text{ MAD}, \delta)$	period-EWMA band + additive floor
output	CUC flags $\{f_c, \text{excess}\}$	anomaly segments $\{[t_1, t_2]\}$
model richness	$K = 1$ (degenerate)	$K > w$ (overcomplete)
threshold prior	none (structure-free)	periodicity-aware

3.7 Scope and Honest Boundaries

Three boundaries of the framework are stated here so they need not be rediscovered in later chapters.

The snapshot detector is outside the framework. Layer 1’s first stage, multi-scale wavelet segmentation and the single-capture bump test of Chapter 5, is a *model-free* detector: it thresholds edge and energy evidence in one observation, with no reference data, no reconstruction, and no residual. It could be forced into the reconstruction-residual mould only by distorting either the detector or the mould. The thesis declines to do so. The two stages of Layer 1 are complementary precisely *because* they are of different classes: the snapshot stage is the fast, assumption-free first line that needs no reference; the temporal stage is the sensitive, framework-based second line that pays for its sensitivity with the baseline requirement.

Deviation from reference, not intent, is detected. A reconstruction-residual detector flags whatever its normality model does not contain. In the RF layer this means an authorised but *new* carrier, or one whose power rose beyond the CFAR floor since the baseline was captured, is flagged exactly as an interferer would be: the temporal stage is a change detector, and its baseline must therefore track the authorised carrier plan (a commissioning capture, refreshed on plan changes). Symmetrically, the telemetry layer flags novel-but-benign behaviour, such as a manoeuvre absent from the training corpus, as readily as a fault. These are properties, not defects, of monitoring against a reference; they are stated so that the performance figures of Chapter 7 are read correctly.

The layers run in parallel; fusion is future work. The two detectors share a framework, not data: they observe different physical quantities and are evaluated on different benchmarks. Correlating an RF interference event with a contemporaneous telemetry anomaly, the natural next step for a deployed system, requires jointly-coupled data that neither benchmark provides, and is accordingly identified as future work in Chapter 8 rather than claimed here.

Chapter 4

System Model and Data Generation

4.1 Role of This Chapter

The framework of Chapter 3 needs three things before any detector’s performance can be measured on equal terms: *observations* to be judged, *reference data* from which normality is modelled, and *ground truth* against which every detection can be scored. The third of these follows from the thesis’s insistence on measurable performance (Objective O1). This chapter constructs all three, for both domains. On the RF side it develops a simulation of an FDMA satellite transponder, explains how the physical quantities, above all the *frequency* of each carrier, are encoded into the generated data, and defines ten scenarios for benchmarking with explicit ground truth. On the telemetry side, it describes the public SMAP/MSL benchmark and the controlled synthetic generator used to validate the pipeline before touching real data.

Two design principles are key in this chapter. First, *reproducibility*: every dataset is produced by seeded, scripted generation, so any number and figure in Chapter 7 can be regenerated from the repository. Second, *separation of generation from detection*: the generator writes its ground truth into *.json* files, and the detectors never read anything but the raw signal. The detector’s output is then compared, at scoring time, against the ground truth recorded in the corresponding *.json* manifest.

4.2 The FDMA Transponder Model

4.2.1 Modelling approach

The simulation models the downlink of a bent-pipe FDMA transponder as seen at a monitoring station: a 36 MHz transponder slot carrying up to sixteen simultaneous legitimate QPSK carriers of different symbol rates (500 ksym/s–5 Msym/s) and powers (spanning up to 18 dB), plus additive channel noise and transponder impairments. The observation point is set *after* the transponder and the downlink channel, at the receiver, where an interference-monitoring system physically sits. What the system observes is the superposition of all carriers and interferers, digitised and read as a power spectrum.

The signal is represented in *complex baseband* (Section 2.4): the monitoring receiver is assumed to have mixed the transponder band down to zero frequency, so each sample is a complex number $x[n] \in \mathbb{C}$ and carrier positions become signed frequency offsets from the transponder centre. The complex sample rate is $f_s = 40$ MS/s, giving an unambiguous observable span of $[-20, +20]$ MHz that comfortably contains the 36 MHz slot with margin at both edges. Each capture is 1.0 s long (4×10^7 complex samples, stored as interleaved 32-bit floats), long enough to cut roughly 4880 non-overlapping frames of 8192 samples. These are the Welch blocks of Section 2.6, called *frames* from here on, and cutting so many of them is what allows the multi-frame integration study of Chapter 5 to synthesise *any* frame count N from a single recording rather than generating a separate dataset per N .

4.2.2 Per-carrier signal generation

Each carrier i is generated independently by the same five-stage chain, and the aggregate is formed only at the end. This is exactly the FDMA principle of Section 2.2, in which carriers share the band by occupying disjoint frequency slots rather than by any coordination in time.

1. **Payload.** A seeded pseudo-random byte stream. The seed is part of the scenario definition, so every dataset is exactly reproducible; the stream length is chosen so that the payload does not repeat within the capture, a point whose importance is a lesson of this thesis and is developed in Section 4.4.1.
2. **Modulation.** The bytes are Gray-mapped onto differential QPSK symbols $a_i[k]$, drawn from the unit-magnitude constellation (Section 2.4).

3. **Pulse shaping.** The symbol stream is upsampled by the integer factor $\text{sps}_i = \lfloor f_s / R_{s,i} \rfloor$ (samples per symbol; where the ratio is not an integer the realised symbol rate sits up to 2.6% above nominal, which moves the bump height of Section 2.7 by at most 0.02 dB) and filtered by a root-raised-cosine pulse with roll-off $\beta = 0.35$ (Section 2.5), producing the band-limited complex envelope $u_i[n]$. At this point the carrier is still centred at zero frequency: $u_i[n]$ occupies the band $|f| \leq R_{s,i}(1 + \beta)/2$, with the flat-top power spectrum derived in Section 2.5.4.
4. **Frequency translation.** The envelope is moved to its assigned slot $f_{c,i}$ by the rotation developed in Section 4.2.3 below, the stage where frequency enters the data.
5. **Power scaling.** The samples are scaled by the amplitude $A_i = 10^{P_i/20}$, where P_i is the carrier's assigned power in decibels. Power is thus controlled per carrier, which is what makes the heterogeneous-power scenarios of Section 4.4 possible.

The transmitted aggregate is the sum $\sum_i x_i[n]$ over all carriers of the scenario, including, in anomaly scenarios, the interferer, which is generated by *the same chain* with its own symbol rate, power, and centre frequency (Section 4.2.5).

4.2.3 Frequency and data correlation

In complex baseband there is no oscillator and no waveform “at” a frequency; there is only a sequence of complex samples. Frequency exists in that sequence in one form: *as a rotation rate of the complex phase from sample to sample*. A complex exponential at frequency f_c , sampled at f_s , advances its phase by a fixed increment every sample:

$$e^{j2\pi f_c n / f_s} = e^{j\varphi n}, \quad \varphi = 2\pi \frac{f_c}{f_s} \text{ rad/sample.} \quad (4.1)$$

Multiplying the zero-centred envelope by this sequence, a *rotator*, produces the translated carrier

$$x_i[n] = A_i u_i[n] e^{j2\pi f_{c,i} n / f_s}. \quad (4.2)$$

Concretely: carrier 5 of the reference scenario sits at $f_c = -3.5$ MHz, so its rotator advances by $\varphi = 2\pi(-3.5/40) = -0.55 \text{ rad} = -31.5^\circ$ per sample. Nothing else distinguishes carrier 5's samples from a carrier at any other frequency, and a negative frequency is simply a clockwise rotation. By the modulation property of the Fourier transform,

$$u_i[n] e^{j2\pi f_{c,i} n / f_s} \xleftrightarrow{\mathcal{F}} U_i(f) * \delta(f - f_{c,i}) = U_i(f - f_{c,i}). \quad (4.3)$$

so the envelope's RRC-shaped spectrum is re-centred at $f_{c,i}$ unchanged. Two consequences organise the whole RF problem:

- **The data smears a spectral line into a band.** An unmodulated carrier would be a single line at $f_{c,i}$; the symbols spread it into a flat-topped band of exactly $R_{s,i}(1+\beta)$: 1.35 MHz for a 1 Msym/s carrier, 405 kHz for a 300 ksym/s CUC. The width and shape of every bump the detector sees are set by the interferer's symbol rate and pulse shaping, not by any property of the receiver.
- **The spectrum estimator undoes the rotation for every frequency at once.** Each DFT bin correlates the signal with a rotation at $-f_k$, so a carrier lights up exactly the bins its band spans, and the Welch-averaged periodogram (Section 2.6) renders the carrier plan with no prior knowledge of where carriers sit. With $N = 8192$ and $f_s = 40$ MS/s the bin spacing is $\Delta f = f_s/N \approx 4.88$ kHz, fine enough that the benchmark's CUC localisation errors (tens of kilohertz, Chapter 7) are limited by the detector, not by the frequency grid.

4.2.4 Channel impairments

A clean superposition of RRC carriers over white noise would flatter the detectors. Each impairment in the model is present because it defeats a specific naive detection strategy, forcing the design choices that Chapter 5 defends.

- **Unequal carrier powers** (up to 18 dB spread). Breaks any single global energy threshold: one set below the weakest legitimate carrier lies far beneath the side-lobes of the strongest. This is the impairment that motivates per-carrier segmentation in the first place.
- **Additive white Gaussian noise.** The thermal noise floor of Section 2.3, generated as complex Gaussian noise of calibrated amplitude. It sets the SNR context in which weak carriers must survive segmentation and against which the periodogram variance argument of Section 2.6 plays out.
- **Transponder gain tilt** (a short FIR whose magnitude slopes 2–6 dB across the band). Real transponders are not flat; a tilted floor defeats any horizontal-line noise-floor estimate and forces the robust tilt-floor fit of Chapter 5. The steep-tilt scenario (E09) exists to stress exactly this.
- **Global frequency offset** (12.5–60 kHz, applied by a final rotator over the aggregate). Models residual Doppler and local oscillator error, so the whole band arrives slightly shifted. The default 12.5 kHz offset is deliberately *not* a multiple of

the 4.88 kHz bin spacing, so no carrier edge falls conveniently on a bin centre; the large offset in E09 stresses the temporal layer’s baseline-alignment step (Section 3.4).

- **A no-guard-band carrier pair** (carriers at +6.0 and +7.0 MHz with overlapping roll-offs). Between these carriers the PSD never returns to the noise floor, so occupancy-based segmentation alone cannot separate them; only edge evidence can. This pair is the standing justification for the wavelet edge detector.

4.2.5 Interference injection: the CUC generated carrier

The Carrier-under-Carrier interferer is not a synthetic artefact painted onto the spectrum; it is a *full QPSK carrier* generated by the same five-stage chain as every legitimate carrier, with three deliberate parameter choices that constitute the threat model of Section 2.7: its centre frequency lies inside a host carrier’s occupied band, its symbol rate is a fraction of the host’s, and its power is 10–22 dB below the host’s. Because it is a genuine modulated carrier, its spectral contribution follows the same flat-top model (Section 2.5.4) as the host, which is precisely why the superposition produces the localised power-density excess

$$\Delta = 10 \log_{10} \left(1 + \frac{P_c}{P_h} \frac{R_{s,h}}{R_{s,c}} \right) \text{ dB}$$

on the host’s flat top, as derived in Section 2.7. The benchmark spans this expression’s difficult range. In the reference scenario, a 300 ksym/s CUC at –13 dB under a 3 Msym/s host at +1 dB gives $P_c/P_h = 10^{-1.4}$ and $R_{s,h}/R_{s,c} = 10$, hence $\Delta \approx 1.5$ dB, which is visible to a snapshot detector. In the dense-narrowband scenario (E02), a 200 ksym/s CUC at –13 dB under a 600 ksym/s host at –3 dB yields a bandwidth ratio of only 3 and $\Delta \approx 1.1$ dB. That bump is larger than the 0.79 dB one of E01’s second interferer, at +6.5 MHz, yet no single spectrum finds it, because the host it sits on is narrow and its own structure is comparable. Absolute bump height is not what decides detection, and that is the reason the temporal layer exists. The generator’s ground truth records every CUC’s centre frequency, symbol rate, and power, so both the detection decision and the localisation error can be scored.

4.3 The Receiver-less Monitoring Model

A deliberate scope decision shapes the entire RF side: *the system contains no communications receiver*. Nowhere in the pipeline is a carrier demodulated: there is no carrier-recovery loop, no symbol-timing recovery, no decision device, and not a single payload

bit is ever decoded.

The omission is a design argument, not a simplification. A receiver that wished to decode carrier 5 would need two synchronisation loops. *Carrier recovery* tracks the true rotation rate, which Doppler and oscillator drift move off the nominal -3.5 MHz, classically with a Costas-type loop [7]. *Symbol-timing recovery* then finds the one sample per symbol at which the RRC cascade is ISI-free (Section 2.5), for example with a Gardner loop [14]. Both run continuously, per carrier, and must be configured with each carrier’s parameters.

The monitoring system needs none of this, for a reason Section 4.2.3 makes visible: the spectrum estimator already performs the de-rotation that a receiver performs, but for all frequencies simultaneously, measuring at each only *how much energy* rotates at that rate, never *what the symbols say*. The detection task is precisely to find where energy sits and whether unexplained energy has appeared; the FFT reveals carrier positions as band locations without being told them, the wavelet stage segments those bands, and the temporal stage’s frequency-alignment step (Section 3.4) absorbs global drift without tracking any individual carrier. Three practical properties follow, and all three are operationally valuable: the monitor requires no cooperation from, or configuration for, the carriers it watches; it is indifferent to encryption, since it never touches payload; and its computational cost is one FFT pipeline for the whole band rather than one synchronised receiver chain per carrier. The price is equally explicit: this system can detect and localise an interferer, but cannot identify or decode it. Attribution is out of scope, stated once here rather than implied anywhere in Chapter 7.

4.4 The RF Benchmark: Scenarios, Ground Truth, and Baselines

A single demonstration dataset shows that a detector *can* work; only a benchmark that varies the conditions shows *when* it works. The benchmark comprises ten scenarios, each designed to stress one specific aspect of the detection problem; together they contain 12 CUC events and one deliberately clean scenario whose only purpose is to punish false alarms. Table 4.4.1 summarises them.

Three aspects of the benchmark’s construction carry methodological weight.

Machine-readable ground truth. Every generated dataset is accompanied by a JSON manifest recording the full carrier plan (for each carrier its centre frequency, symbol rate, occupied bandwidth, power, and an explicit `is_cuc` flag) together with the sample

Table 4.4.1: The ten benchmark scenarios. Each varies the carrier layout, power spread, or channel impairments to stress a different aspect of the detector; E10 contains no CUC and exists solely to measure false-alarm behaviour.

Scenario	Layout	CUCs	Stresses
E01 baseline	14 heterogeneous carriers	2	reference layout
E02 dense narrowband	16 carriers, 600 ksym, 2 MHz pitch	1	segmentation resolution; weak bump
E03 sparse wideband	4 carriers, 5 Msym	1	wide-carrier edges
E04 uniform control	8 identical 1 Msym carriers	1	false-alarm control; weak bump
E05 power spread	powers +6 to −12 dB	1	floor estimate, weak carriers
E06 deep CUC	CUC 22 dB below a strong host	1	sensitivity limit
E07 wide CUC	700 ksym CUC in 1 Msym host	1	broad, host-like bump
E08 multi-CUC	3 simultaneous CUCs	3	multi-anomaly handling
E09 tilt + offset	6 dB tilt, 60 kHz offset	1	floor tracking, alignment
E10 clean complex	14 carriers, moderate tilt	0	must flag nothing

rate, duration, and impairment parameters. The scoring harness reads the manifest; the detectors read only the raw samples. Detection is scored per CUC (detected within a frequency tolerance, or missed) and false alarms are counted as flagged excesses matching no ground-truth CUC, so every headline number of Chapter 7 traces to an unambiguous, versioned definition of correctness.

Baseline twins as the framework’s reference data. Each scenario is generated *twice*: the anomaly capture, and a *baseline twin* with the same carrier plan and the same impairments, but with every CUC removed and with different payload seeds. The twin is the reference data $\mathcal{V}_{\text{ref}}$ of the framework (Section 3.2), from which the temporal detector’s single-template model is built. The different seeds are essential: were the payloads identical, the baseline would share not only the carriers’ deterministic spectral shape but their data realisation, and differencing would succeed for a trivial reason that no operational baseline could reproduce. With different data, only what is *structurally* common, the authorised carrier plan and the channel response, cancels. This is exactly the operational premise (a commissioning capture taken at a different time from the monitored traffic).

4.4.1 The non-repeating payload requirement

One generation parameter deserves its own subsection, because getting it wrong produced the single most instructive failure of this work. Early datasets fed each modulator from a 10^4 -byte payload buffer that repeated for the whole capture. A repeating payload makes the transmitted symbol sequence *periodic*, and a periodic signal does not have the smooth, continuous RRC spectrum of Section 2.5.4: its spectrum is a *comb* of discrete lines at multiples of the repetition rate, shaped by the RRC envelope. Under Welch averaging this fine structure does not behave like noise. It is deterministic, identical in

every frame, and therefore *does not average down*. On baseline differencing against a different-seed twin, the combs fail to cancel. Their lines sit at the same positions, since the spacing is set by the buffer length and the symbol rate, but the line amplitudes depend on the data and so differ between the two captures. The residual floor this leaves is 0.29 dB, enough to bury the 0.23 dB density excess of the deepest benchmark CUC (E06), which was missed for exactly this reason and initially misread as a sensitivity limit of the method.

The repair is a generation parameter, not a detector change: the payload length is raised to 4×10^6 bytes, exceeding the data transmitted over the capture even by the fastest (5 Msym/s) carrier, so no carrier’s symbol sequence repeats within the recording. The spectrum of each carrier then converges under averaging to its smooth continuous shape, the differencing residual floor drops to 0.07 dB, and E06’s CUC clears the threshold, giving 12 of 12 at zero false alarms (Chapter 7). The fix is also the *realistic* configuration: operational traffic does not loop a 10 kB buffer. The episode is reported at length because it is a clean instance of the failure mode that Section 3.6.3 elevates to a framework-level principle: deterministic structure survives averaging, and whatever the detector must cancel, its reference must genuinely share.

Figure 4.4.1 shows both halves of the effect on E06, regenerated for this purpose with each payload length in turn.

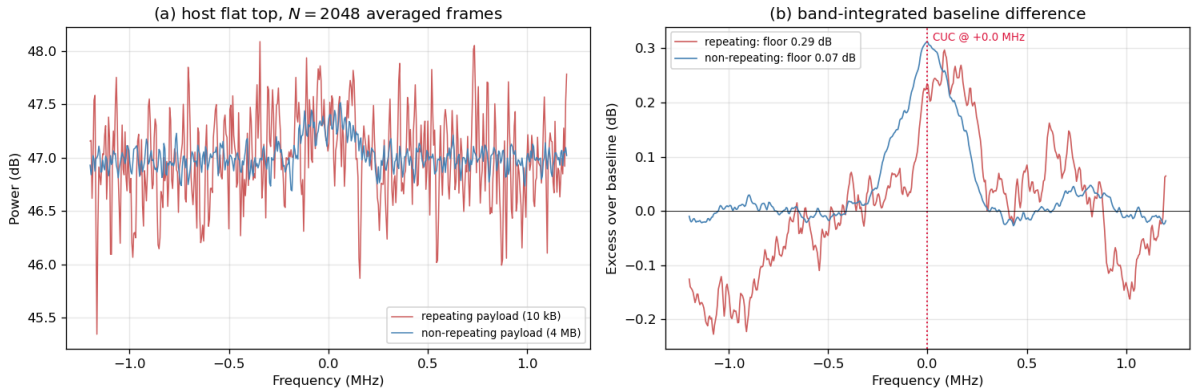


Figure 4.4.1: The payload-repetition failure and its repair, on scenario E06 at $N = 2048$ averaged frames. (a) Inside the host’s flat top, a 10 kB looping payload (red) produces a comb of discrete lines that averaging does not remove, because the comb is identical in every frame; a 4 MB non-repeating payload (blue) converges to the smooth continuous RRC spectrum. (b) Differencing each capture against its own clean twin. With the repeating payload the two combs sit at the same line positions but carry different amplitudes, so they fail to cancel and leave a residual floor of 0.29 dB, which swallows the CUC’s 0.23 dB excess entirely. With the non-repeating payload the floor falls to 0.07 dB and the same interferer stands four times clear of it at 0.31 dB.

4.5 The Telemetry Datasets

4.5.1 The SMAP/MSL public benchmark

The telemetry layer is evaluated on the public benchmark released with the LSTM-NDT work [21]: real housekeeping telemetry from two NASA missions, the SMAP orbiter and the MSL rover. Its label file lists 82 rows, 55 for SMAP and 27 for MSL, but these cover 81 distinct channels: the SMAP channel P-2 appears twice, with slightly different anomaly windows. The evaluation therefore runs on 81 channels, 54 SMAP and 27 MSL. Each channel provides exactly the split the framework requires: an anomaly-free *training* sequence, and a *test* sequence with expert-labelled anomalous intervals recorded as index ranges. Each sample is a vector whose first component is the telemetry value and whose remaining components are one-hot encoded command flags; the labelled anomalies are all contiguous index *ranges* rather than isolated samples, which is the extent the windowed method of Section 2.10 targets. The release additionally classes each of the 105 ranges as *point* (62) or *contextual* (43). That class describes whether the deviation leaves the nominal range or merely looks wrong for its context, not how many samples it spans. The choice of this dataset over alternatives is deliberate, on three grounds. First, *structural fit*: the anomaly-free training split matches the dictionary-learning requirement exactly, and the labelled anomaly *ranges* match the window-level granularity of the method (Section 2.14.2). Second, *published comparators*: the LSTM-based detector that the reference K-SVD-ADMM paper compares against was published *on this very dataset*, so Chapter 7 can compare against published numbers rather than against a re-implementation of unknown fidelity. Third, *known pitfalls*: the community’s scoring conventions on this benchmark, point-adjusted evaluation in particular, are documented to inflate results [19], and using the standard benchmark is what makes engaging that critique possible, via the dual point-wise/point-adjusted reporting motivated in Section 2.14.2. The benchmark’s contents have been criticised too: Wu and Keogh [40] report that many of its anomalies yield to a one-line rule, and that channels D-2, M-1 and M-2 mark more than half of their test split as anomalous. Those flaws are inherited here.

From each channel, observations are built exactly as Section 3.5 prescribes: values are standardised using statistics of the *training* split only (the test split must not influence the model), and windowed into the matrix $S \in \mathbb{R}^{w \times L}$ by a sliding window. Window-level scores are spread back to sample level for scoring against the labelled intervals, so that metrics are computed in the units the labels are given in.

4.5.2 The synthetic validation generator

Real telemetry is the evaluation target, but it is a poor debugging instrument: its anomalies are few, uncontrolled, and of unknown difficulty. The telemetry pipeline is therefore validated first on a synthetic generator whose every property is chosen. This is the telemetry-side analogue of the RF scenario generator, built on the same reproducibility principle: seeded, scripted, ground truth by construction.

The generator produces multi-channel periodic telemetry (a sum of a few harmonics with light noise, mimicking the thermal and orbital cycles of real housekeeping data) with an anomaly-free training split and a test split carrying injected *collective* anomalies of controlled type, position, and length: amplitude spikes, frequency shifts, and, critically, a *frozen-sensor* segment in which the signal holds constant. The frozen case is included by design: it is the anomaly that produces an *abnormally low* reconstruction residual (a flat segment is too easy to reconstruct), and it is the documented reason the threshold must be two-sided (Section 2.13). Phase 1 of the implementation, which used a one-sided threshold, missed exactly this channel. The theory’s prediction appeared on schedule in the controlled setting, before any real data was involved (Chapter 6). An optional generator mode adds a benign once-per-period transient with per-period amplitude jitter, reproducing the reference paper’s premise that periodic telemetry yields recurring nominal residual peaks; it is kept off by default and documented as an option, because tuning the synthetic data until it manufactures a desired comparison outcome would make the validation circular.

4.6 Summary: The Data in Framework Terms

Table 4.6.1 maps every dataset of this chapter onto the framework roles of Section 3.2.

Table 4.6.1: The datasets of this chapter in the roles defined by the framework of Chapter 3.

Framework role	RF domain	Telemetry domain
observations	scenario captures (I/Q, 1 s)	test-split windows $\mathbf{S}$
reference data $\mathcal{Y}_{\text{ref}}$	baseline twins (CUC-free, new seeds)	anomaly-free train split
ground truth	JSON manifests (is_cuc , f_c , R_s , P)	labelled anomaly intervals
validation sandbox	clean scenario E10	synthetic generator

Chapter 5

Layer 1: Wavelet-Based Interference Detection

5.1 Overview: A Two-Stage Detector

The interference layer consists of two detection stages of deliberately different character, run against the same capture. The *snapshot stage* operates on a single Welch-averaged spectrum with no prior knowledge whatsoever: it segments the band into carriers using multi-scale wavelet edge evidence and tests each carrier’s flat top for the localised bump of a Carrier-under-Carrier. It is fast, requires no reference data, and, being a model-free detector, sits outside the reconstruction-residual framework, as Section 3.7 established. The *temporal stage* is the framework’s degenerate instance (Section 3.4): it differences the integrated spectrum against a clean baseline twin and thresholds the residual with a CFAR rule, recovering the weak and narrow CUCs that no single-capture method can reach, at the price of requiring that a baseline exist.

This chapter develops both stages in the order the work developed them, because the order carries the argument: the snapshot stage is built first and pushed to its limit (Section 5.2); the limit is then measured and diagnosed rather than assumed (Section 5.3); and the temporal stage is introduced as the repair that the diagnosis dictates (Section 5.4). Section 5.5 assembles the two stages into the operational whole. Throughout, the detector operates on the observation defined in Chapter 4: the N -frame Welch PSD of the 40 MS/s complex-baseband capture, 8192 bins over $[-20, +20]$ MHz.

5.2 The Snapshot Stage

5.2.1 Estimating and denoising the spectrum

The spectrum estimate itself is the N -frame Welch average of 8192-point *Blackman*-windowed periodograms (Sections 2.6 and 4.2.3). The window choice is dictated by the band’s 18 dB power spread: spectral leakage through a strong carrier’s sidelobes can bury a weak neighbour outright, and the Blackman window’s ≈ 58 dB sidelobe suppression buys immunity from this at the price of a broader main lobe, a resolution loss of a few bins that no downstream stage, all of which operate at $\gtrsim 0.1$ MHz granularity, can perceive. The trade goes the right way because the detector’s enemy is leakage, not resolution.

Even after Welch averaging, the PSD estimate carries residual estimation ripple that any bin-level edge or peak logic would mistake for structure. The snapshot stage therefore first denoises the dB spectrum with a discrete wavelet transform: a level-3 decomposition on the `sym4` wavelet, soft thresholding of the detail coefficients at the universal (VisuShrink) threshold $\sigma\sqrt{2\ln N_{\text{bins}}}$, with the noise scale σ estimated robustly from the finest-level coefficients via the median absolute deviation (Section 2.8). The choice replaces the five-point moving average of the reference method [34], and the reason is the same asymmetry that motivates wavelets throughout this thesis: a moving average buys smoothness by blurring exactly the sharp roll-off edges the next stage must localise, whereas soft-thresholding in a wavelet basis suppresses the incoherent ripple, which projects onto many small coefficients, while leaving the edges, which project onto few large ones, essentially intact. The approximation band is never thresholded, so the carrier plateaus themselves pass through unchanged.

5.2.2 Multi-scale edge evidence

Carrier boundaries are located by the continuous wavelet transform of the denoised spectrum with the first-derivative-of-Gaussian atom (`gaus1`), evaluated at nine dyadic scales $s_j = 2^{j/2}$, $j = 0, \dots, 8$. Because `gaus1` is a smoothed differentiator, its response fixes the sign convention left open in Section 2.8: a *rising* power transition, a carrier start, produces a *negative* coefficient, and a *falling* transition, a carrier end, a *positive* one. The scale range is a hardening of the reference method’s $j \leq 6$: the widest benchmark carriers (5 Msym/s, 6.75 MHz occupied) have roll-offs so gradual that only the largest scales produce a compact response to them, and capping the range at $s_6 = 8$ left wide-carrier edges supported by smeared, low-confidence evidence only.

At each scale j the coefficients $c_j(f)$ are converted into binary edge votes by a robust,

scale-adaptive threshold. The coefficient noise scale is estimated as $D_j = 1.4826 \text{ med}[c_j - \text{med}(c_j)]$ (the MAD estimate, exploiting the near-zero median of a signed edge response), and the threshold is $T_j = k_j D_j$ with $k_j = k_0(0.8 + 0.4 s_j^2 / s_{j_{\max}}^2)$, $k_0 = 2$, slightly stricter at large scales, where the smoothing that makes the response robust also correlates neighbouring coefficients and would otherwise inflate the vote runs. The votes preserve polarity in two separate maps,

$$B_j^{\text{start}}(f) = \mathbf{1}[c_j(f) < -T_j], \quad B_j^{\text{end}}(f) = \mathbf{1}[c_j(f) > +T_j], \quad (5.1)$$

rather than voting on $|c_j|$: polarity is what later allows a start to be paired with its own end, the role played by the separate start/end edge maps of the reference method.

5.2.3 Cross-scale weighting: adaptive versus fixed

The nine per-scale vote maps must be fused into one edge-evidence function per polarity. The reference method uses fixed triangular weights peaking at mid-scale, $w_j = 1 - |j - j_{\text{mid}}|/j_{\max}$, a built-in prior that mid-scales are most trustworthy. This thesis replaces the prior with a measurement: the Shannon entropy H_j of each scale's normalised coefficient-energy distribution (Section 2.9) is low precisely when that scale's response is *focused*, with energy concentrated at a few frequencies, the signature of genuine localised edges, and high when the response is diffuse ripple. The weights

$$w_j = \frac{1/\sqrt{H_j}}{\sum_i 1/\sqrt{H_i}} \quad (5.2)$$

therefore trust whichever scales the *data* says are informative. The square root is a deliberate softening: weighting by $1/H_j$ outright crushes the large-scale votes on which wide carriers depend, re-introducing by the back door the mid-scale bias the adaptive scheme was meant to remove. The fused evidence is the weighted average $V^{\text{start}}(f) = \sum_j w_j B_j^{\text{start}}(f)$, and likewise for V^{end} ; both weighting schemes are implemented behind the same interface, which is what makes the controlled comparison of Chapter 7 possible. The honest finding there, anticipated in Section 2.14.1, is that the two are equivalent across all ten benchmark scenarios, returning segment boundaries that agree bin for bin.

5.2.4 Segmentation: occupancy skeleton, vote-refined edges

The reference method derives carrier boundaries purely from the fused votes: declare an edge wherever $V(f)$ exceeds a threshold. On the adverse band of Chapter 4 this proved non-robust, and the failure is worth recording. The *gaus1* response fires not once per

boundary but repeatedly along every RRC roll-off and on residual flat-top ripple; on the reference scenario the vote-threshold rule yielded of the order of sixty candidate edges for fourteen carriers, with unbalanced start/end counts, so any sequential pairing of starts with ends desynchronised within a few carriers. Rather than tune thresholds until the failure hid, the segmentation was restructured around a division of labour:

1. **Noise floor.** The transponder tilt (Chapter 4) makes any horizontal floor estimate wrong at one end of the band, and windowed percentile estimates fail in the crowded band centre, where a window sits entirely on neighbouring carriers and reports a “floor” tens of dB above the truth. The floor is instead fit as a single straight line by an iteratively-reweighted least-squares scheme [20]: fit a line, keep only bins within 3 dB above it, refit, and iterate. Carrier energy is discarded from the fit no matter how densely the band is packed, and the line tracks the tilt by construction.
2. **Occupancy skeleton.** Bins more than 6 dB above the fitted floor are marked occupied; contiguous occupied runs closer than 0.15 MHz are merged (bridging the shallow notch of the no-guard-band pair’s overlap). This yields one reliable run per carrier: energy occupancy is robust about *existence* but imprecise about *boundaries*, since the crossing point depends on the margin.
3. **Vote refinement.** Each boundary of a run is then snapped to the strongest fused wavelet vote within ± 0.30 MHz. The votes are imprecise about existence (they fire too often) but precise about *position*, the complementary strength. Runs whose refined width falls outside $[0.2, 12]$ MHz are discarded as noise slivers or merged aggregates.

The floor fit of step 1 is compact enough to exhibit, because it shows the carrier-rejecting mechanism directly (`carrier_cuc_analysis._estimate_tilt_floor`, abridged):

```

1  fit = np.full(n, np.percentile(power_db, 15))    # start at a low level
2  for _ in range(6):                             # 6 iterations
3      below = power_db < fit + keep_db            # bins near the current
          ↪ floor
4      m, b = np.polyfit(x[below], power_db[below], 1)
5      fit = m * x + b                            # refit the line to them

```

Each pass keeps only the bins within 3 dB of the current line and refits; carrier bins, tens of dB above, never enter the fit, so the line converges onto the tilted noise floor regardless of how much of the band the carriers occupy, which is the property the windowed-percentile approach lacked.

The result, on the benchmark, is 100% carrier segmentation coverage across all ten scenarios, including the no-guard-band pair and the 18 dB power spread. Figure 5.2.1 shows the stage mid-flight on the reference scenario: the denoised spectrum with its segmented carriers, and beneath it the signed cross-scale vote functions from which the boundaries were refined.

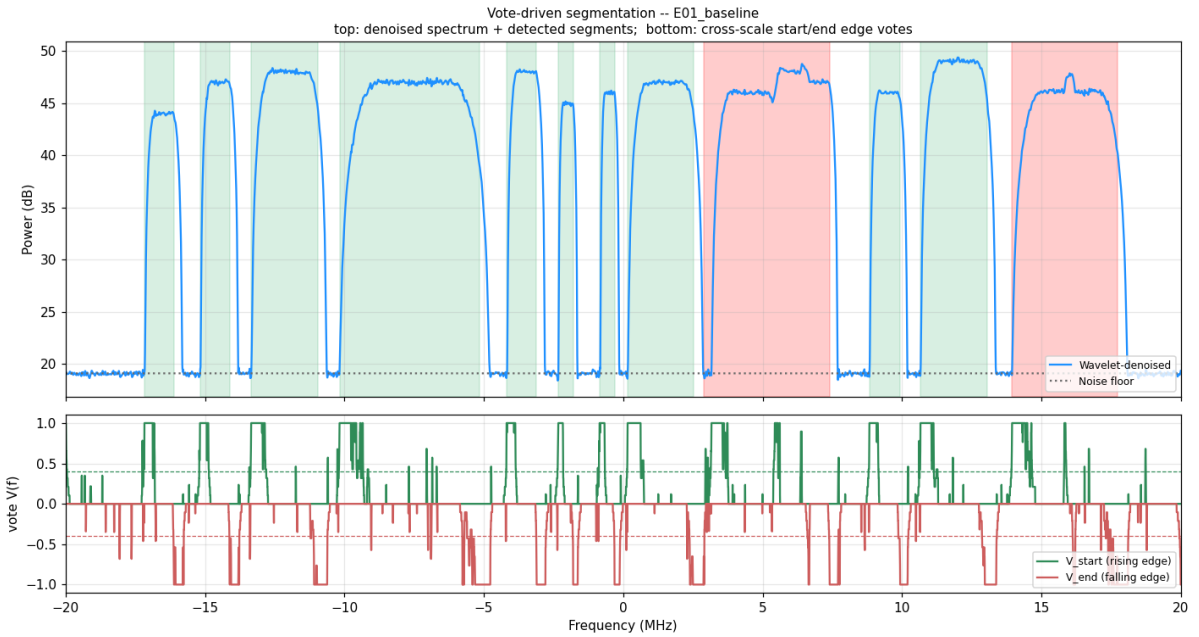


Figure 5.2.1: The snapshot stage on the reference scenario (E01). Top: wavelet-denoised Welch spectrum with the fitted noise floor and the segmented carriers (shaded); the two shaded-red segments are the carriers flagged as CUC hosts, each showing the localised bump of the buried interferer on its flat top. Bottom: the fused, signed cross-scale edge evidence V^{start} (positive, green) and V^{end} (negative, red) used to refine the segment boundaries; note the repeated firing along roll-offs that makes pure vote-thresholded segmentation non-robust (Section 5.2.4).

5.2.5 The flat-top bump test

Each segmented carrier is now examined for the CUC signature derived in Section 2.7: a localised power-density excess on the flat top. The inner 70% of the carrier (trimming 15% at each edge to exclude the roll-offs) is taken from the *denoised* spectrum, its median is the host's flat-top level, and a peak detector searches for excursions satisfying three conditions simultaneously: height at least h dB above the median, prominence at least

p dB, and width within a band $[w_{\min}, w_{\max}]$ MHz. The width band is the decisive discriminator, and it is dictated by the physics of Chapter 4: a genuine CUC's bump width is its occupied bandwidth $R_{s,c}(1 + \beta)$, hundreds of kHz for the threat model, whereas the two false-positive populations that rival a real CUC in height are both outside the band: residual estimation ripple is *too sharp*, and the plateau step where two merged carriers meet is *too broad*. A test on height and prominence alone cannot separate these; the width band does.

One candidate discriminator was tried and rejected, and the rejection is reported because it is instructive: flagging hosts whose flat-top *variance* is an outlier among the segmented carriers. The CUC hosts are not variance outliers, since a sub-dB bump on a wide flat top barely moves the variance, while the genuinely high-variance carriers are the weak, narrow ones whose dB trace is noisy near the floor: the variance test flags exactly the wrong population. It survives in the implementation as a reported diagnostic only.

Algorithm 5.1 consolidates the full stage.

Algorithm 5.1 – Snapshot CUC detection

Require: capture; frame count N ; preset $(h, p, [w_{\min}, w_{\max}])$

- 1: $P(f) \leftarrow N$ -frame Welch PSD (8192-bin Blackman), in dB
- 2: $\tilde{P}(f) \leftarrow$ sym4 soft-threshold denoise of P (VisuShrink) ▷ §5.2.1
- 3: **for** scales $j = 0, \dots, 8$ **do** ▷ §5.2.2
- 4: $c_j \leftarrow$ CWT($\tilde{P}$, gauss1, $2^{j/2}$); votes $B_j^{\text{start}}, B_j^{\text{end}}$ by threshold $T_j = k_j D_j$
- 5: **end for**
- 6: $V^{\text{start}}, V^{\text{end}} \leftarrow$ entropy-weighted vote averages, Eq. (5.2) ▷ §5.2.3
- 7: floor $\leftarrow$ iterative-linear tilt fit; occupied $\leftarrow \tilde{P} > \text{floor} + 6$ dB ▷ §5.2.4
- 8: carriers $\leftarrow$ merged occupied runs, each boundary snapped to the strongest vote within ± 0.3 MHz; keep widths in $[0.2, 12]$ MHz
- 9: **for** each carrier **do** ▷ §5.2.5
- 10: flat top $\leftarrow$ inner 70% of $\tilde{P}$ over the carrier; $m \leftarrow$ its median
- 11: find peaks with height $\geq m + h$, prominence $\geq p$, width $\in [w_{\min}, w_{\max}]$
- 12: flag the carrier as CUC host if any peak survives; report the peak frequency
- 13: **end for**

Ensure: carrier list; CUC flags with frequency localisation

The three thresholds $(h, p, [w_{\min}, w_{\max}])$ form the detector's operating point, and three calibrated presets are exposed (Table 5.2.1). The presets are not arbitrary sensitivity knobs: they trace the detector's ROC on the benchmark, and the *conservative* point, the operational default, is the most sensitive setting that still raises no false alarm anywhere

in the benchmark. What the looser presets buy, and what they cost, is quantified in Chapter 7; what none of them can buy is the subject of the next section.

Table 5.2.1: The three calibrated operating points of the flat-top bump test, and their benchmark scores (12 CUCs across the ten scenarios of Table 4.4.1).

Preset	h (dB)	p (dB)	width band (MHz)	detected	false alarms
conservative	0.9	0.80	[0.16, 0.70]	7 / 12	0
balanced	0.7	0.45	[0.12, 0.90]	8 / 12	1
high	0.5	0.45	[0.12, 0.90]	10 / 12	1

5.3 The Limit of the Snapshot Stage

The conservative preset misses five of the twelve benchmark CUCs. Their density excesses run from 0.35 to 1.14 dB, and that range *overlaps* the excesses of the CUCs it finds: E02’s 1.14 dB bump is missed while E01’s 0.79 dB bump is detected. Absolute bump height is therefore not what decides the outcome. What decides it is the bump’s height against the host’s own flat-top structure at that point in the band. Before designing a remedy, the limit was measured, and the measurement rules out the intuitive fix.

The intuitive fix is more integration: average more frames, reduce the ripple, and the bump should emerge. A direct probe of bump height against flat-top ripple as a function of the frame count N shows why this fails. On the uniform-control scenario (E04, CUC bump ≈ 0.6 dB), the ripple falls with N as Welch theory predicts, reaching ≈ 0.13 dB at $N = 450$ for a bump-to-ripple ratio near 5, yet the bump still fails the conservative test, because 0.6 dB is below the *absolute* 0.9 dB height threshold, and that threshold cannot be lowered without admitting false alarms from the host structure that averaging does *not* remove. The dense-narrowband scenario (E02) is worse: its host spans only about a hundred flat-top bins, and the “ripple” there stalls at ≈ 0.8 dB regardless of N , because it is not statistical fluctuation at all but the host’s own deterministic shape, the residual RRC curvature (Section 2.5.4) and implementation ripple, which every frame shares. Averaging reduces variance; it cannot reduce a bias. For contrast, the detected reference CUC (E01, +16 MHz) has a measured 1.1 dB bump (the ideal value of Equation (2.31) is 1.46 dB) over 0.3 dB of structure, comfortably clear, which is precisely why the snapshot stage finds it.

The diagnosis, then, is the failure mode that Section 3.6.3 raised to a framework principle: *the obstacle is deterministic, and deterministic structure does not average down*. A sub-dB bump cannot be separated from sub-dB host structure by any amount of single-capture integration, because both are constants of the capture. But a constant is reproducible, and what is reproducible can be *subtracted*. That is the temporal stage.

5.4 The Temporal Baseline-Differencing Stage

5.4.1 Method

The temporal stage instantiates the framework of Chapter 3 with the degenerate single-template model of Section 3.4. The observation is the N -frame-averaged spectrum of the monitored capture; the reference is the equally-averaged spectrum b of the baseline twin (Section 4.4): same carrier plan, same channel, no CUC, different payloads. The detector computes the residual $r(f) = y(f) - b(f)$ in dB and flags localised positive excesses. Because the host’s deterministic shape is present identically in both spectra, it cancels in the difference; what survives is the CUC’s density excess over a residual floor of order 0.05 dB set by the two captures’ independent noise and data realisations. That floor, unlike the host structure, genuinely falls with N .

Three refinements make this operational rather than idealised.

- **Frequency alignment.** The baseline is captured at a different time, so oscillator and Doppler drift shift the two spectra relative to each other, and a raw difference turns every carrier edge into a spurious residual spike. The baseline is therefore shifted, on an interpolated grid searched in 0.5 kHz steps over ± 60 kHz, to the offset that minimises the median absolute deviation of the difference over the occupied bins, a robust criterion that ignores the very excesses being hunted.
- **Linear detrending.** Residual gain and tilt drift between the captures appear as a slope and offset in the difference; both are removed by subtracting a least-squares line fit to the residual. These two operations are the “nuisance registration” of Section 3.4: a localised CUC bump survives a linear detrend; broad reference drift does not.
- **Band integration.** The aligned, detrended residual is averaged over a sliding window of 0.30 MHz, a rectangular matched filter of roughly the threat model’s CUC bandwidth, trading frequency resolution the detector does not need for per-bin noise suppression it does. The gain follows from an asymmetry between the two things being averaged. A CUC’s excess is deterministic and spread across its whole bandwidth, so every bin beneath it carries the same offset and the average of M such bins reproduces that offset undiminished. The residual noise is random, so its scatter falls as the square root of the number of *independent* bins averaged. That number is smaller than M , because the Blackman window couples neighbouring bins: its equivalent noise bandwidth is 1.73 bins [17], and adjacent power estimates correlate at about 0.57. At $\Delta f \approx 4.88$ kHz the 0.30 MHz window spans

$M \approx 61$ bins, which behave as about 26 independent ones, so the bump-to-noise ratio improves about fivefold at no cost to the bump itself. The window width is what makes this a *matched* filter rather than generic smoothing: widening it beyond the CUC bandwidth dilutes the excess with bins that contain only noise, the same dilution that reads a 1.5 dB core-band bump as a fraction of that when measured over too wide a span, while narrowing it leaves available signal-to-noise unclaimed.

The alignment step (`_frequency_align` in `multiframe_cuc.py`) must be computed on a capture that may contain the very anomaly being hunted:

```

1  occ = cur_db > (cur_db.max() - 20.0)    # occupied bins (within 20 dB of
    ↪ peak)
2  best_v, best_s = np.inf, 0.0
3  for sk in np.arange(-max_khz, max_khz + step_khz, step_khz):
4      sh = np.interp(freqs - sk / 1000.0, freqs, base_db)
5      d = (cur_db - sh)[occ]
6      v = np.median(np.abs(d - np.median(d)))    # MAD of the difference
7      if v < best_v:
8          best_v, best_s = v, sk

```

A criterion sensitive to the CUC's excess would shift the baseline to partially cancel the bump. The median absolute deviation cannot be moved by the few bins carrying a genuine excess, so the alignment is blind to the anomaly by construction. Each of the 241 candidate shifts costs one vectorised pass, so the brute-force search gives up nothing to a gradient or correlation estimator and avoids their local minima. Algorithm 5.2 consolidates the stage.

Algorithm 5.2 – Temporal baseline-differencing CUC detection

Require: capture, baseline twin; frames N ; $k = 6$; floor $\delta = 0.20$ dB; band $W = 0.30$ MHz

- 1: $y(f), b(f) \leftarrow N$ -frame Welch PSDs of capture and baseline (dB)
- 2: $b \leftarrow$ frequency-align b to y (MAD-minimising shift, ± 60 kHz)
- 3: $d \leftarrow y - b$; subtract the least-squares line from d ▷ gain/tilt detrend
- 4: $r \leftarrow$ moving average of d over W ▷ matched band integration
- 5: $\tau \leftarrow \text{med}(r) + \max(k \cdot \text{MAD}(r), \delta)$ ▷ CFAR, Eq. (3.4)
- 6: **for** each contiguous excursion of $r > \tau$ **do**
- 7: report detection at $\arg \max r$: frequency, excess (dB), margin (MAD units)

8: **end for**

Ensure: detection list

The integrated residual is thresholded by the CFAR rule of Equation (3.4): $\tau = \text{med}(r) + \max(k \text{MAD}(r), \delta)$ with $k = 6$ and $\delta = 0.20$ dB. The adaptive term keeps the false-alarm behaviour constant as the integrated floor falls with N ; the additive floor δ takes over in the deep-integration regime where the MAD becomes vanishingly small and a multiplicative rule alone would flag the residual's own numerical texture. This is the same degenerate regime, met for the same reason, as the telemetry threshold repair of Section 3.6.2. Each excursion above τ is reported with its centre frequency, excess, and margin in MAD units.

5.4.2 What integration now buys

With the deterministic component cancelled, integration finally pays. Figure 5.4.1 shows the progression on the uniform-control scenario: in a single frame the spectrum is ~ 5.6 dB of periodogram noise and the CUC is invisible; after averaging all 4882 frames the host is rendered cleanly but the 0.6 dB bump remains buried in host structure; in the baseline difference the host vanishes and the bump stands alone, fifteen times the residual floor. Every CUC the snapshot stage misses is recovered this way, and Chapter 7 scores the stage across the full benchmark.

The operating point is not fragile. The zero-false-alarm score holds for any additive floor in $[0.16, 0.30]$ dB, and $\delta = 0.20$ dB sits inside that range with margin either side. Below 0.16 dB false alarms begin to appear on E05 and E06; above 0.30 dB the deepest CUC is eventually lost. The residual and threshold on the reference scenario are shown in Figure 5.4.2; among the two events it isolates is the +16 MHz CUC that Figure 1.3.2(b) introduced as leaving no visible trace in the raw spectrum.

5.4.3 Robustness and operational caveats

A detector that differences against a reference is only as good as the reference's validity, so the stage's tolerance to baseline drift was measured directly by perturbing the baseline relative to the capture. With the alignment and detrending steps active, the benchmark score survives gain mismatch beyond 4 dB, spectral tilt through the full 6 dB tested, and frequency drift beyond 50 kHz, against a bare 2 kHz tolerance for raw differencing; the registration steps, not the differencing itself, carry the operational robustness. The stress scenario E09 (6 dB tilt, 60 kHz offset) confirms the same conclusion inside the benchmark proper.

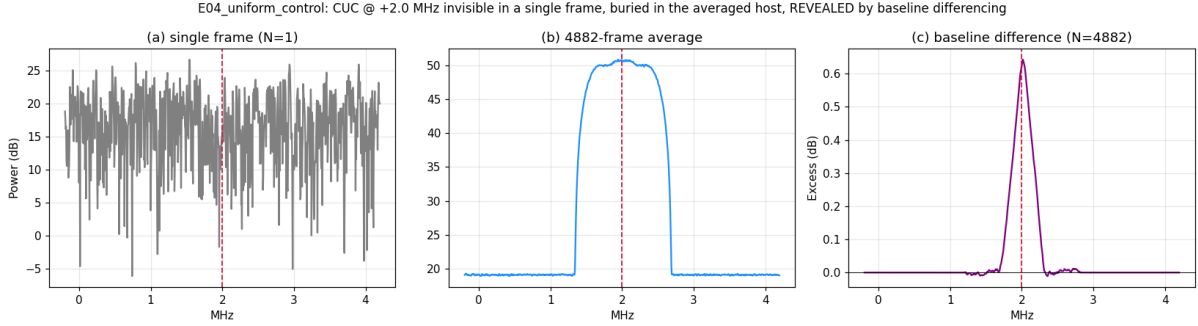


Figure 5.4.1: The progression that motivates the temporal stage, on scenario E04 (CUC at +2.0 MHz, bump ≈ 0.6 dB). (a) A single frame is dominated by periodogram noise. (b) Averaging all 4882 frames renders the host cleanly, but the bump remains of the order of the host’s own deterministic structure, so more averaging cannot help. (c) Differencing against the equally-averaged clean baseline cancels the host; the CUC’s density excess stands alone over a ~ 0.05 dB floor.

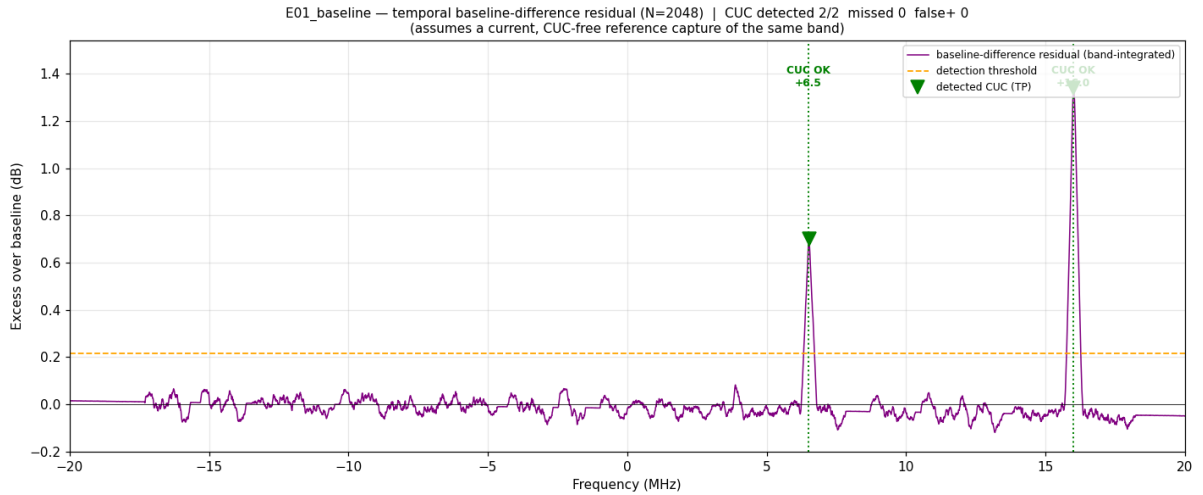


Figure 5.4.2: The temporal stage on the reference scenario (E01, $N = 2048$): band-integrated baseline-difference residual, CFAR threshold (dashed), and the two ground-truth CUCs, both detected with no false alarms. The residual floor across the remaining 36 MHz of band stays within ± 0.1 dB, because the host carriers, spanning 11 dB of power spread, have cancelled.

Two caveats travel with these numbers, both consequences of the framework’s deviation-not-intent boundary (Section 3.7). First, the stage flags *any* added power: an authorised carrier that is new, or $\gtrsim 0.2$ dB stronger than the baseline recorded, is flagged exactly as an interferer would be. The baseline must therefore track the authorised carrier plan: operationally, a commissioning capture refreshed on plan changes, the cost anticipated in Section 4.4. Second, the stage needs dwell: its sensitivity is bought with N averaged frames (2048 frames ≈ 0.4 s of signal at the benchmark’s configuration), so it is a monitoring detector, not an instantaneous one.

5.5 The Two Stages as a System

The stages are complementary by construction, and the complementarity is of kind, not merely of degree.

Table 5.5.1: The two stages of the interference layer.

	Snapshot stage	Temporal stage
detects	CUCs above host structure	weak/narrow CUCs
needs	one capture, no prior	clean baseline twin, dwell
detector class	model-free segmentation	framework instance ($K=1$)
latency	one averaged spectrum	$N = 2048$ frames
benchmark	7/12, 0 false alarms	12/12, 0 false alarms
fails when	bump $\lesssim$ host structure	baseline stale or absent

Operationally the two run as first and second line. The snapshot stage is always available: it needs nothing but the capture, reacts within one averaged spectrum, and its conservative preset raises no false alarms, so whatever it flags can be trusted immediately. The temporal stage is engaged where its precondition is met, a maintained baseline for the transponder, and supplies the sensitivity floor the snapshot stage cannot: detection down to ~ 0.2 dB of added spectral density, at the cost of dwell and of baseline upkeep. Neither subsumes the other: without a baseline the temporal stage does not exist, and with one the snapshot stage still provides the instant, assumption-free first response and a cross-check whose false-alarm behaviour is independent of baseline validity. Computationally the whole layer is undemanding by design: beyond the FFTs of the Welch estimate, every stage of Algorithms 5.1 and 5.2 is a linear-time pass over the 8192 spectrum bins, and nothing iterates to convergence, a deliberate contrast with the telemetry layer, whose reconstruction is itself an iterative optimisation (Chapter 6).

Chapter 6

Layer 2: Sparse-Representation Telemetry Anomaly Detection

6.1 Overview: The Rich Instance of the Framework

The telemetry layer implements the K-SVD–ADMM anomaly detection method of He et al. [18], reviewed in Section 2.14.2, as the learned-dictionary instance of the framework of Chapter 3. Like the reference method, it operates in two phases. *Offline*, a dictionary of normal local patterns is learned by K-SVD from anomaly-free training telemetry; this builds the normality model $\mathcal{M}$. *Online*, each batch of test windows is jointly decomposed by an ADMM solver into a sparse reconstruction over that dictionary plus a column-sparse residual matrix that absorbs whatever the dictionary cannot explain. This is the reconstruction and residual-scoring stage in one optimisation. A period-referenced adaptive threshold then turns residual scores into detections.

This chapter develops each stage at three levels: the *logic* (why the stage is formulated as it is), the *mathematics* (the objective and the derived update rules, building on the tools of Sections 2.11–2.12), and the *implementation* (how each equation becomes NumPy code in `src/telemetry_anomaly/`, including the numerical guards that the equations alone do not reveal). The implementation is deliberately pure NumPy, with no optimisation library called, both so that every update in the solver corresponds line-for-line to a derived equation, and because the joint (X, E) ADMM solve is not a stock algorithm that a library provides. Throughout, the notation matches Chapter 3: windows $\mathbf{S} \in \mathbb{R}^{w \times L}$, dictionary $\mathbf{D} \in \mathbb{R}^{w \times K}$, codes $\mathbf{X} \in \mathbb{R}^{K \times L}$, residual $\mathbf{E} \in \mathbb{R}^{w \times L}$. The generic signal dimension n and

atom count m of Chapter 2 specialise here to the window length w and dictionary size K .¹

6.2 Problem Formulation

6.2.1 Offline: dictionary learning

The training telemetry is standardised (using training statistics only) and cut into overlapping windows of length w , forming $\mathbf{S}_{\text{train}} \in \mathbb{R}^{w \times L_1}$ (Equation (2.41)). The offline problem is the dictionary learning of Equation (2.47): find $\mathbf{D}$ and sparse codes $\mathbf{X}$ minimising $\|\mathbf{S}_{\text{train}} - \mathbf{DX}\|_F^2$ subject to $\|\mathbf{x}_j\|_0 \leq T_0$ for every window. Because the training split is labelled anomaly-free, whatever $\mathbf{D}$ learns to reconstruct is treated as normality. The model’s claim to detect anomalies therefore rests on that labelling being right (Section 4.5.1): an anomaly left unlabelled in the training split would be learned as normal, and mislabelling of exactly this kind is documented for these benchmarks [40].

6.2.2 Online: joint reconstruction and residual

The online model is the decisive formulation choice, inherited from the reference method. Rather than reconstructing each test window and measuring the leftover after the fact, the residual is made a *variable of the model*:

$$\mathbf{S}_{\text{test}} = \mathbf{DX} + \mathbf{E}, \quad (6.1)$$

with $\mathbf{E}$ constrained to be column-sparse: an anomaly affects a whole window, so anomaly evidence should occupy whole *columns* of $\mathbf{E}$, and only a few of them. Relaxing the two combinatorial sparsity constraints to their convex surrogates (Sections 2.11 and 2.12.2) yields the working objective of the layer:

$$\min_{\mathbf{X}, \mathbf{E}} \underbrace{\frac{1}{2} \|\mathbf{S} - \mathbf{DX} - \mathbf{E}\|_F^2}_{\text{explain everything}} + \underbrace{\alpha \|\mathbf{X}\|_{1,1}}_{\text{tax on atom usage}} + \underbrace{\beta \|\mathbf{E}\|_{2,1}}_{\text{toll on declaring anomalies}}. \quad (6.2)$$

The annotations are the right way to read Equation (6.2): the three terms compete. The data term wants the observation fully explained; the α term makes explanation through

¹Two errors in the reference paper are corrected here, so that the code matches the mathematics rather than the printed symbols: the data-fidelity term of the online objective retains the factor $\frac{1}{2}$ and the square of the Frobenius norm (the paper’s eq. 7 drops both, which would break the least-squares form of the $\mathbf{X}$ -update), and the code matrix has dimensions $K \times L$, not $w \times L$ (a dimension typo in the paper’s Algorithm 2 output).

the dictionary, cost something per active coefficient, forcing the code to stay sparse; and the β term makes explanation through $\mathbf{E}$, that is *declaring a window anomalous*, cost β per unit of column norm. A window ends up in $\mathbf{E}$ only when the dictionary cannot explain it more cheaply than the toll, which is why β is, quite literally, the detector's sensitivity knob: the analysis of how the optimiser enforces this is the block-thresholding step of Section 6.4. The anomaly score of window j is the norm of its column of evidence, $r_j = \|\mathbf{E}(:, j)\|_2$.

6.3 K-SVD as Implemented

The theory of K-SVD, the atom-by-atom rank-one updates and their optimality via the Eckart–Young theorem, was developed in Section 2.11.6. What remains for this chapter is the concrete algorithm actually run, which differs from the textbook loop in three engineering respects, and the way its inner sparse-coding step is coded.

6.3.1 The sparse-coding step: OMP with support masking

Every K-SVD iteration starts by coding each training window with the OMP of Equations (2.44)–(2.45). The implementation (`sparse_model.omp`) is the greedy loop exactly as derived (correlate atoms with the current residual, select the best, re-fit all active coefficients by least squares, update the residual) with one guard the equations do not show:

```

1  proj = np.abs(D.T @ residual)      # correlate atoms with residual
2  if idx:
3      proj[idx] = -np.inf             # never reselect an active atom
4      idx.append(int(np.argmax(proj)))
5      Ds = D[:, idx]
6      coef, *_ = np.linalg.lstsq(Ds, y, rcond=None) # re-fit active set
7      residual = y - Ds @ coef

```

In exact arithmetic the masking line is redundant: after the least-squares re-fit the residual is orthogonal to every active atom, so no active atom can win the next correlation. In floating point, however, the correlations of active atoms are tiny-but-nonzero, and when the remaining residual is small an active atom can be re-selected, silently stalling the support below T_0 . The mask guarantees the support grows to exactly T_0 atoms, which matters downstream, because the K-SVD atom update partitions work by which windows *use* which atoms, and degenerate supports starve atoms of users.

6.3.2 The dictionary update, line for line

The rank-one update of Section 2.11.6 carries over line for line into `sparse_model.py`; the correspondence between Equations (2.49)–(2.51) and the code is one-to-one:

```

1  using = np.nonzero(X[k, :])[0]           # omega_k: windows using atom k
2  X[k, using] = 0.0                       # remove atom k from the model
3  E_k = S_train[:, using] - D @ X[:, using] # E_k^R: error without atom k
4  U, s, Vt = np.linalg.svd(E_k, full_matrices=False)
5  D[:, k] = U[:, 0]                       # d_k      <- u_1
6  X[k, using] = s[0] * Vt[0, :]           # x_R^k     <- sigma_1 * v_1

```

Zeroing row k before forming the error matrix is what realises the “atom k removed” term E_k of Equation (2.48) without building the explicit sum over $j \neq k$; restricting columns to `using` is the support restriction E_k^R that preserves sparsity; and the two assignment lines are the Eckart–Young optimum of Equation (2.51) verbatim. Because u_1 is a unit vector, the atom renormalisation that other dictionary-update schemes need is automatic.

6.3.3 Engineering departures from the textbook loop

Algorithm 6.1 – K-SVD dictionary learning, as implemented

Require: train windows $S \in \mathbb{R}^{w \times L}$, atoms K , sparsity T_0 , iterations n_{it} , cap $L_{\max}$

- 1: if $L > L_{\max}$: subsample $L_{\max}$ columns of S at random
- 2: $D \leftarrow K$ random columns of S , ℓ_2 -normalised
- 3: **for** n_{it} iterations **do**
- 4: **for** each window j **do** ▷ sparse coding
- 5: $X(:, j) \leftarrow \text{OMP}(D, S(:, j), T_0)$
- 6: **end for**
- 7: **for** each atom $k = 1, \dots, K$ **do** ▷ dictionary update
- 8: $\omega_k \leftarrow \{j : X(k, j) \neq 0\}$
- 9: **if** $\omega_k = \emptyset$ **then**
- 10: $D(:, k) \leftarrow$ worst-reconstructed window, normalised ▷ re-seed dead atom
- 11: **else**
- 12: rank-one SVD update of (d_k, x_R^k) on E_k^R ▷ Eq. (2.51)
- 13: **end if**
- 14: **end for**
- 15: **end for**

Ensure: dictionary $\mathbf{D}$

Three departures from the textbook loop (Algorithm 6.1) are deliberate. *Dead-atom re-seeding* (line 10): an atom that no window selected receives no update and would stay frozen forever; it is instead re-seeded onto the currently worst-reconstructed training window, so dictionary capacity migrates to where the model is failing. *Training-set capping* (line 1): the cost of one iteration is dominated by L runs of OMP, and consecutive overlapping windows are nearly identical, so on long channels the training windows are subsampled to $L_{\max} = 2000$ before learning. The dictionary needs coverage of the normal patterns, not every copy of them. *Initialisation from data*: the dictionary starts as random training windows (normalised), not random noise, so even iteration one reconstructs plausibly. This is also exactly the Phase-1 “naive dictionary” baseline (Section 6.7), making the ablation between a naive and a learned dictionary a pure measurement of what the K-SVD iterations add.

6.4 The ADMM Solver

6.4.1 The choice of ADMM, and the splitting

The online objective (6.2) is convex, but it resists direct minimisation: it couples a smooth quadratic to *two* non-smooth penalties acting on *two* different variables. No closed-form solution exists, and generic subgradient descent would be slow and would never produce the exact zeros, in the code and, critically, in the residual columns, that the detection logic depends on. What makes the problem tractable is the observation, from Section 2.12, that each of its pieces is easy *in isolation*: a quadratic in $\mathbf{X}$ is a linear solve; an $\ell_{1,1}$ penalty has the element-wise soft-threshold as its proximal map (2.53); an $\ell_{2,1}$ penalty has the block soft-threshold (2.55). ADMM is the machinery that lets each piece be solved in isolation while a dual variable stitches the pieces back into one problem.

The splitting is chosen so that the non-smoothness lands on its own variables. The code matrix is duplicated into a copy $\mathbf{Z}$ carrying the $\ell_{1,1}$ penalty, with the constraint $\mathbf{Z} = \mathbf{X}$ re-imposing consistency:

$$\min_{\mathbf{X}, \mathbf{Z}, \mathbf{E}} \frac{1}{2} \|\mathbf{S} - \mathbf{D}\mathbf{X} - \mathbf{E}\|_F^2 + \alpha \|\mathbf{Z}\|_{1,1} + \beta \|\mathbf{E}\|_{2,1} \quad \text{s.t.} \quad \mathbf{Z} = \mathbf{X}. \quad (6.3)$$

The augmented Lagrangian, the constrained problem’s objective plus a multiplier term and a quadratic penalty on the constraint violation (cf. Equation (2.57)), is

$$\mathcal{L}_\mu = \frac{1}{2} \|\mathbf{S} - \mathbf{D}\mathbf{X} - \mathbf{E}\|_F^2 + \alpha \|\mathbf{Z}\|_{1,1} + \beta \|\mathbf{E}\|_{2,1} + \langle \mathbf{M}, \mathbf{Z} - \mathbf{X} \rangle + \frac{\mu}{2} \|\mathbf{Z} - \mathbf{X}\|_F^2, \quad (6.4)$$

with multiplier matrix $\mathbf{M} \in \mathbb{R}^{K \times L}$ and penalty weight $\mu > 0$. One ADMM iteration minimises $\mathcal{L}_\mu$ over each primal variable in turn with the others fixed, then takes a gradient-ascent step on the multiplier. Each of the three minimisations now has a closed form, derived next.

6.4.2 Deriving the three updates

X-update: a regularised least-squares solve. With $\mathbf{Z}, \mathbf{E}, \mathbf{M}$ fixed, $\mathcal{L}_\mu$ is smooth and quadratic in $\mathbf{X}$. Setting its gradient to zero,

$$\nabla_{\mathbf{X}} \mathcal{L}_\mu = -\mathbf{D}^\top (\mathbf{S} - \mathbf{D}\mathbf{X} - \mathbf{E}) - \mathbf{M} - \mu(\mathbf{Z} - \mathbf{X}) = 0,$$

and collecting the $\mathbf{X}$ terms gives the normal equations

$$(\mathbf{D}^\top \mathbf{D} + \mu \mathbf{I}) \mathbf{X} = \mathbf{D}^\top (\mathbf{S} - \mathbf{E}) + \mathbf{M} + \mu \mathbf{Z}. \quad (6.5)$$

The system matrix is symmetric positive definite for any $\mu > 0$, because $\mu \mathbf{I}$ regularises the Gram matrix $\mathbf{D}^\top \mathbf{D}$, which for an overcomplete dictionary is rank-deficient on its own. The solve is therefore unique and stable. This is where the $\frac{1}{2}$ and the squared norm dropped by the paper's eq. 7 matter: without them the right-hand side acquires a spurious factor and Equation (6.5) is no longer the minimiser.

Z-update: element-wise soft-thresholding. With $\mathbf{X}$ fixed, the terms involving $\mathbf{Z}$ are $\alpha \|\mathbf{Z}\|_{1,1} + \langle \mathbf{M}, \mathbf{Z} \rangle + \frac{\mu}{2} \|\mathbf{Z} - \mathbf{X}\|_F^2$; completing the square in $\mathbf{Z}$ shows this is exactly the proximal problem (2.52) of the ℓ_1 norm at the point $\mathbf{P} = \mathbf{X} - \mathbf{M}/\mu$ with parameter α/μ , whose solution is the soft-threshold (2.53):

$$\mathbf{Z} = \eta_{\alpha/\mu}(\mathbf{X} - \mathbf{M}/\mu) = \text{sign}(\mathbf{P}) \odot \max(|\mathbf{P}| - \alpha/\mu, 0), \quad (6.6)$$

applied entry by entry. This is the step that produces exact zeros in the code: coefficients the data does not support are not merely small, they are removed.

E-update: block soft-thresholding, the detector. With $\mathbf{X}$ fixed, write $\mathbf{Q} = \mathbf{S} - \mathbf{D}\mathbf{X}$ for the part of the data the current reconstruction leaves unexplained. The $\mathbf{E}$ terms of $\mathcal{L}_\mu$ are $\frac{1}{2} \|\mathbf{Q} - \mathbf{E}\|_F^2 + \beta \|\mathbf{E}\|_{2,1}$, which separates over columns into the proximal problem of the column- ℓ_2 norm; by Equation (2.55),

$$\mathbf{E}(:, j) = \max\left(0, 1 - \frac{\beta}{\|\mathbf{Q}(:, j)\|_2}\right) \mathbf{Q}(:, j), \quad j = 1, \dots, L. \quad (6.7)$$

This one line is the detector, and it repays reading slowly. For each window, the solver looks at the unexplained energy $\|\mathbf{Q}(:, j)\|_2$ and makes a hard decision: below the toll β , the column is set exactly to zero and the window is declared normal, and whatever small misfit remains is left for the data term to absorb as noise; above β , the column survives into $\mathbf{E}$, shrunk by the toll, and the window carries anomaly evidence. Detection is therefore not a post-processing step applied to residuals: it happens *inside* the optimisation, jointly with reconstruction, each influencing the other across iterations. The group structure of the penalty (Section 2.12.2) is what makes the decision per-window rather than per-sample, which is precisely the collective-anomaly modelling choice.

Dual ascent and penalty escalation. The multiplier absorbs the remaining constraint violation, $\mathbf{M} \leftarrow \mathbf{M} + \mu(\mathbf{Z} - \mathbf{X})$, and the penalty is escalated geometrically, $\mu \leftarrow \rho\mu$ with $\rho = 1.25$, tightening the $\mathbf{Z} = \mathbf{X}$ coupling as the iterations proceed.

6.4.3 The algorithm, and its convergence caveat

Algorithm 6.2 – ADMM solve of the online objective

Require: $\mathbf{S} \in \mathbb{R}^{w \times L}$, $\mathbf{D}$, weights α, β , penalty μ_0 , growth $\rho > 1$, cap $\mu_{\max}$, tolerance ϵ

- 1: $\mathbf{X}, \mathbf{Z}, \mathbf{M} \leftarrow \mathbf{0}^{K \times L}$; $\mathbf{E} \leftarrow \mathbf{0}^{w \times L}$; $\mu \leftarrow \mu_0$
- 2: precompute $\mathbf{D}^\top \mathbf{D}$ and $\mathbf{D}^\top \mathbf{S}$
- 3: **for** up to n_{it} iterations **do**
- 4: $\mathbf{X} \leftarrow (\mathbf{D}^\top \mathbf{D} + \mu \mathbf{I})^{-1} (\mathbf{D}^\top (\mathbf{S} - \mathbf{E}) + \mathbf{M} + \mu \mathbf{Z})$ ▷ Eq. (6.5)
- 5: $\mathbf{Z} \leftarrow \eta_{\alpha/\mu}(\mathbf{X} - \mathbf{M}/\mu)$ ▷ Eq. (6.6)
- 6: $\mathbf{Q} \leftarrow \mathbf{S} - \mathbf{D}\mathbf{X}$; block soft-threshold each column at β into $\mathbf{E}$ ▷ Eq. (6.7)
- 7: $\mathbf{M} \leftarrow \mathbf{M} + \mu(\mathbf{Z} - \mathbf{X})$
- 8: **if** $\|\mathbf{Z} - \mathbf{X}\|_F / \|\mathbf{X}\|_F < \epsilon$ **then break**
- 9: **end if**
- 10: $\mu \leftarrow \min(\rho\mu, \mu_{\max})$
- 11: **end for**

Ensure: scores $r_j = \|\mathbf{E}(:, j)\|_2$ and evidence $\mathbf{E}$

One caveat remains: this is a *three-block* ADMM (blocks $\mathbf{X}$, $\mathbf{Z}$, $\mathbf{E}$), and the classical convergence guarantee for ADMM [2] covers the two-block form of Equation (2.56), and it does not carry over: Chen et al. [5] exhibit a convex three-block problem on which the direct extension diverges. The implementation therefore treats convergence as an empirical matter and hardens the loop accordingly: the penalty escalation $\rho > 1$ (the

reference method's choice) strongly damps oscillation between blocks in practice; the penalty is capped at $\mu_{\max} = 10^4$ so the $\mathbf{X}$ -solve stays well-conditioned; and the loop exits early when the relative primal residual $\|\mathbf{Z} - \mathbf{X}\|_F / \|\mathbf{X}\|_F$ falls below 10^{-4} . Across every channel of the evaluation the loop behaved uneventfully, but the thesis claims observed convergence, not guaranteed convergence.

6.4.4 From equations to code

The solver, `sparse_model.admm_residual_scores`, is the loop of Algorithm 6.2 in eleven lines, shown in full because the one-to-one correspondence with the derivation is the point of writing the solver by hand:

```

1  for _ in range(n_iter):
2      # X-update:  $(D^T D + \mu I) X = D^T(S - E) + \text{Mult} + \mu Z$ 
3      X = np.linalg.solve(DtD + mu * I, DtS - D.T @ E + Mult + mu * Z)
4      # Z-update: element-wise soft-threshold at  $\alpha/\mu$ 
5      P = X - Mult / mu
6      Z = np.sign(P) * np.maximum(np.abs(P) - alpha / mu, 0.0)
7      # E-update: column-wise block soft-threshold at  $\beta$  on  $Q = S - D X$ 
8      Q = S - D @ X
9      norms = np.linalg.norm(Q, axis=0) + 1e-12
10     E = Q * np.maximum(0.0, 1.0 - beta / norms)
11     # dual ascent + penalty escalation, with early exit
12     primal = Z - X
13     Mult = Mult + mu * primal
14     if np.linalg.norm(primal) / (np.linalg.norm(X) + 1e-12) < tol:
15         break
16     mu = min(mu * rho, mu_max)
17 return np.linalg.norm(E, axis=0), E

```

Reading the code against the mathematics, several implementation-level decisions become visible that the equations leave implicit:

- **All L windows at once.** Every update is written on whole matrices, so the solver processes a channel's entire test matrix in one call: the column-separability of Equations (6.6) and (6.7) becomes vectorised NumPy rather than a Python loop over windows.

- **Guards.** The $+10^{-12}$ on the column norms makes the block threshold well-defined on exactly-zero columns (a perfectly reconstructed window must yield $\mathbf{E}(:, j) = 0$, not a division-by-zero); the same guard on $\|\mathbf{X}\|_F$ keeps the stopping test meaningful at the first iterations, when $\mathbf{X}$ is still near zero.

6.5 Adaptive Thresholding as Implemented

6.5.1 The period-referenced EWMA band

The residual scores of a healthy periodic channel are not flat: the dictionary reconstructs some phases of the cycle better than others, so the score itself inherits a periodic texture, and any threshold referenced to the *immediately* preceding samples mistakes every recurring peak for an onset. The reference method’s answer (Section 2.13) is to reference the running statistic one *period* back:

$$U_t = \lambda U_{t-p} + (1 - \lambda) a_t, \quad (6.8)$$

where a_t is the per-point score, p the period, and $\lambda = 0.8$ the memory. U_t thus tracks what the score *normally* is at this phase of the cycle. A two-sided band around it,

$$\tau_t^{\text{up}} = U_t(1 + \eta) + \delta, \quad \tau_t^{\text{down}} = U_t(1 - \gamma) - \delta, \quad (6.9)$$

flags a_t outside $[\tau_t^{\text{down}}, \tau_t^{\text{up}}]$. Both sides are load-bearing. The upper bound catches novelty: patterns the dictionary cannot represent. The lower bound catches the opposite failure: a signal reconstructed *too* easily, the signature of a frozen or stuck sensor whose flat windows any dictionary fits almost exactly; dropping the lower bound on the assumption that residuals are one-sided loses precisely this fault class, a prediction the validation of Section 6.7 confirms experimentally. On a flag, the implementation applies the reference method’s reset rule $U_t \leftarrow U_{t-p}$, so that the anomaly itself does not contaminate the running statistic and desensitise the detector immediately after an event.

6.5.2 The additive floor δ : a discovered repair

The band of Equation (6.9) as published has $\delta = 0$: it is purely multiplicative, its width proportional to U_t . That design carries a hidden assumption, that the healthy-score baseline stays bounded away from zero, as it visibly does in the reference paper’s own case study. The assumption fails for this implementation’s well-trained dictionaries: the ADMM E-step zeroes the columns of normal windows *exactly* (Equation (6.7)), so

on clean data the score series is near-binary, long stretches at zero with sparse peaks, and $U_t \rightarrow 0$ drags both band edges to zero, flagging the baseline’s numerical texture wholesale. The repair is the additive margin δ in Equation (6.9): a band width the threshold cannot tighten below, defaulting to half the robust spread of the score series ($\delta = \frac{1}{2} \cdot 1.4826 \text{ MAD}$) and reducing to the published rule at $\delta = 0$. The default needs one guard that the formula alone does not show. On a well-reconstructed channel the score series is not merely small but mostly *exactly* zero, so its median and its MAD are both zero and δ would default to zero, collapsing the repair in precisely the regime that motivates it. Measured on the benchmark, four of the first eight channels have between 94 and 99 per cent exact zeros. The implementation therefore falls back to half the standard deviation whenever the MAD vanishes. The RF layer’s CFAR threshold needs no such fallback, because its floor δ is a fixed 0.20 dB rather than a fraction of the score’s own spread. As Section 3.6.2 argued, this additive robust margin is exactly the CFAR term of the RF layer’s threshold.

One point of scope belongs here rather than in the results. Two thresholds appear in this thesis, and the reported figures use only one of them. The evaluation of Chapter 7 fixes a per-channel scalar operating point by sweeping it for best point-adjusted F_1 , the protocol common in this literature, and every number reported there comes from that rule. The band of Equation (6.9) is the layer’s own threshold and is assessed separately in Section 7.2.4, because its periodic prior is a claim about the *data* rather than about the detector, and SMAP/MSL is largely command-driven rather than periodic. That ablation finds the prior a liability on the channels that carry no period and of no measurable benefit on those that do, which is why the headline configuration uses the structure-free rule.

Algorithm 6.3 – Period-referenced EWMA detection with additive floor

Require: point scores $a_{1..n}$, period p , memory λ , band (η, γ) , floor δ

```

1: for  $t = 1, \dots, n$  do
2:    $\text{ref} \leftarrow U_{t-p}$  if  $t > p$  else last available  $U$  (else  $a_t$ )
3:    $U_t \leftarrow \lambda \text{ref} + (1 - \lambda) a_t$ 
4:   flag if  $a_t > U_t(1 + \eta) + \delta$  or  $a_t < U_t(1 - \gamma) - \delta$ 
5:   if flagged then  $U_t \leftarrow \text{ref}$  ▷ do not poison the trend
6:   end if
7: end for
```

6.6 Assembling the Pipeline

Three pieces of glue turn the components into the detector evaluated in Chapter 7, and each embodies a small decision.

Window scores to point scores. Scores live on windows; labels live on samples. Each window's score is spread over the w samples it covers, and a sample takes the *maximum* over all windows containing it: a point is as anomalous as the most anomalous context it appears in. The maximum (rather than mean) keeps short events from being diluted by the majority-normal windows that overlap them. Its cost, a halo of elevated scores around each true event and roughly a window length wide, is a known precision tax quantified in Chapter 7.

Period estimation. The period p of Equation (6.8) is estimated from the *training* split as the first dominant peak of the autocorrelation beyond a minimum lag, consistent with the rule that nothing about the model may depend on test data. On channels with no meaningful periodicity the autocorrelation decays without a later peak, so its maximum sits at the minimum lag itself, $p = 5$. The band then references a sample five steps back rather than the same phase of a cycle, and Section 7.2.4 measures what that costs.

Standardisation. Channel values are standardised by the training split's mean and deviation before windowing, the same statistics applied to train and test, so dictionary atoms and thresholds operate on a scale set by normal behaviour alone.

6.7 Validation before Evaluation

The pipeline was validated in staged form on the synthetic generator of Section 4.5.2 before touching the benchmark, and the two stages each produced a finding worth the discipline.

Phase 1: naive dictionary, OMP, fixed threshold. With the dictionary a mere random sample of training windows and a constant CFAR threshold on the scores, the pipeline already detected the spike and frequency-shift anomaly channels with full recall, and scored *zero* recall on the frozen-sensor channel. This is not a bug but the theory arriving on schedule: a frozen segment yields an abnormally *low* residual, invisible from above; a one-sided threshold cannot see it (Section 6.5). The miss reproduces, in a controlled setting, exactly the motivation the reference paper gives for its two-sided band, before that band was implemented here.

Phase 2: K-SVD, ADMM, EWMA band. With the full machinery, all four synthetic channels are detected, including the frozen one (recall $0 \rightarrow 0.57$): the learned atoms are oscillatory, so a flat segment becomes expensive to represent and the evidence lands in E. The learned dictionary fixes from below what the two-sided band fixes from above. The same experiments surfaced the multiplicative-band degeneration and its additive-floor repair described above. One negative decision is recorded deliberately: the synthetic generator was *not* iterated until the paper’s adaptive-versus-fixed improvement re-materialised on it, because tuning the validation data until it flatters the method under test would make the validation circular. The generator’s benign-transient mode, which reproduces the regime where the periodic band genuinely outperforms a fixed threshold, exists as a documented option and defaults to off.

6.8 Configuration and Summary

Table 6.8.1: Layer-2 configuration used in Chapter 7, against the reference method’s case study [18].

Parameter	Reference	This work	Note
window w	60 / 100	60	their setting for most anomalies
atoms K	1500	128	see discussion below
OMP sparsity T_0	4	4	
K-SVD iterations	≤ 50	10–12	converged RMSE in practice
train-window cap	–	2000	subsampling, Alg. 6.1
ADMM α, β	1, 2.6	1, 2.6	adopted unchanged
ADMM μ_0, ρ	1, 1.25	1, 1.25	+ cap $\mu_{\max} = 10^4$
ADMM iterations	≤ 100	≤ 100	early exit at 10^{-4}
EWMA λ	0.8	0.8	
band η, γ	swept	swept	not the rule behind the Ch. 7 figures
floor δ	– (0)	$\frac{1}{2}$ robust spread	this work’s repair

The one deliberate departure in scale is the dictionary size: the reference case study learns $K = 1500$ atoms from long antenna-component telemetry, whereas SMAP/MSL channels provide a few thousand training samples each. 1500 atoms against ≤ 2000 (capped) training windows would make the dictionary nearly as large as its training set, memorising rather than generalising. $K = 128$ with $w = 60$ keeps the dictionary overcomplete ($K > w$, the regime the theory requires) while leaving atoms an order of magnitude fewer than training windows.

Chapter 7 measures what the *model* half of this machinery buys, against the PCA midpoint of the model-richness axis, under two scoring protocols that turn out to disagree about the winner.

Chapter 7

Performance Analysis

7.1 Layer 1 Evaluation: RF Interference Detection

This section evaluates the wavelet-based interference detector of Chapter 5 on the ten-scenario FDMA benchmark of Section 4.4. Both stages are scored on the same captures: the snapshot stage, which reads a single Welch-averaged spectrum with no reference data, and the temporal stage, which differences an integrated spectrum against a clean baseline twin. The comparison is controlled in the same sense as the Layer 2 comparison of Section 7.2. The two detectors see identical data and differ only in how much prior knowledge each is allowed to use.

7.1.1 Benchmark and scoring protocol

The benchmark comprises ten scenarios, E01 to E10, carrying twelve Carrier-under-Carrier events between them. E10 is a clean control with no CUC at all, included so that false alarms can be counted on a capture where every flag is by definition wrong. Each scenario ships with a JSON manifest recording every carrier's centre frequency, symbol rate, occupied bandwidth and power, together with the flag marking which carriers are interferers.

Scoring reads the manifest; the detectors read only the raw samples. A CUC counts as detected when a flagged excess falls within a frequency tolerance of its true centre, and a flagged excess matching no ground-truth CUC counts as a false alarm. Recall is taken over the twelve CUCs, and false alarms are counted across all ten scenarios rather than

per scenario, so one spurious flag anywhere in the benchmark is visible in the headline figure.

The tolerance is 1.0 MHz for both stages, and no result below depends on it: re-scoring at 0.10 MHz, ten times tighter, leaves every count in this section unchanged.

The interferers span the difficult part of the threat model. Their powers run from 10 to 22 dB below their hosts and their occupied bandwidths from 0.270 to 0.945 MHz, which by Equation (2.31) places their density excesses between 0.35 and 2.53 dB. Eleven of the twelve fall below 1.5 dB.

7.1.2 Segmentation and the snapshot stage

Segmentation is a precondition rather than a result, since the bump test can only examine flat tops the segmenter has found. Across all ten scenarios every ground-truth carrier falls inside a reported segment, for 100% coverage. This holds in the two configurations built to break it. The first is the no-guard-band pair at +6.0 and +7.0 MHz, whose roll-offs overlap so that the spectrum never returns to the noise floor between them. In E01 the neighbouring +4.5 MHz carrier overlaps in turn, so three carriers are reported as a single merged segment, which is why E01's fourteen carriers yield twelve segments. The second is the 18 dB power spread of E05. Coverage counts whether every carrier falls inside some segment, not whether segments and carriers correspond one to one.

Table 7.1.1 reports the snapshot stage at its three calibrated presets, and Figure 7.1.1 plots them as an operating curve.

Table 7.1.1: The snapshot CUC detector at its three presets, over twelve CUCs in ten scenarios. Thresholds are the minimum bump height h , the minimum prominence p , and the admissible width band $[w_{\min}, w_{\max}]$ of Section 5.2.5. False alarms are counted across the whole benchmark.

Preset	h (dB)	p (dB)	$w_{\min}$ (MHz)	$w_{\max}$ (MHz)	TP	FN	FA	Recall (%)	F_1
Conservative	0.90	0.80	0.16	0.70	7	5	0	58.3	0.737
Balanced	0.70	0.45	0.12	0.90	8	4	1	66.7	0.762
High	0.50	0.45	0.12	0.90	10	2	1	83.3	0.870

The conservative preset detects seven of the twelve CUCs and raises no false alarm anywhere in the benchmark. Loosening to the high preset recovers three more for one false alarm, which falls on the clean control E10.

Which CUCs resist detection is more informative than the count. Two are missed at every preset, and they are the two cases Section 5.3 predicts.

The two are E02 and E07, and they are the *two most powerful interferers in the benchmark relative to their hosts*, at 10 and 11 dB below respectively. Every weaker CUC is found at

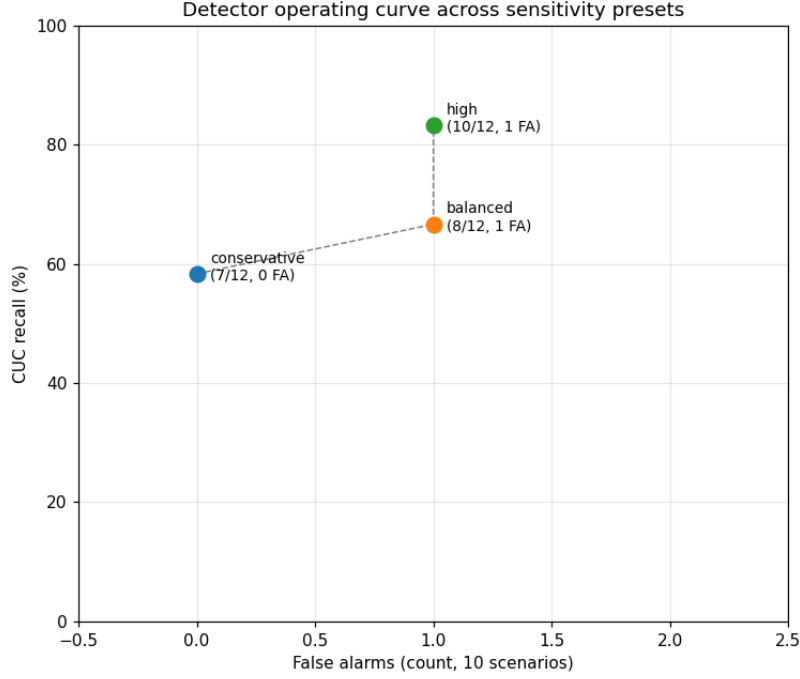


Figure 7.1.1: The snapshot stage’s operating curve across the three presets. Recall is over the twelve benchmark CUCs; false alarms are counted over all ten scenarios. The conservative preset is the operational default and is the most sensitive point that still raises no false alarm.

some preset, including the one 22 dB down. No single observation shows more clearly that snapshot detectability is not governed by the power ratio. Appendix C lists all twelve events with their powers, bandwidths and outcomes.

What defeats each is its geometry. E07 carries the widest interferer at 0.945 MHz, giving a bandwidth ratio near unity, so its excess spreads evenly across the whole host top and produces a uniform lift with no localised feature for a bump test to find. E02 fails for the opposite reason. Its interferer raises the host by 1.14 dB, more than four of the seven CUCs the stage does detect, but its host spans only about a hundred flat-top bins and the deterministic ripple across them reaches 0.8 dB. That bump clears the height threshold and fails on prominence, because there is too little clean flat top around it to stand out from.

7.1.3 The temporal stage

At $N = 2048$ frames the temporal stage detects **all twelve CUCs with zero false alarms**, across all ten scenarios including the clean control. Figure 7.1.2 sets the two stages side by side, scenario by scenario.

The five CUCs the conservative snapshot preset misses are all recovered, including the

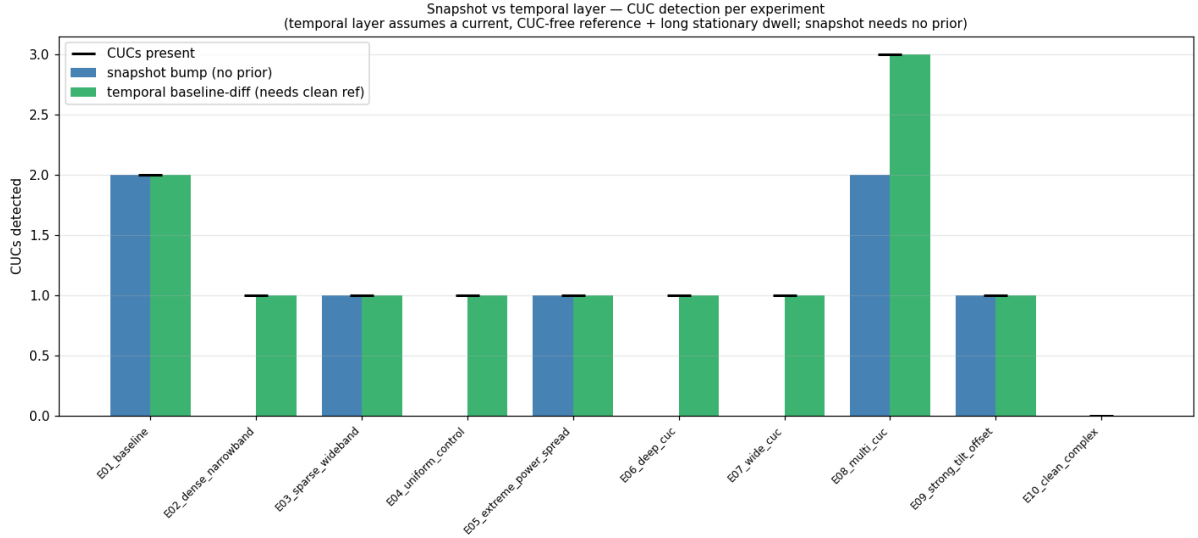


Figure 7.1.2: CUC detections per scenario for the two stages of Layer 1. Black bars mark the number of CUCs actually present. The snapshot stage (blue, conservative preset) needs no reference data; the temporal stage (green, $N = 2048$) requires a maintained clean baseline and about 0.4 s of dwell. The five CUCs the snapshot stage misses are exactly those whose excess does not clear their host’s own deterministic structure.

two that no preset reaches. E06, the deepest interferer at 22 dB below its host, and E07, the widest, are both detected. The mechanism is the one Section 5.4.1 derives. Differencing against a baseline twin removes the host’s deterministic structure, which averaging alone cannot do. The residual noise that remains then falls as $1/\sqrt{N}$ while the CUC’s excess does not.

This is not a like-for-like ranking of two competing designs. The temporal stage buys its sensitivity with a maintained baseline and with dwell, and it is blind without them; the snapshot stage needs one spectrum and no prior at all. They are the first and second lines of defence set out in Section 5.5, which is why both are reported rather than only the stronger.

The two stages are also scored at different frame counts, 400 for the snapshot stage and 2048 for the temporal stage, and the asymmetry runs in the snapshot stage’s favour. Re-running the snapshot pipeline unchanged at the temporal stage’s own frame count returns five of twelve at $N = 1024$ and five at $N = 2048$, against seven at $N = 400$. Integration does not merely fail to help the snapshot stage, as Section 5.3 predicts it cannot; past a few hundred frames it costs detections. The seven of twelve quoted here is therefore the snapshot stage at its best operating point, and the gap to twelve of twelve understates rather than exaggerates what the reference data buys.

7.1.4 Localisation accuracy

Detecting a CUC is only half of what an operator needs. The other half is knowing where it sits, so that the affected host carrier can be named.

Across all twelve CUCs the temporal stage places the interferer within 0.002 to 0.066 MHz of its true centre frequency, with a mean error of 0.023 MHz. For the seven it also detects, the snapshot stage manages 0.015 to 0.086 MHz with a mean of 0.058 MHz.

Two comparisons make these numbers meaningful. Against the 4.88 kHz bin spacing of the Welch estimate, the temporal errors run from well under one bin to about fourteen, so localisation is limited by the detector and not by the frequency grid, as Section 4.2.3 anticipated. Against the interferers' own occupied bandwidths of 0.270 to 0.945 MHz, the error is between half a per cent and twenty per cent of the width of the signal being located.

The temporal stage is the more accurate of the two by a factor of about 2.5 in the mean, and it is more accurate on six of the seven CUCs both stages find. The mechanism is the one that gives the stage its sensitivity. Band integration over $M \approx 61$ bins suppresses the per-bin noise that jitters a peak's position. Baseline differencing then removes the host's own sloped structure, so the residual maximum is not pulled off centre by the shape it sits on.

The worst temporal case is E09, which carries 6 dB of tilt and a 60 kHz frequency offset. It is the only scenario where the temporal stage (0.066 MHz) localises less accurately than the snapshot stage (0.042 MHz). The alignment grid does not explain it, since its 0.5 kHz step is over a hundred times smaller than the error. The same interferer at the same frequency in E01, without those impairments, is placed within 0.017 MHz, so the extra error comes with E09's impairments, but no mechanism in the detector has been isolated for it and it is recorded as an observation.

7.1.5 Tolerance to baseline drift

The temporal stage's headline result assumes a baseline that still describes the band. Real baselines age, so the stage's tolerance was measured directly by perturbing the baseline against the capture on E04 at $N = 2048$ and re-running the detector. Table 7.1.2 reports the result with the registration steps of Section 5.4.1 disabled (*raw*) and enabled (*robust*). The two drift axes fail in different ways, and the difference matters operationally.

Frequency drift is the dangerous one for raw differencing. A misaligned baseline turns every carrier edge into a spurious residual spike, and from 2 kHz of drift onward the raw mode reports seven to nine false alarms while still finding the CUC. The alignment step removes this completely: the robust mode holds zero false alarms out to 50 kHz, a

Table 7.1.2: Detection and false alarms as the baseline is perturbed against the capture (E04, $N = 2048$, one CUC present). *Raw* differences the two spectra directly; *robust* adds the frequency alignment and linear detrending of Section 5.4.1. Gain offsets are omitted from the table because both modes detect the CUC with no false alarm at every offset tested up to 4 dB.

Baseline drift	Raw		Robust	
	CUC found	FA	CUC found	FA
Tilt 0.2 dB	yes	0	yes	0
Tilt 0.5 dB	no	0	yes	0
Tilt 1.0 dB	no	0	yes	0
Tilt 2.0 dB	no	0	yes	0
Tilt 4.0 dB	no	0	yes	3
Tilt 6.0 dB	no	0	yes	0
Freq 1 kHz	yes	0	yes	0
Freq 2 kHz	yes	9	yes	0
Freq 5 kHz	yes	7	yes	0
Freq 20 kHz	yes	7	yes	0
Freq 50 kHz	yes	7	yes	0

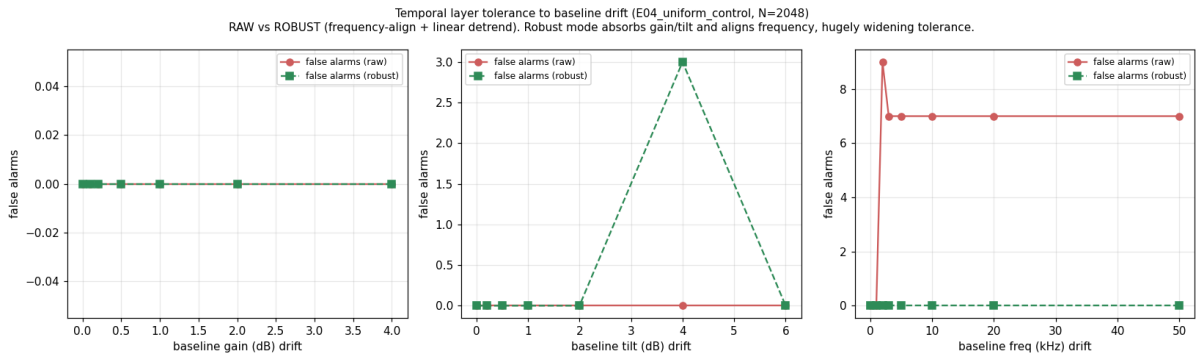


Figure 7.1.3: False alarms as the baseline drifts in gain, tilt and frequency (E04, $N = 2048$). Note that the raw mode's flat zero under tilt is not robustness: Table 7.1.2 shows it has stopped detecting the CUC at all beyond 0.2 dB, so it has nothing left to raise a false alarm with.

tolerance twenty-five times wider than the point at which raw differencing breaks down. Tilt drift fails silently instead. Beyond 0.2 dB the raw mode stops detecting the CUC altogether, and its false-alarm count stays at zero only because it is no longer flagging anything. Reading the left and centre panels of Figure 7.1.3 without the detection column would invert this conclusion, which is why both are tabulated. The detrending step keeps the CUC detected through 6 dB of tilt, the largest perturbation tested.

One result is reported without a clean explanation. At exactly 4 dB of tilt the robust mode raises three false alarms, while both 2 dB and 6 dB raise none. The behaviour is not monotonic in the perturbation and no mechanism in the detrending step accounts for it, so it is recorded as an observation rather than explained away.

Plan changes were stressed on the same scenario. An authorised carrier whose power rises by 0.2 dB above the recorded baseline is flagged, as is a newly appeared carrier. Both are correct behaviour for a change detector and both are false alarms for an interference monitor, which is the deviation-not-intent boundary of Section 3.7 appearing as a measurement. It sets the operational requirement on the baseline: it must be refreshed whenever the authorised carrier plan changes.

7.1.6 Cross-scale weighting: entropy against fixed triangular

Section 5.2.3 replaces the reference method’s fixed triangular cross-scale weights with data-adaptive entropy weights $w_j \propto 1/\sqrt{H_j}$. Both are implemented behind one interface, so the substitution can be measured rather than argued.

It makes no difference. Across all ten scenarios the two weightings produce segmentations that agree bin for bin, identical CUC flags, and the same seven of twelve detections at zero false alarms. Not one scenario separates them.

The reason is not that the two weightings agree, because they do not. Averaged over the benchmark the per-scale entropies fall monotonically from 6.71 nats at the coarsest scale to 4.64 nats at the finest, so the entropy weights rise monotonically towards the fine scales, from 0.093 to 0.147. The triangular weights instead peak in the middle, running from 0.077 at the extremes to 0.154 at mid-scale. The two schemes emphasise opposite ends of the scale range, with comparable spread, and still return the same boundaries. That isolates where the segmentation’s robustness actually comes from, and it is not the cross-scale weighting. The votes only refine boundaries that the occupancy skeleton has already placed (Section 5.2.4), and that skeleton is fixed by the iteratively-reweighted tilt floor and the energy gate, neither of which the weighting touches.

This is a negative result and it is reported as one. Section 2.14.1 anticipated it. The entropy weighting is retained because it is a measurement rather than an assumption, and because Section 5.2.3 argues it should degrade more gracefully on a band where

one scale genuinely is dominated by noise. That claim is untested here.

7.1.7 Threats to validity

Four caveats bound these numbers.

First, the benchmark is simulated. The transponder model of Chapter 4 carries the impairments that matter to this detector: additive noise, gain tilt, frequency offset and a no-guard-band pair. It omits the non-linear effects a real transponder exhibits near saturation, among them intermodulation products and AM/AM compression. A CUC hidden beneath an intermodulation product is not represented here.

Second, the presets are calibrated on this benchmark. The three operating points were chosen with these twelve events visible, so the recall figures are not out-of-sample. What that costs is bounded by the form the calibration takes. Every number in this section comes from one fixed configuration applied unchanged to all ten scenarios: the conservative preset $(h, p, [w_{\min}, w_{\max}])$ for the snapshot stage, and $k = 6$ with $\delta = 0.20$ dB for the temporal stage. Nothing is swept per scenario, and the conservative preset was fixed before the stress tests were run. A rule of that kind transfers to a new capture as it stands. The Layer 2 figures of Section 7.2 are instead produced by a threshold swept per channel against that channel's own labels, which transfers nowhere, and Section 7.2.4 reports what that layer achieves once the tuning is removed.

Third, the temporal stage's twelve-of-twelve depends on a baseline twin generated under the same conditions as the capture. Section 7.1.5 measures how far that assumption can be stretched along three axes, but a real commissioning capture drifts along more axes than three, and the benchmark cannot say what happens when several drift at once.

Fourth, no competing detector is run on this benchmark. Section 7.1.6 substitutes the reference method's cross-scale weighting into the pipeline developed here, but that is a component swap rather than a contest between methods. The comparisons here are internal: they establish what reference data buys over a single snapshot and what each design choice contributes, not superiority over the method this layer extends. The temporal stage is the part that is genuinely new, and the reference method has no counterpart to it, so no like-for-like contest is available even in principle; a fair external comparison would need a published detector that also consumes multiple frames against a clean reference.

7.2 Layer 2 Evaluation: Telemetry Anomaly Detection

This section evaluates the sparse-representation telemetry anomaly detector of Chapter 6 (a K-SVD–learned dictionary, an ADMM sparse reconstruction, and a residual-based decision, after He et al. [18]) on a public spacecraft-telemetry benchmark. Where Layer 1 is the framework’s single-reference instance, Layer 2 is its learned-dictionary instance. To isolate what the learned, overcomplete and *sparse* dictionary contributes, the K-SVD–ADMM detector is compared against a baseline that differs only in the model, a dense linear Principal Component Analysis (PCA) reconstruction, under an identical pipeline, scoring and operating-point protocol.

7.2.1 Benchmark and protocol

The evaluation uses the NASA Soil Moisture Active Passive (SMAP) and Mars Science Laboratory (MSL) telemetry benchmark released by Hundman et al. [21], the dataset on which the LSTM-based baseline cited in the reference paper was validated. The evaluation runs on its 81 distinct channels, 54 SMAP and 27 MSL (Section 4.5.1). Each channel provides an anomaly-free *train* split and a *test* split annotated with the index ranges of its anomalies, each a contiguous subsequence rather than an isolated sample. This train/test structure maps directly onto the offline/online split of the method: the dictionary is learned from the anomaly-free train split, and detection is performed on the labelled test split.

Each channel’s primary telemetry value is standardised using train-split statistics only (no test-split leakage) and segmented into overlapping windows of length w (shift 1), stacked as the columns of a sample matrix. Two normality models are then fitted to the train windows:

- **K-SVD–ADMM:** an overcomplete dictionary learned with K-SVD [1]; test windows are sparse-reconstructed by an ADMM solver [2] that separates a sparse code from a column-sparse residual matrix E , and the per-window anomaly score is the residual column norm $\|E(:, j)\|_2$.
- **PCA reconstruction (baseline):** the top- k principal directions of the train windows; the per-window score is the norm of the residual after projecting the window onto that subspace.

Both models score a window by how badly the normal model reconstructs it; they differ only in that K-SVD–ADMM is overcomplete and sparse whereas PCA is dense, linear

and orthogonal. The comparison therefore isolates the contribution of sparsity and over-completeness rather than confounding it with a change of detection principle. Window scores are mapped back to per-time-step scores by assigning each point the maximum score of any window covering it. Unless stated otherwise the operating parameters are $w = 60$, 128 dictionary atoms, sparsity $T_0 = 4$, 10 K-SVD iterations, ADMM penalties $\alpha = 1$, $\beta = 2.6$, and $k = 10$ PCA components.

7.2.2 Scoring methodology

Two complementary scoring conventions are reported, because they answer different questions and, as the results show, can disagree on which detector is preferable.

- **Point-wise.** Each time step is scored independently against its label. This rewards *precise localisation*, flagging the correct time steps and few others, and is the stricter measure.
- **Point-adjusted** [37]. If any time step inside a labelled anomaly segment is flagged, the entire segment is counted as detected. This is the convention under which the published baselines on SMAP/MSL were scored, so it is required for any comparison against them.

For each channel and method a per-method operating point is chosen by sweeping the decision threshold and retaining the value that maximises point-adjusted F_1 , mirroring the “threshold at the ROC point closest to the ideal” protocol of the reference paper. As discussed in Section 7.2.6, this “best- F_1 ” selection uses label knowledge and is therefore optimistic. It is the usual way to score detectors that ship no threshold rule of their own [37], and the same protocol is applied to both methods so the comparison between them remains fair. Precision, recall, F_1 and the false-positive rate (FPR) are micro-averaged over channels, both overall and per spacecraft.

7.2.3 Results

Table 7.2.1 reports the overall comparison across all 81 channels under both scoring conventions. The two methods rank *oppositely* depending on the metric: K-SVD–ADMM attains the higher point-wise F_1 (0.522 vs. 0.437) together with higher point-wise precision and a lower false-positive rate, whereas PCA attains the higher point-adjusted F_1 (0.841 vs. 0.786), driven almost entirely by a very high point-adjusted recall (0.950).

Table 7.2.2 breaks the point-wise and point-adjusted F_1 down per spacecraft and places them beside a published external reference: the LSTM-NDT detector of Hundman et al.

Table 7.2.1: Overall telemetry anomaly-detection performance on SMAP/MSL (81 channels, micro-averaged). Best value per column in bold.

Method	Point-wise			Point-adjusted			
	P	R	F_1	P	R	F_1	FPR
K-SVD–ADMM	0.647	0.437	0.522	0.771	0.802	0.786	0.034
PCA	0.542	0.367	0.437	0.754	0.950	0.841	0.044

[21], as re-evaluated under point-adjusted scoring by Su et al. [37]. On point-adjusted F_1 both present methods fall below LSTM-NDT on SMAP, by 0.045 for PCA and 0.105 for K-SVD–ADMM, and both *exceed* it on MSL, by 0.250 and 0.224 respectively. These published figures come from LSTM-NDT’s own label-free threshold rather than from one tuned against the labels, and are therefore offered as a reference band, not a controlled comparison; the only strictly controlled comparison here is K-SVD–ADMM versus PCA.

Table 7.2.2: Per-spacecraft F_1 and comparison with the published LSTM-NDT reference [21], point-adjusted scores as reported by Su et al. [37]; context only, as the protocol differs.

Spacecraft	Method	Point-wise F_1	Point-adjusted F_1
SMAP	K-SVD–ADMM	0.532	0.785
	PCA	0.443	0.845
	LSTM-NDT (publ.)	–	0.890
MSL	K-SVD–ADMM	0.458	0.788
	PCA	0.397	0.814
	LSTM-NDT (publ.)	–	0.564

Figure 7.2.1 summarises both findings. The left panel makes the metric-dependent reversal explicit; the right panel situates the point-adjusted scores against the published reference per spacecraft. Figure 7.2.2 shows a representative channel, SMAP E-1, the same channel whose frozen-sensor fault opened this thesis in Figure 1.3.2(d). It plots the telemetry trace, the per-window residual scores of both methods, and the selected thresholds, with the labelled anomaly shaded.

7.2.4 The threshold prior: second design axis

The comparison above varies the normality model and holds the threshold fixed, which instruments Axis 1 of Section 3.3. Axis 2 is the prior built into the threshold, and it needs the opposite experiment: hold the model fixed and vary the rule. That experiment is reported here, for two reasons. The framework claims two axes and had measured only one. The figures above are also produced by a swept scalar threshold rather than by the period-referenced band of Section 6.5, so the band’s own contribution was nowhere

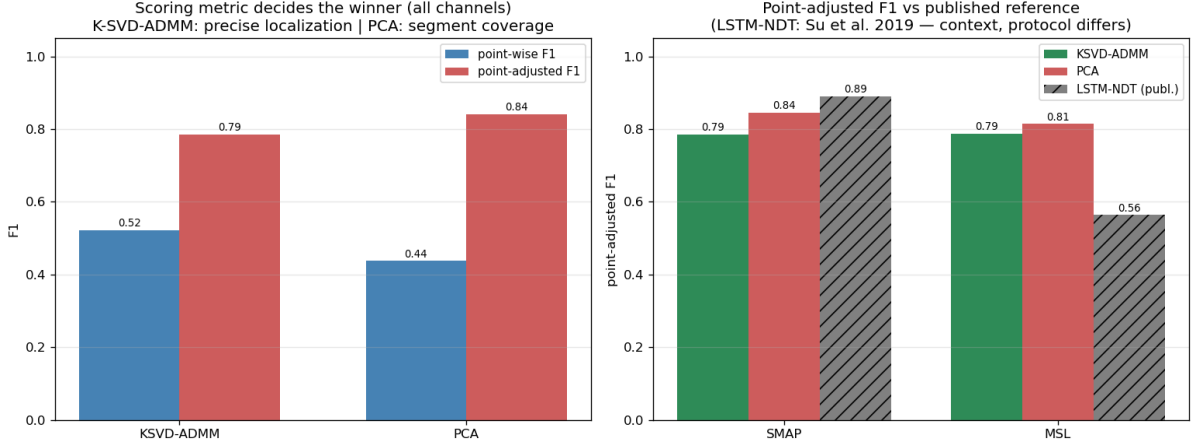


Figure 7.2.1: Telemetry anomaly-detection comparison. *Left*: the scoring metric determines the apparent winner: K-SVD–ADMM leads on point-wise F_1 (precise localisation), PCA leads on point-adjusted F_1 (segment coverage). *Right*: point-adjusted F_1 per spacecraft against the published LSTM-NDT reference (hatched; protocol differs).

quantified.

Five rules are run on *identical* K-SVD–ADMM scores, over the same 81 channels, under the same best- F_1 protocol. Only the threshold differs, so any difference between them is attributable to the prior alone.

- **Constant (oracle).** A scalar threshold swept for best point-adjusted F_1 . This is the rule behind every other number in this chapter. It reads the labels and is therefore not deployable.
- **CFAR (swept).** $\text{med}(r) + k \hat{\sigma}$ with k swept, the structure-free end of Axis 2 and the rule Layer 1 uses.
- **CFAR (fixed).** The same rule at $k = 4$, with no sweep at all. No labels are read, so this is the only deployable operating point in the comparison.
- **EWMA (plain).** The band of Equation (6.9) with $p = 1$: adaptive, but with no periodicity prior.
- **EWMA (period).** The same band with the estimated period. This is the reference method’s rule and the far end of Axis 2.

The plain variant matters because the band changes two things at once relative to a constant threshold. It becomes adaptive, and it becomes periodic. Setting $p = 1$ separates the two.

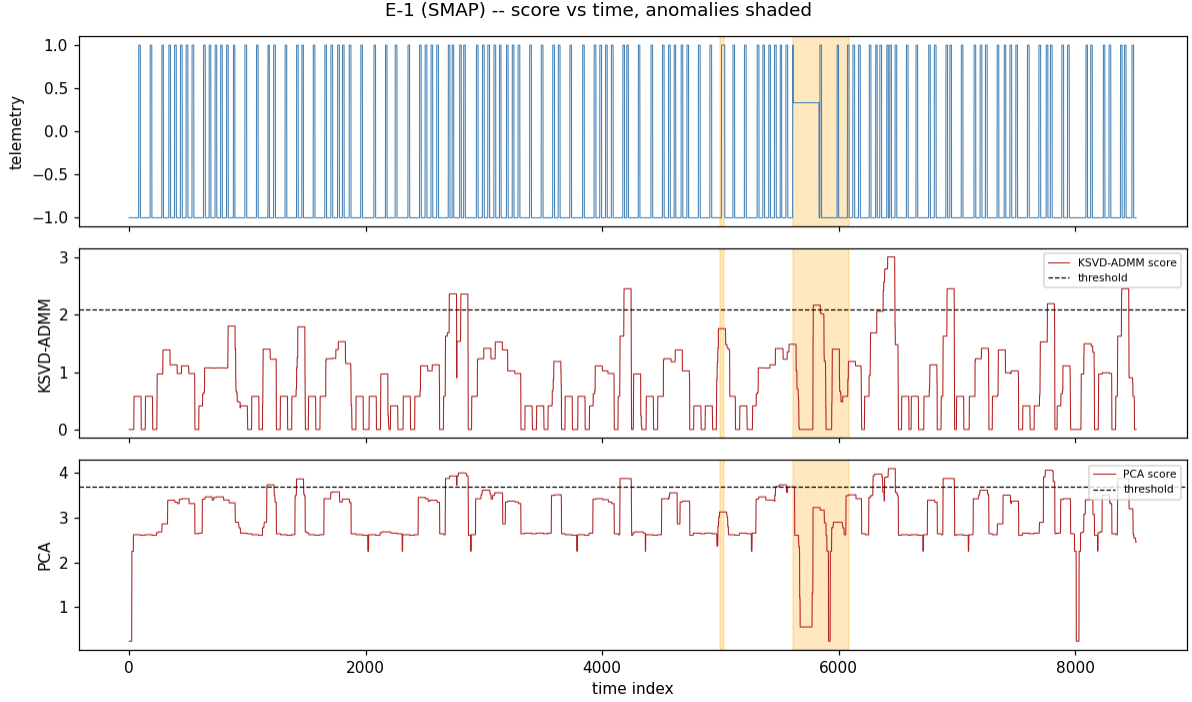


Figure 7.2.2: Representative SMAP channel: telemetry trace (top) and per-window residual scores with the selected threshold for each method, with the labelled anomaly segment shaded.

The periodicity prior earns its place, but only where a period exists. Comparing the two EWMA rows isolates the prior from the adaptivity. Averaged per channel, the period-referenced band beats the plain one by 0.152 in point-adjusted F_1 on the 21 channels where the autocorrelation finds a genuine period, winning on ten and losing on two. On the 60 channels where it does not, the margin falls to 0.073. The prior is doing real work, and it does more of it where its assumption holds.

The framework’s prediction is half confirmed. Section 3.3 claims the two ends of Axis 2 trade generality against structure: the periodic band should win where the process is periodic and lose where it is not. Comparing the two ends within each group of channels, by per-channel mean point-adjusted F_1 :

	Periodic (21)	Aperiodic (60)
CFAR (swept)	0.601	0.684
EWMA (period)	0.609	0.625

The means order as predicted in both groups, but the per-channel counts show that only one half of the prediction holds. On the 60 aperiodic channels the structure-free limit wins on 30 and loses on 11, the rest tying, a split a sign test puts at $p \approx 0.004$. All sixteen

Table 7.2.3: Axis-2 ablation: five threshold rules on identical K-SVD–ADMM scores, 81 channels, micro-averaged. Only the threshold differs.

Threshold	Point-wise			Point-adjusted			
	P	R	F_1	P	R	F_1	FPR
Constant (oracle)	0.647	0.437	0.522	0.771	0.802	0.786	0.034
CFAR (swept)	0.540	0.418	0.471	0.690	0.794	0.738	0.051
CFAR (fixed, $k = 4$)	0.473	0.135	0.210	0.724	0.395	0.511	0.022
EWMA (plain)	0.455	0.244	0.318	0.640	0.520	0.574	0.042
EWMA (period)	0.530	0.273	0.361	0.733	0.665	0.697	0.035

channels of the fourth caveat in Section 7.2.6 fall in this group, and dropping them leaves the direction intact at 21 wins to 10, but the split then carries $p \approx 0.07$, so the effect is consistent rather than resolved at this sample size. On the 21 periodic channels the band’s 0.008 lead in the mean comes with 5 wins, 8 losses and 8 ties ($p \approx 0.58$), so there the two rules are indistinguishable. The periodic prior therefore costs performance where its assumption fails and shows no measurable gain where it holds, although the band is tuned over a 231-point grid and the limit over a single multiplier. Axis 2 is still a real axis: moving along it changes performance, and the change depends on the data. On this benchmark it favours the structure-free end.

A deployable operating point. Every other figure in this chapter, and in the literature it compares against, comes from a threshold tuned against the labels. CFAR at a fixed $k = 4$ is the exception. It reaches a point-adjusted F_1 of 0.511 at a false-positive rate of 0.022, with precision 0.724, having read no ground truth at all. It detects roughly two in five labelled anomaly segments while raising false alarms on about two per cent of samples. That is far below the oracle rules, as it must be, but it is the only number here that would survive being deployed. It is also the same family of rule as Layer 1’s threshold, a median plus a robust scale multiple, so the unification argued in Section 3.6 is exercised in both domains rather than only argued.

Limits of this comparison. Three. The constant rule sweeps a single free scalar directly against the labels, which makes it close to the best achievable constant threshold and a demanding baseline for a rule whose shape is constrained. The band is swept over the reference method’s own grid, $\eta \in [0, 2]$ and $\gamma \in [0, 1]$ in steps of 0.1; widening that grid tenfold changed the outcome by less than 0.001, so the band is limited by its form rather than by its tuning. Finally, only 21 of the 81 channels carry a detectable period at all, so the null result on the periodic side rests on a quarter of the benchmark and could hide a modest effect.

7.2.5 Discussion

The central, controlled finding is that *the sparse, overcomplete dictionary buys localisation precision*. Under the strict point-wise measure K-SVD–ADMM improves F_1 by 0.085 over the dense-linear PCA baseline and lowers the false-positive rate (0.034 vs. 0.044): it flags the correct time steps and few others. That margin is the conservative reading. On the 65 channels whose training split lets a model be learned at all, it grows to 0.114, for the reason the fourth caveat of Section 7.2.6 gives. PCA’s advantage appears only under point-adjusted scoring, and its mechanism is visible in the underlying rates: a near-saturated point-adjusted recall (0.950) combined with a *lower* point-wise precision than K-SVD–ADMM. PCA fires broadly enough to touch some point inside almost every anomaly segment, and point-adjusted scoring then credits the entire segment for that single hit. This is precisely the inflation behaviour that Kim et al. [23] identify as a weakness of point-adjusted evaluation on these benchmarks, under which even simple detectors score highly. The two detectors optimise different objectives, precise localisation versus segment coverage, and a single headline metric would conceal the trade. For an operational telemetry monitor, where the value lies in identifying *when* and *where* a fault occurs rather than merely that something happened somewhere in a long window, the point-wise behaviour, on which the sparse method leads, is the more relevant criterion.

Placed against the published LSTM-NDT reference, both methods trail it on SMAP and exceed it on MSL, with a markedly simpler model and no deep network. The protocol difference runs in their favour: LSTM-NDT’s threshold reads no labels, and Su et al. [37] found it scores no differently from that method’s best achievable one, while the tuned threshold here is worth 0.275 in point-adjusted F_1 to the sparse detector (0.786 against 0.511 label-free). The comparison places the two detectors within the range of a published deep-learning result; it does not rank them above it.

7.2.6 Threats to validity

Three caveats bound the interpretation of these results. First, every headline figure uses a decision threshold selected to maximise point-adjusted F_1 against the labels. This “best- F_1 ” protocol is standard in this literature for detectors that have no threshold rule of their own. The threshold could not be set in flight, where no labels exist, so the figures are an upper bound on achievable performance rather than an achievable operating point. The comparison *between* methods is unaffected, since the identical protocol is applied to both. Second, only one baseline (PCA) is run under fully controlled conditions; the LSTM-NDT figures are context, for the reason given in the discussion above.

Third, 16 of the 81 channels carry a training split of constant value. Standardisation maps every training window to zeros, so K-SVD returns a dictionary of zero atoms and the PCA fit returns the first ten canonical axes rather than anything learned from the channel. Neither model then holds information about the data, and both detectors reduce to the magnitude of the standardised test window. Both score well on exactly those channels: five reach a point-adjusted F_1 of 1.000 and fourteen reach a point-adjusted recall of 1.000. This is the triviality Wu and Keogh [40] report for the NASA archive, reached here through the model rather than through the labels. Removing the sixteen lowers the point-wise F_1 from 0.522 to 0.422 for K-SVD–ADMM and from 0.437 to 0.308 for PCA, while both point-adjusted figures move by less than 0.01. The controlled comparison survives that removal and widens, from 0.085 to 0.114 point-wise, because the degenerate channels flatter the dense baseline more than the sparse one. The headline figures cover all 81 channels, since the published comparators are scored on the same file.

Chapter 8

Conclusions and Future Work

8.1 What Was Built

This thesis set out to monitor a satellite link from two directions at once, and to do so under one principle rather than two unrelated toolboxes. Four objectives were stated in Section 1.4, and each is met.

O1, a reproducible environment and benchmark. A scenario-driven GNU Radio model of a 36 MHz FDMA transponder generates 40 MS/s complex baseband captures carrying up to sixteen simultaneous QPSK carriers, with controllable gain tilt, frequency offset, noise and a deliberately overlapping carrier pair. Ten scenarios carry twelve Carrier-under-Carrier events between them, spanning density excesses from 0.35 to 2.53 dB, each with a JSON manifest recording the full carrier plan and an explicit interferer flag, and each with a clean twin generated under the same plan with different payload seeds. Scoring reads the manifest and the detectors read only the samples, so every number in Chapter 7 traces to a versioned definition of correctness.

O2, reliable CUC detection. The interference layer segments the band with a multi-scale wavelet edge detector and covers every ground-truth carrier in all ten scenarios. Its snapshot stage detects seven of twelve interferers at zero false alarms across the whole benchmark. Its temporal stage, differencing an integrated spectrum against the clean twin, detects all twelve at zero false alarms at $N = 2048$ frames, and places each within 0.002 to 0.066 MHz of its true centre frequency. Both figures come from one fixed configuration applied unchanged to every scenario.

O3, telemetry anomaly detection. The telemetry layer learns a 128-atom dictionary by K-SVD from anomaly-free windows and reconstructs test windows by an ADMM solver

whose residual matrix is column-sparse, so the detection decision is taken inside the optimisation rather than after it. On 81 channels of the public SMAP/MSL benchmark it reaches a point-wise F_1 of 0.522 against 0.437 for a Principal Component Analysis baseline that differs only in the model, with higher precision and a lower false-alarm rate.

O4, unification and honest evaluation. Both detectors are shown to be instances of one reconstruction-residual scheme, positioned in a two-dimensional design space, and both axes of that space are sampled experimentally. The evaluation reports point-wise scores alongside the point-adjusted convention the literature uses, and states where the two disagree.

8.2 What the Results Establish

Three findings carry beyond the particular detectors built here.

Reference data buys more than integration does. The snapshot stage's limit is not noise. Averaging more frames does not lift its recall and past a few hundred frames it lowers it, from seven of twelve at $N = 400$ to five at $N = 2048$, because what limits it is the host carrier's own deterministic structure, which averaging preserves. Differencing against a capture that *shares* that structure removes it, and the residual then falls as $1/\sqrt{N}$ while the interferer's excess does not. The two CUCs no preset ever reaches are precisely the two the theory predicts: one too wide to produce a localised feature, one sitting on a host whose own ripple matches its excess.

The choice of scoring metric can decide the apparent winner. Under point-wise scoring the sparse dictionary leads the dense linear baseline by 0.085 in F_1 , and by 0.114 once the channels with a constant training split are set aside (Section 7.2.6). Under point-adjusted scoring the ordering reverses, and the mechanism is visible in the underlying rates: the baseline reaches a near-saturated point-adjusted recall of 0.950 while holding a *lower* point-wise precision, which is to say it fires broadly enough to touch some point inside almost every labelled segment and is then credited with the whole of it. A single headline number would have concealed the trade entirely.

Model richness and threshold prior are separable, and both matter. The design space is not decorative. Holding the threshold fixed and varying the model gives the sparse dictionary its advantage in localisation precision. Holding the model fixed and varying the threshold shows the periodicity prior costing performance where no period exists and gaining nothing measurable where one does, so on this benchmark the structure-free end of the axis is the better choice.

8.3 The Unexpected Results

Four results ran against expectation, and each turned out to be more informative than the confirmation would have been. Three concern components or conventions inherited from elsewhere. The first is a null on a proposal made here.

Data-adaptive entropy weighting of the cross-scale votes was introduced in Section 5.2.3 to replace the reference method's fixed triangular weights with a measurement rather than an assumption. It makes no measurable difference: the two schemes return segment boundaries that agree bin for bin across all ten scenarios, and the same seven detections at the same zero false alarms. This is the one refinement proposed here that does not change the outcome, and it is reported as such. The finding is still informative, because the two schemes weight the scales quite differently and yet produce identical output, which locates the segmentation stage's robustness elsewhere: in the iteratively-reweighted tilt floor and the energy gate, neither of which the weighting touches. Nothing else is affected: the interference layer's substantive contribution is the temporal stage, which takes detection from seven of twelve to twelve of twelve, and no part of that result depends on the weighting.

A repeating payload defeats baseline differencing outright. A 10 kB looping buffer gives each carrier a line spectrum that is identical in every frame and therefore survives averaging, leaving a residual floor of 0.29 dB that swallowed the deepest benchmark interferer whole. The cure was a generation parameter rather than a detector change, and the episode is the clearest instance in this work of the principle that whatever a detector must cancel, its reference must genuinely share.

The purely multiplicative threshold band of the reference telemetry method degenerates exactly where that method succeeds best. A well-trained dictionary reconstructs normal windows so completely that the ADMM step zeroes their residual columns outright, the tracked statistic collapses toward zero, and a band proportional to it collapses with it. The additive floor introduced as a repair is the Constant-False-Alarm-Rate margin the interference layer already used, which is how the two layers arrived at a common threshold without being designed to.

Finally, the periodicity prior does not transfer to this benchmark. It is a liability on the sixty channels with no detectable period and draws level with the structure-free rule on the twenty-one that have one, so the configuration reported throughout uses the structure-free rule. The framework predicted the first half of this. The second half means the reference method's band showed no advantage anywhere on SMAP/MSL.

8.4 Limitations

The full set is stated with the results, in Sections 7.1.7 and 7.2.6. Two bound the conclusions most tightly. The interference layer is evaluated on simulated captures only, and no competing detector is run against it, so its results establish what each design choice contributes without establishing superiority over the method it extends. The telemetry layer’s headline figures use a per-channel threshold selected against the labels, which is common in this literature but is not an operating point a spacecraft could adopt; Section 7.2.4 reports what remains when that tuning is removed.

8.5 Future Work

Cross-layer fusion. The two layers share a framework but not data. Realising the cross-validation that Section 1.3 gives as a central motivation requires jointly-coupled RF and telemetry captures that neither benchmark provides.

Cross-channel telemetry modelling. The telemetry layer models one channel at a time. A fault that leaves every channel individually plausible while breaking the relationship between them, a velocity inconsistent with the accelerations recorded beside it, is outside its reach. The SMAP/MSL release already carries per-sample command flags that the present pipeline discards, which makes a multivariate extension the most direct next step.

A deployable operating point. Fixing the CFAR multiplier at $k = 4$ gives the telemetry layer a threshold that reads no labels, reaching a point-adjusted F_1 of 0.511 at a false-alarm rate of 0.022. Choosing k on held-out channels would make that figure defensible as an operating point rather than an illustration.

Closing the gaps in Layer 1’s evaluation. The interference layer has never seen a real transponder. Recorded captures would test it against the intermodulation products and amplifier compression the model omits, and against baselines that age in more ways than the three axes measured here. It also has no external comparator; a published multi-frame detector working against a clean reference would supply the head-to-head comparison the telemetry layer already has.

Appendix A

Acronyms

ADMM Alternating Direction Method of Multipliers

AWGN Additive White Gaussian Noise

CFAR Constant False Alarm Rate

CUC Carrier-under-Carrier

CWT Continuous Wavelet Transform

DFT Discrete Fourier Transform

DWT Discrete Wavelet Transform

EWMA Exponentially Weighted Moving Average

FDMA Frequency-Division Multiple Access

FFT Fast Fourier Transform

FPR False-Positive Rate

I/Q In-phase and Quadrature

ISI Intersymbol Interference

JSON JavaScript Object Notation

K-SVD Dictionary-learning algorithm generalising K -means by singular value decomposition

LSTM Long Short-Term Memory

LSTM-NDT Long Short-Term Memory with Nonparametric Dynamic Thresholding

MAD Median Absolute Deviation

MSL Mars Science Laboratory

NASA National Aeronautics and Space Administration

OMP Orthogonal Matching Pursuit

PCA Principal Component Analysis

PSD Power Spectral Density

QPSK Quadrature Phase-Shift Keying

RC Raised Cosine

RF Radio Frequency

ROC Receiver Operating Characteristic

RRC Root-Raised Cosine

SMAP Soil Moisture Active Passive

SNR Signal-to-Noise Ratio

SVD Singular Value Decomposition

Appendix B

Code and Reproducibility

Objective O1 of Section 1.4 claims that detection performance in this thesis can be measured objectively. That claim rests on every reported number being regenerable from a public repository rather than transcribed by hand. This appendix records which script produces which result, and where each writes its output, so that any figure or table in Chapters 5 to 7 can be traced to the code and data that produced it.

The repository carries the code for both detection layers and the GNU Radio transponder simulation behind the RF benchmark. It also holds the CSV files and figures used to produce every table and plot in this thesis:

<https://github.com/dcharistes/A-Dual-Layer-Framework-for-Interference-and-Anomaly-Detection-in-Sattelite-Systems>

The captures themselves are not there. Ten scenarios and their clean twins at 40 MS/s complex64 are past what a repository will hold, so data/ is left out and the scenario generator stands in for it: rerunning `experiment_scenarios.py` rebuilds both the captures and the manifests that describe them. The telemetry layer reads NASA's public SMAP/MSL release, which is not redistributed here either.

All commands are run from the repository root. The interference layer requires GNU Radio for data generation only; the detectors themselves depend on NumPy, SciPy, PyWavelets and Matplotlib.

Data generation

`transponder_gnuradio/experiment_scenarios.py`

Generates the ten benchmark captures E01–E10, their clean baseline twins, and one JSON manifest per scenario recording the full carrier plan, the interferer flags and the

impairment settings. Output: `data/experiments/`. This is the source of Table 4.4.1 and of the ground truth every score in Chapter 7 is measured against.

`transponder_gnuradio/fdma_tsp.py`

The transponder flowgraph itself: per-carrier RRC shaping, frequency translation, power scaling, gain tilt, global frequency offset and noise. Its `data_len` parameter is the payload length whose effect Figure 4.4.1 shows.

Layer 1: interference detection

`src/rf_interference/studies/generate_reports.py`

Runs the snapshot stage over all ten scenarios at each sensitivity preset. Writes `summary_metrics.csv` (per-scenario coverage, detections and localisation), `cuc_detail.csv` (one row per true CUC), `sensitivity_roc.csv` and `roc_by_experiment.csv`. Source of Table 7.1.1 and Figure 7.1.1.

`src/rf_interference/studies/multiframe_study.py`

Runs the temporal stage. Writes `multiframe_progression.csv` (the frame-count sweep), `baseline_benchmark.csv` (the twelve-of-twelve scoreboard) and `temporal_cuc_detail.csv` (per-CUC localisation). Source of Figure 7.1.2 and of the localisation figures in Section 7.1.4.

`src/rf_interference/studies/baseline_stress.py`

Perturbs the baseline against the capture in gain, tilt and frequency, with the registration steps enabled and disabled. Writes `baseline_stress.csv` and `baseline_stress_planchange.csv`. Source of Table 7.1.2 and Figure 7.1.3.

`src/rf_interference/studies/ablation_weighting.py`

Compares entropy against fixed triangular cross-scale weighting. Writes `ablation_weighting_benchmark.csv`. Source of Section 7.1.6.

Layer 2: telemetry anomaly detection

`src/telemetry_anomaly/evaluate.py`

Learns a dictionary per channel, reconstructs the test split by ADMM, and scores both the K-SVD–ADMM detector and the PCA baseline under point-wise and point-adjusted conventions. Writes `per_channel.csv` and `summary.csv`. Source of Tables 7.2.1 and 7.2.2.

`src/telemetry_anomaly/ablation_threshold.py`

Holds the model fixed and varies the threshold rule across the five points of Section 7.2.4. Writes `threshold_ablation.csv` and `threshold_ablation_per_channel.csv`. Source of Table 7.2.3.

`src/telemetry_anomaly/make_comparison.py`

Assembles `comparison.csv`, which places the two detectors beside the published LSTM-NDT figures.

Outputs

Metrics are written to `results/metrics/rf/` and `results/metrics/telemetry/`; figures to `results/figures/`. The figures reproduced in this document are copied into `thesis/doc/figures/`. Every table in Chapters 5 and 7 is transcribed from one of the CSV files named above, and each of those files is overwritten in full by the script that owns it, so a complete re-run reproduces the reported numbers rather than merging with them.

Two caveats apply to exact reproduction. The transponder flowgraph draws its payloads from seeded generators, so regenerating the captures reproduces the scenarios but not the identical sample stream; residual floors and localisation errors therefore move in the third decimal place. And the telemetry evaluation selects each operating point by sweeping against the labels, as Section 7.2.6 states, so its figures are reproducible but are not an operating point a deployed system could adopt.

Appendix C

Per-Event Detection Detail

With only twelve interference events in the benchmark, the pattern of which are missed carries more weight than the aggregate recall. Table C.0.1 gives that pattern in full: every Carrier-under-Carrier event in the benchmark, its parameters, the density excess those parameters imply through Equation (2.31), and the outcome under each stage of Layer 1. Powers are quoted relative to the host the interferer hides beneath, which is the quantity that enters the bump expression. Localisation is the distance between the reported centre and the manifest's true centre.

Table C.0.1: Every CUC in the benchmark, with the outcome under the snapshot stage at its conservative preset and under the temporal stage at $N = 2048$. A tick marks detection; localisation errors are in MHz. The two events no preset ever reaches, E02 and E07, are the two strongest relative to their hosts.

Scen.	f_c (MHz)	$P_c - P_h$ (dB)	BW (MHz)	Δ (dB)	Snapshot det. err.		Temporal det. err.	
E01	+16.0	-14	0.405	1.46	✓	0.015	✓	0.010
E01	+6.5	-13	0.338	0.79	✓	0.086	✓	0.017
E02	+1.0	-10	0.270	1.14	–	–	✓	0.043
E03	+13.5	-12	0.540	2.53	✓	0.024	✓	0.029
E04	+2.0	-13	0.405	0.67	–	–	✓	0.019
E05	+13.0	-15	0.473	1.04	✓	0.071	✓	0.012
E06	+0.0	-22	0.405	0.35	–	–	✓	0.002
E07	+5.0	-11	0.945	0.47	–	–	✓	0.041
E08	-8.0	-17	0.405	1.02	✓	0.084	✓	0.008
E08	+4.0	-13	0.338	1.46	✓	0.081	✓	0.017
E08	+14.0	-16	0.405	0.67	–	–	✓	0.013
E09	+6.5	-13	0.338	0.79	✓	0.042	✓	0.066
Total					7/12, 0 FA		12/12, 0 FA	

Three readings follow from the table that the aggregate counts conceal.

Detection by the snapshot stage does not order by power. The two events it never finds sit at -10 and -11 dB relative to their hosts, while the one at -22 dB is recovered at the high preset. What separates them is the density excess Δ and the geometry that produces it, not the power ratio alone.

Nor does it order by Δ cleanly. E02's 1.14 dB excess is missed while E01's 0.79 dB excess is found, because Δ measures the bump against the host's flat top and says nothing about the host's own structure. E02's host spans only about a hundred flat-top bins, and its deterministic ripple across them reaches 0.8 dB (Section 7.1.2).

The temporal stage's localisation is better than the snapshot stage's on six of the seven events both detect, and its single worst case is E09, the scenario carrying 6 dB of tilt and a 60 kHz offset (Section 7.1.4).

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